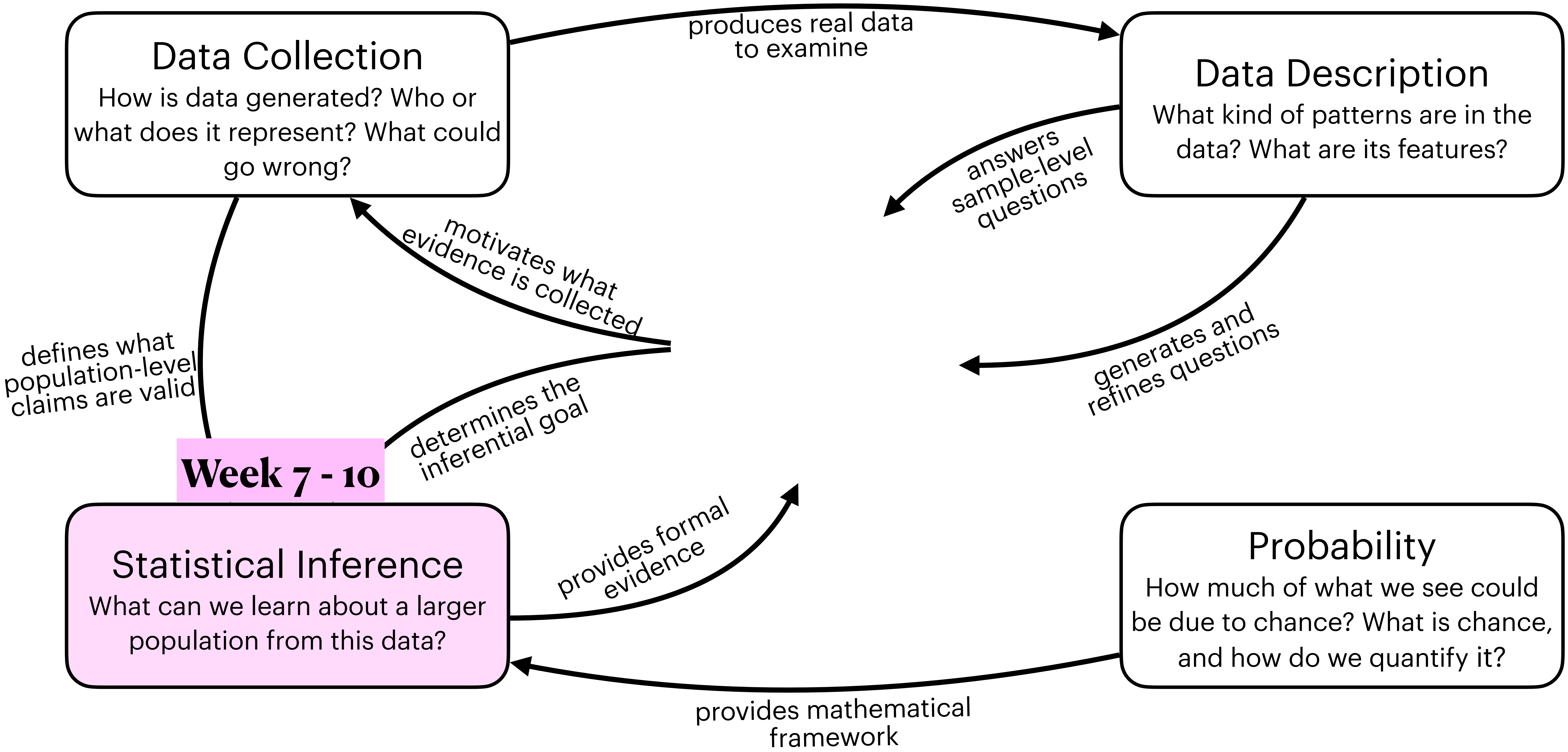
Inference for One Population

STATS 8, Topic 7
(Week 8/9)


Data Collection
How is data generated? Who or what does it represent? What could go wrong?
produces real data to examine
Data Description
What kind of patterns are in the data? What are its features?
answers sample-level questions
motivates what evidence is collected
defines what population-level claims are valid
generates and refines questions
determines the inferential goal
Week 7 - 10
Statistical Inference
What can we learn about a larger population from this data?
provides formal evidence
Probability
How much of what we see could be due to chance? What is chance, and how do we quantify it?
provides mathematical framework

Statistical Inference

- We will learn two types of statistical inference:
  - **Confidence intervals** — when you want to estimate the numeric value of a parameter, with some uncertainty around it. *What values of the parameter are plausible?*
  - **Hypothesis testing** — when you want to provide evidence for or against a claim about a parameter. *Is this one claimed value plausible?*

★<u>Remember</u>: the idea behind inference is that there is a population we are interested in…

  - but we cannot collect data on the whole population…
  - so we have to rely on a sample from that population to learn about it.

- <u>The uncertainty comes entirely from the fact that we are taking a random sample from a larger population rather than getting information from the whole population.</u>

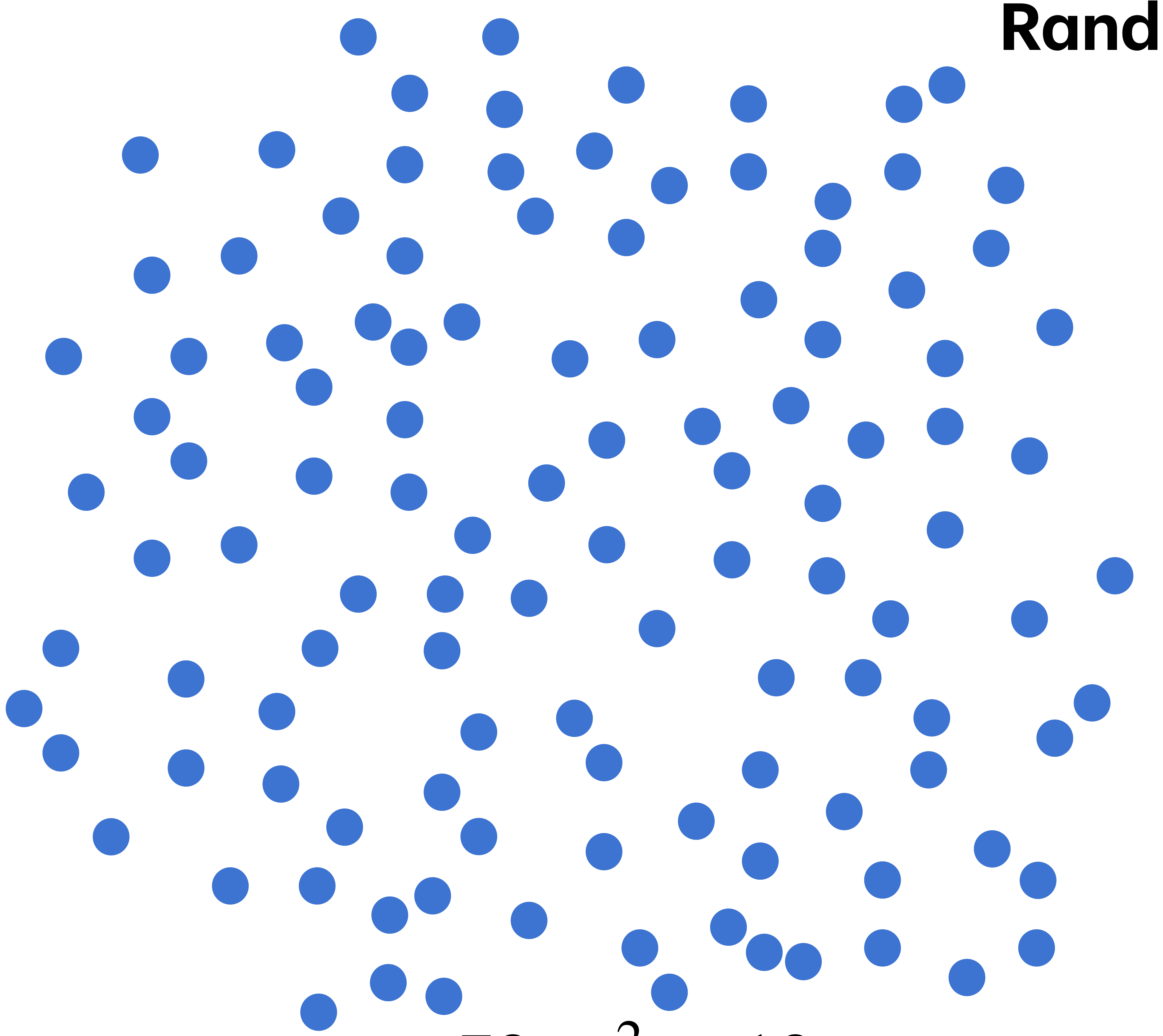
In these slides:

- Confidence intervals (and related topics: confidence level, margin of error, standard error, etc.) for a mean (μ) and a proportion (p)
- Hypothesis testing (and related topics: null and alternative hypotheses, p-values, significance level, types of error) for a mean (μ) and a proportion (p)
  - Both two-sided tests and one-sided tests

★This is a **lot** of material! These slides are meant to walk you through understanding, and procedural summaries will be given separately.

Sampling from a Population

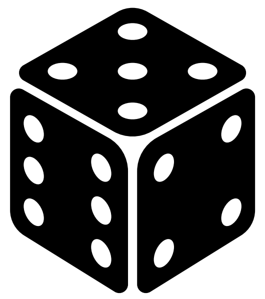
Random Sampling & CLT



$\mu = 72, \sigma^2 = 12$

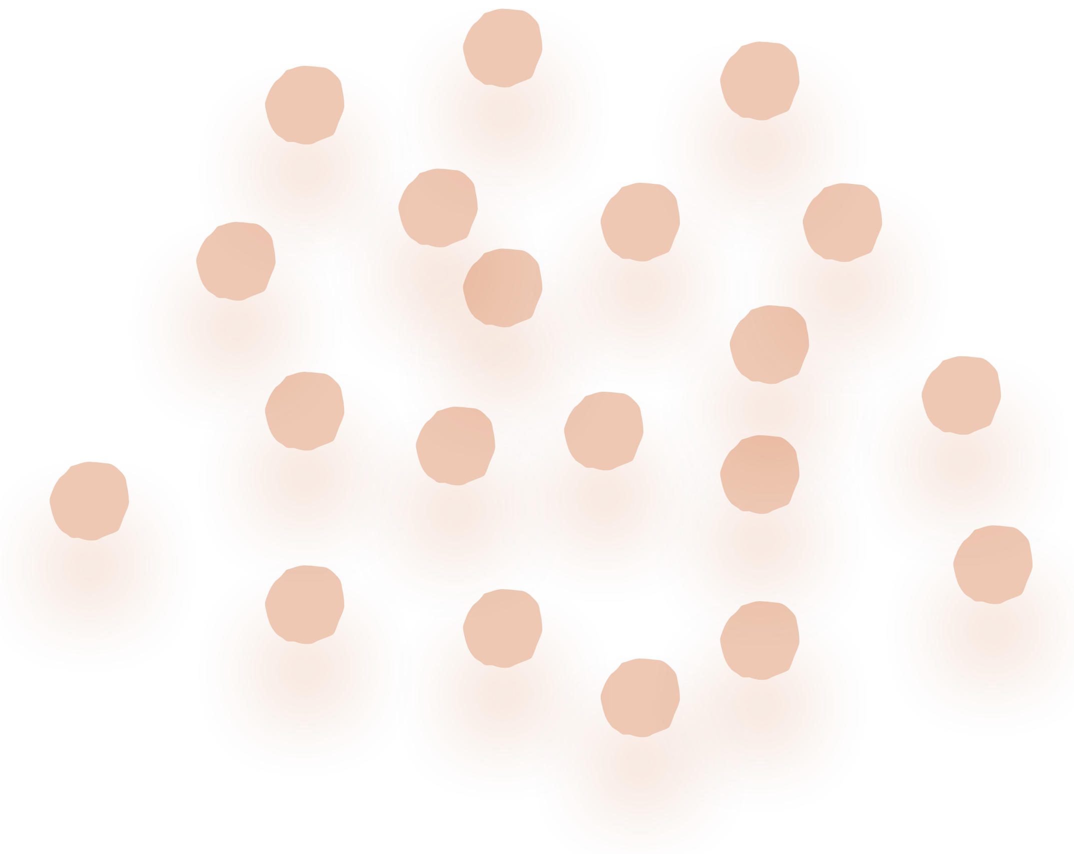
(mean and variance from the whole population)

Take a random sample of 20



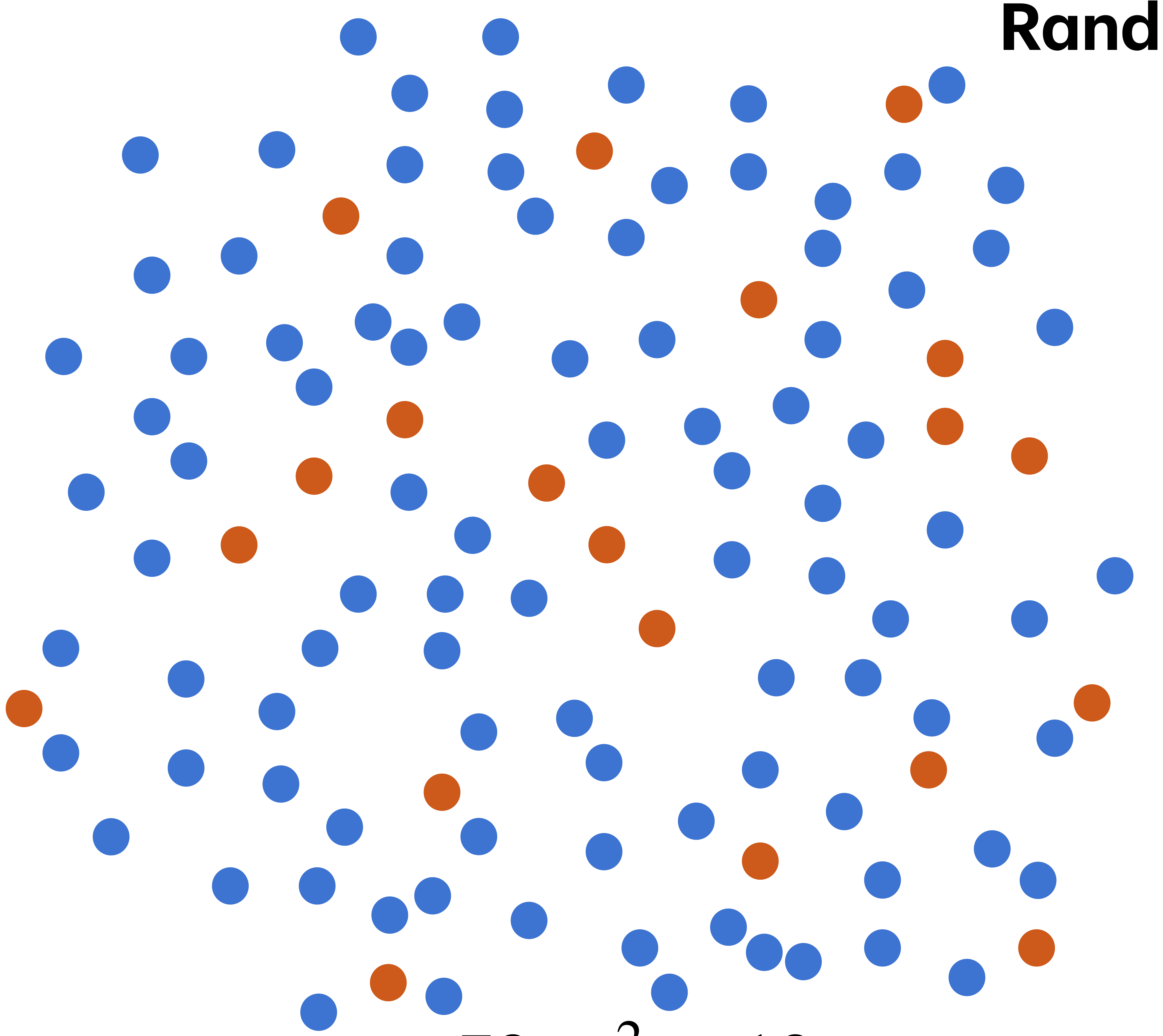
$$\bar{X} \sim N\left(72, \frac{12}{20}\right)$$

(assuming conditions for CLT apply!!)



Sampling from a Population

Random Sampling & CLT



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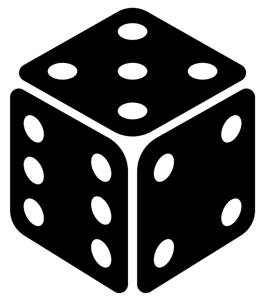

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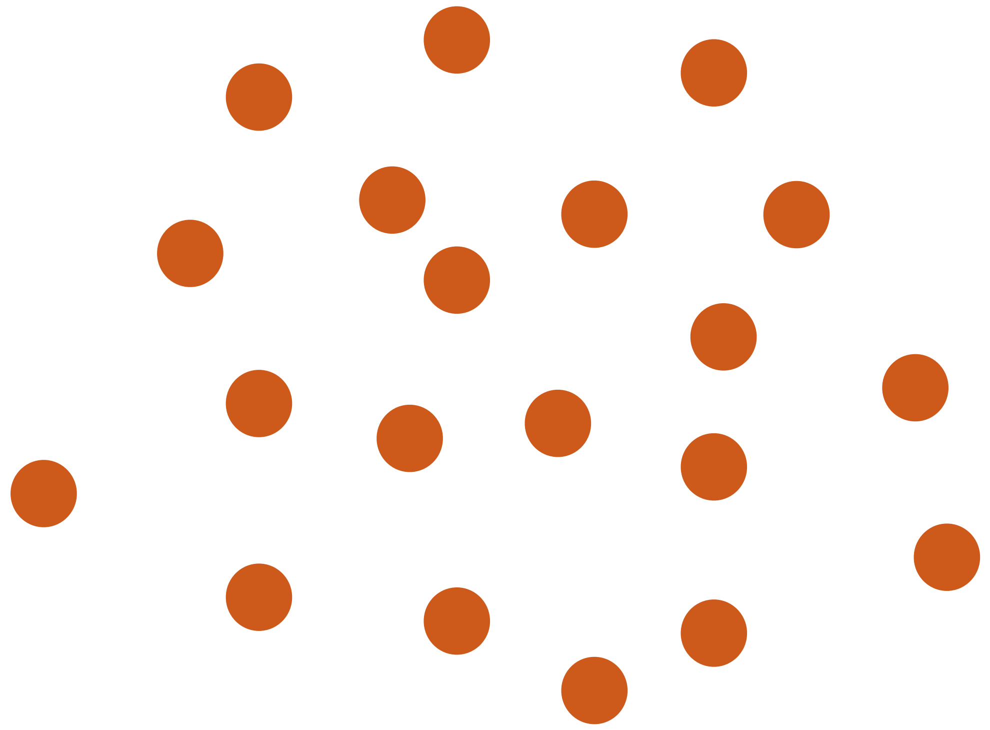
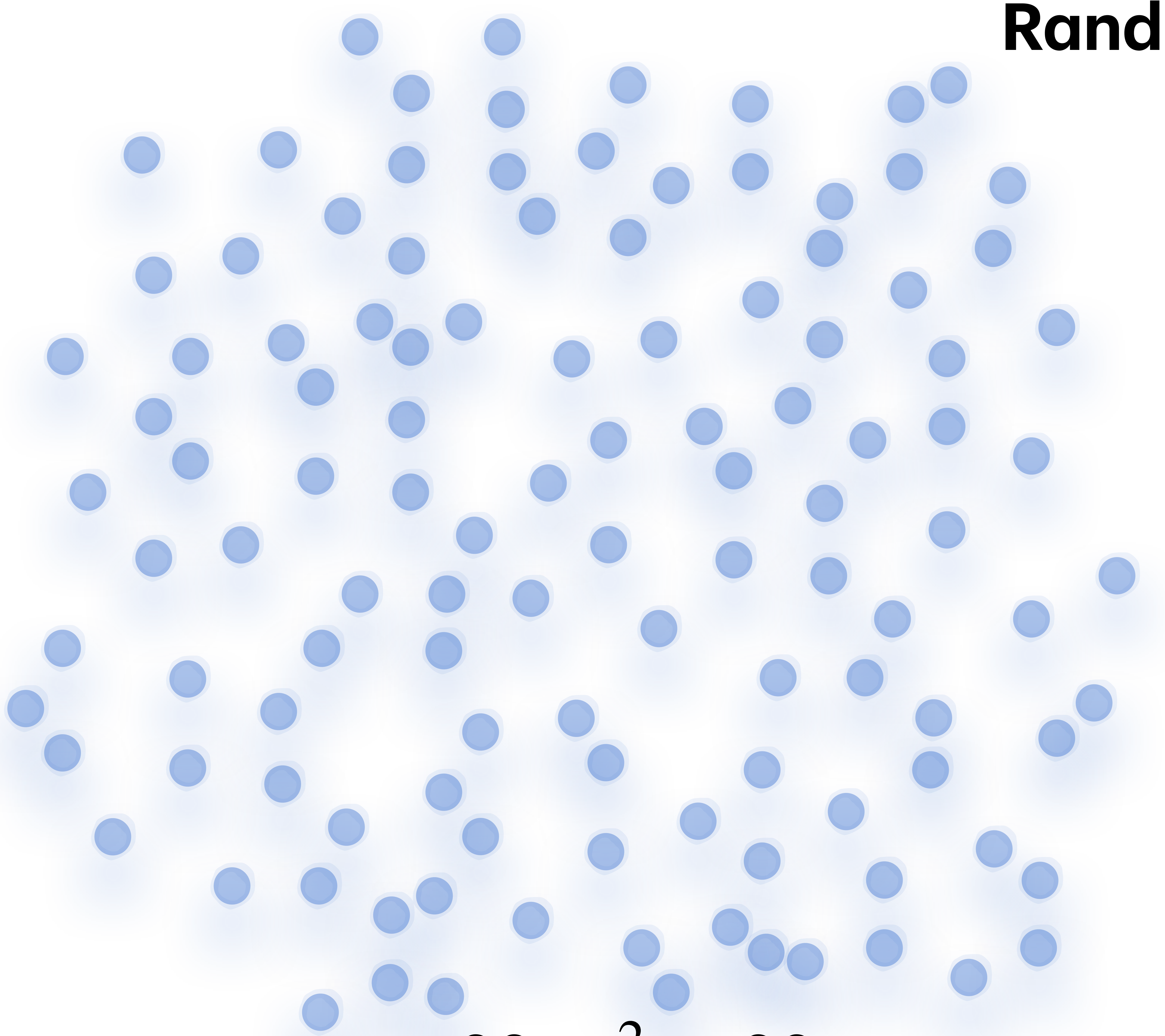


$$\bar{x} = 68,\ s^2 = 11$$

(mean and variance from the collected sample)

Statistical Inference

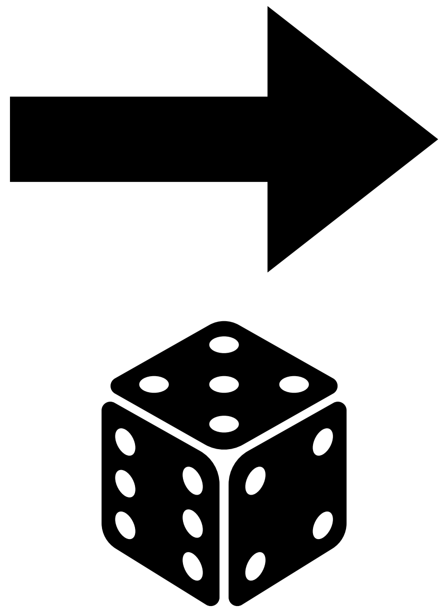
Random Sampling & CLT



$\mu = ??, \sigma^2 = ??$

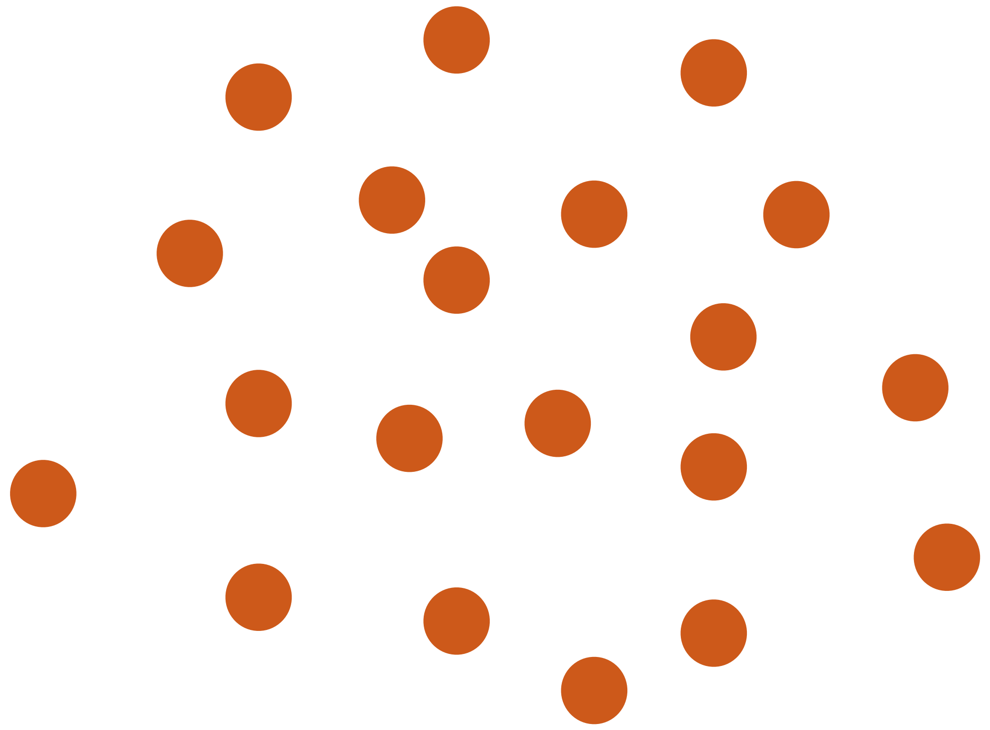
(mean and variance from the whole population)

Take a random sample of 20



$$\bar{X} \sim N\left(\mu, \frac{\sigma^2}{20}\right)$$

(assuming conditions for CLT apply!!)



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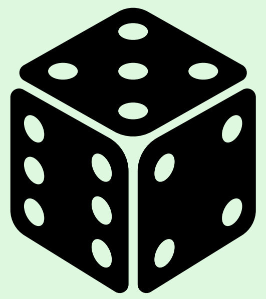
Confidence Interval Intuition

Take a random sample of 20



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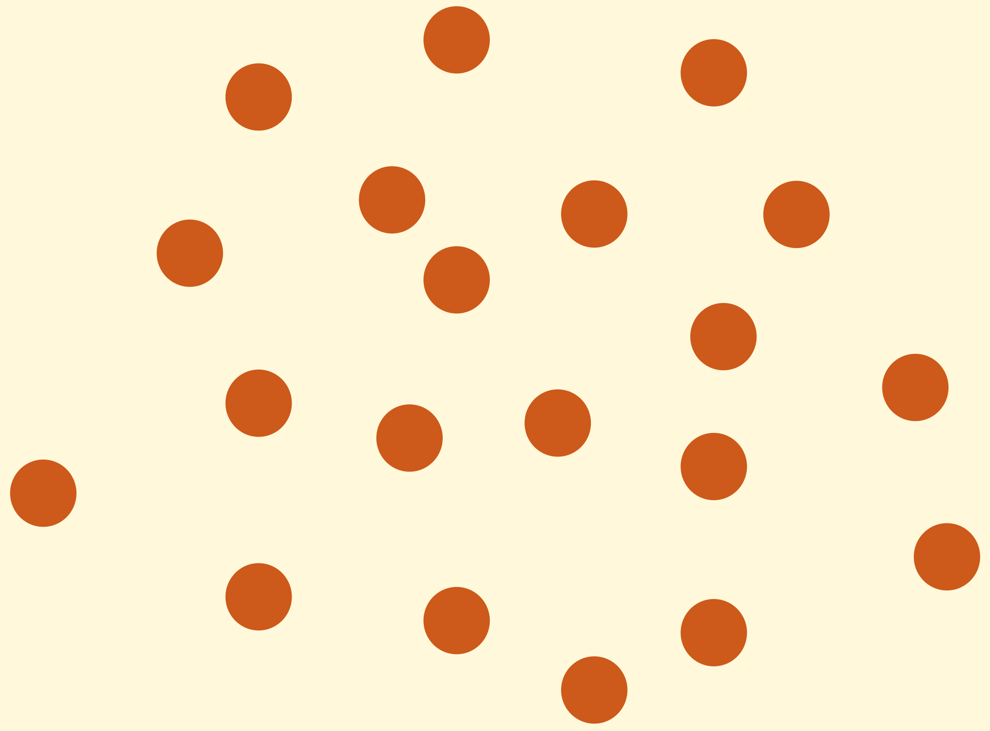


$$\bar{x} = 68,\ s^2 = 11$$

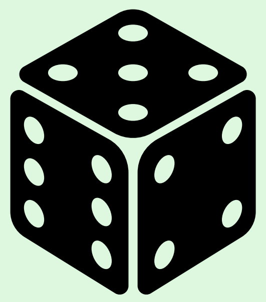
(mean and variance from the collected sample)

This is the random process → that creates this sample when the randomness is resolved

Confidence Interval Intuition

PollEv.com/stats8

Take a random sample of 20



$$\bar{X} \sim N\left(\mu, \frac{\sigma^2}{20}\right)$$

(assuming conditions for CLT apply!!)

What is the (approximate) probability that a random sample from this process has a mean between:

$$\mu - 2\frac{\sigma}{\sqrt{20}}$$

and

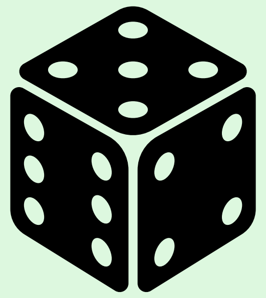
$$\mu + 2\frac{\sigma}{\sqrt{20}}?$$

Hint: empirical rule

Confidence Interval Intuition

PollEv.com/stats8

Take a random sample of 20



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What is the (approximate) probability that a random sample from this process has a mean between:

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and

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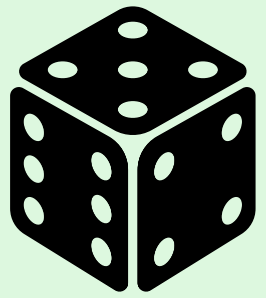
Hint: empirical rule

95%, or 0.95

Confidence Interval Intuition

Take a random sample of 20



$$\bar{X} \sim N\left(\mu, \frac{\sigma^2}{20}\right)$$

(assuming conditions for CLT apply!!)

To be more precise, the probability that $\bar{X}$ is between

$$\mu - 1.96\frac{\sigma}{\sqrt{20}}$$

and

$$\mu + 1.96\frac{\sigma}{\sqrt{20}}$$

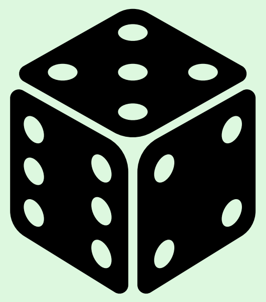
is 95%. Using our frequentist definition of probability, this means that **in many many samples of this size from this population, 95% of them will be within the above (unknown) interval.**

→use this to design a procedure to create an interval based on $\bar{x}$ that captures the unknown population mean 95% of the time?!

Confidence Intervals

Confidence Interval Intuition

Take a random sample of 20



$$\bar{X} \sim N\left(\mu, \frac{\sigma^2}{20}\right)$$

(assuming conditions for CLT apply!!)

From before we have:

$$P\left(\mu - 1.96\frac{\sigma}{\sqrt{20}} \leq \bar{X} \leq \mu + 1.96\frac{\sigma}{\sqrt{20}}\right) \approx 0.95$$

Rearranging, using some algebra:

$$P\left(\bar{X} - 1.96\frac{\sigma}{\sqrt{20}} \leq \mu \leq \bar{X} + 1.96\frac{\sigma}{\sqrt{20}}\right) \approx 0.95$$

This means that if we build an interval using the <u>random endpoints</u> $\bar{X} - 1.96\frac{\sigma}{\sqrt{20}}$ and $\bar{X} + 1.96\frac{\sigma}{\sqrt{20}}$, we have a 95% chance of capturing the unknown population mean

Confidence Interval

Definition and Procedure

- We would like to find out the value of a $\overset{\mu}{\underline{\text{population parameter}}}$ but we can't collect data on the entire population. We collect $\overset{n}{\underline{\text{sample data}}}$ and use the $\overset{\bar{x}}{\underline{\text{sample statistic}}}$ as our best guess for the value of the parameter, along with an assessment of the uncertainty around this guess.

- A **confidence interval** is an interval of values computed from sample data that is likely to include the unknown value of a population parameter.

- A **level 95% confidence interval for a population mean** μ with known population standard deviation σ is given by:

$$\left(\bar{x} - 1.96\frac{\sigma}{\sqrt{n}}, \bar{x} + 1.96\frac{\sigma}{\sqrt{n}}\right)$$

(assuming conditions for CLT apply!!)

Confidence Interval

Definition and Procedure

- We would like to find out the value of a population parameter (μ) but we can't collect data on the entire population. We collect sample data (n) and use the sample statistic ($\bar{x}$) as our best guess for the value of the parameter, along with an assessment of the uncertainty around this guess.

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How often do you think we want to estimate the population mean but we somehow know the population standard deviation?

- Ans: **Never**. For now, we use this just for the sake of building up an understanding.

Confidence Interval

Definition and Procedure

- A **level 95% confidence interval for a population mean** μ with known population standard deviation σ is given by:

$$\left(\bar{x} - 1.96\frac{\sigma}{\sqrt{n}}, \bar{x} + 1.96\frac{\sigma}{\sqrt{n}}\right)$$

- **Ex:** Suppose we are interested in the mean resting heart rate in a large population where we know that the population standard deviation is 12. We randomly sample 49 people from this population, and their sample mean resting heart rate is 75. Create a 95% confidence interval for the population mean.

Confidence Interval

Definition and Procedure

- A **level 95% confidence interval for a population mean** μ with known population standard deviation σ is given by:

$$\left(\bar{x} - 1.96\frac{\sigma}{\sqrt{n}}, \bar{x} + 1.96\frac{\sigma}{\sqrt{n}}\right)$$

μ

- **Ex:** Suppose we are interested in the mean resting heart rate in a large population

σ n

where we know that the population standard deviation is 12. We randomly sample 49

$\bar{x}$

people from this population, and their sample mean resting heart rate is 75. Create a

μ

95% confidence interval for the unknown population mean.

Confidence Interval

Definition and Procedure

- A **level 95% confidence interval for a population mean** μ with known population standard deviation σ is given by:

$$\left(\bar{x} - 1.96\frac{\sigma}{\sqrt{n}}, \bar{x} + 1.96\frac{\sigma}{\sqrt{n}}\right) \to \left(75 - 1.96\frac{12}{\sqrt{49}}, 75 + 1.96\frac{12}{\sqrt{49}}\right) = (71.64,78.36)$$

μ

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  σ n

  where we know that the population standard deviation is 12. We randomly sample 49

  $\bar{x}$

  people from this population, and their sample mean resting heart rate is 75. Create a

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  95% confidence interval for the unknown population mean.

Confidence Interval

Interpretation

- A **level 95% confidence interval for a population mean** μ with known population standard deviation σ is given by:

$$\left(\bar{x} - 1.96\frac{\sigma}{\sqrt{n}}, \bar{x} + 1.96\frac{\sigma}{\sqrt{n}}\right) \rightarrow \left(75 - 1.96\frac{12}{\sqrt{49}}, 75 + 1.96\frac{12}{\sqrt{49}}\right) = (71.64, 78.36)$$

- How do we interpret this interval, (71.64, 78.36)?
    - Remember: idea is we are creating an interval that we hope captures the unknown population mean

Confidence Interval

Interpretation

- A **level 95% confidence interval for a population mean** μ with known population standard deviation σ is given by:

$$\left(\bar{x} - 1.96\frac{\sigma}{\sqrt{n}}, \bar{x} + 1.96\frac{\sigma}{\sqrt{n}}\right) \rightarrow \left(75 - 1.96\frac{12}{\sqrt{49}}, 75 + 1.96\frac{12}{\sqrt{49}}\right) = (71.64, 78.36)$$

- How do we interpret this interval, (71.64, 78.36)?
    - PollEv.com/stats8: Consider: can we say that the probability that the unknown population mean is between 71.64 and 78.36 is 95%? Why or why not?
        - Hint: what, exactly, is random in this setup?

Confidence Interval

Interpretation

- A **level 95% confidence interval for a population mean** μ with known population standard deviation σ is given by:

$$\left(\bar{x} - 1.96\frac{\sigma}{\sqrt{n}}, \bar{x} + 1.96\frac{\sigma}{\sqrt{n}}\right) \rightarrow \left(75 - 1.96\frac{12}{\sqrt{49}}, 75 + 1.96\frac{12}{\sqrt{49}}\right) = (71.64, 78.36)$$

- How do we interpret this interval, (71.64, 78.36)?
  - PollEv.com/stats8: Consider: can we say that the probability that the unknown population mean is between 71.64 and 78.36 is 95%? Why or why not?
    - **No! Nothing is random. The endpoints *were* random before we observed the data, but now they are fixed numbers.**

🚨Interpretation of a Confidence Interval🚨

Please try to understand this it will make all statisticians very happy

- When we use a sample to learn about the population, **the only thing that is random is the process of sampling!**
- When we collect data and build a confidence interval…
  - The <u>sample mean</u>, once collected and computed, is **not random**
  - The <u>population mean</u> is **not random**
  - ‣ We can't talk in probabilities about a final confidence interval!
- What we **can** say is that the **procedure we use to create a confidence interval has a certain probability** of capturing the population parameter

🚨Interpretation of a Confidence Interval🚨

Please try to understand this it will make all statisticians very happy

- When we use a sample to learn about the population, **the only thing that is random is the process of sampling!**

$$\left(\bar{X} - 1.96\frac{\sigma}{\sqrt{n}}, \bar{X} + 1.96\frac{\sigma}{\sqrt{n}}\right)$$

before we collect the sample, these are random (and they are the only things in the whole procedure that are random!!)

🚨 Interpretation of a Confidence Interval 🚨

Please try to understand this it will make all statisticians very happy

- When we use a sample to learn about the population, **the only thing that is random is the process of sampling!**

$$\left(75 - 1.96\frac{12}{\sqrt{49}},\ 75 + 1.96\frac{12}{\sqrt{49}}\right)$$

after we collect the sample, these are fixed values (and nothing at all is random!!)

🚨 Interpretation of a Confidence Interval 🚨

Please try to understand this it will make all statisticians very happy

- When we use a sample to learn about the population, **the only thing that is random is the process of sampling!**
- If we can't use probability, how do we interpret an individual confidence interval?

(71.64, 78.36)

- We have special language for this: "confidence"
- **"We are 95% confident that the population mean resting heart rate is between 71.64 and 78.36 beats per minute."**
  - We say "95% confident" because we can no longer accurately say "probability" — but we want to impart the idea that the *procedure* to create the interval has 95% probability of capturing the parameter

Confidence Interval

PollEv.com/stats8

- A **level 95% confidence interval for a population mean** μ with known population standard deviation σ is given by:

$$\left(\bar{x} - 1.96\frac{\sigma}{\sqrt{n}}, \bar{x} + 1.96\frac{\sigma}{\sqrt{n}}\right)$$

- We are interested in the mean full-term birth weight of U.S.-born babies, and we know the population standard deviation of U.S. birth weights is 1.1 lbs. We take a sample of 36 U.S. babies, and find that in the sample the average birth weight is 6.5 lbs.

- Find a 95% confidence interval for the population mean birth weight of U.S.-born babies.

Confidence Interval

PollEv.com/stats8

- A **level 95% confidence interval for a population mean** μ with known population standard deviation σ is given by:

$$\left(\bar{x} - 1.96\frac{\sigma}{\sqrt{n}}, \bar{x} + 1.96\frac{\sigma}{\sqrt{n}}\right)$$

- We are interested in the mean full-term birth weight of U.S.-born babies, and we know the population standard deviation of U.S. birth weights is 1.1 lbs. We take a sample of 36 U.S. babies, and find that in the sample the average birth weight is 7.0 lbs.

- A 95% confidence interval for μ is **(6.64, 7.36)**. Interpret this interval in context.

Confidence Interval

PollEv.com/stats8

- A **level 95% confidence interval for a population mean** μ with known population standard deviation σ is given by:

$$\left(\bar{x} - 1.96\frac{\sigma}{\sqrt{n}}, \bar{x} + 1.96\frac{\sigma}{\sqrt{n}}\right)$$

- We are interested in the mean full-term birth weight of U.S.-born babies, and we know the population standard deviation of U.S. birth weights is 1.1 lbs. We take a sample of 36 U.S. babies, and find that in the sample the average birth weight is 7.0 lbs.

- A 95% confidence interval for μ is **(6.64, 7.36)**. We are 95% confident that the population mean birth weight of U.S.-born babies is between 6.64 and 7.36 lbs.

Confidence Level

What if we want something other than 95%?

- A **level 95% confidence interval for a population mean** μ with known population standard deviation σ is given by:

$$\left(\bar{x} - 1.96\frac{\sigma}{\sqrt{n}}, \bar{x} + 1.96\frac{\sigma}{\sqrt{n}}\right)$$

- We can choose a different confidence level depending on how sure we want to be that we capture the population mean.

- PollEv.com/stats8: For some population with a given sample from that population, if we want to change the confidence level, what part of the above formula do you think changes?

Confidence Level

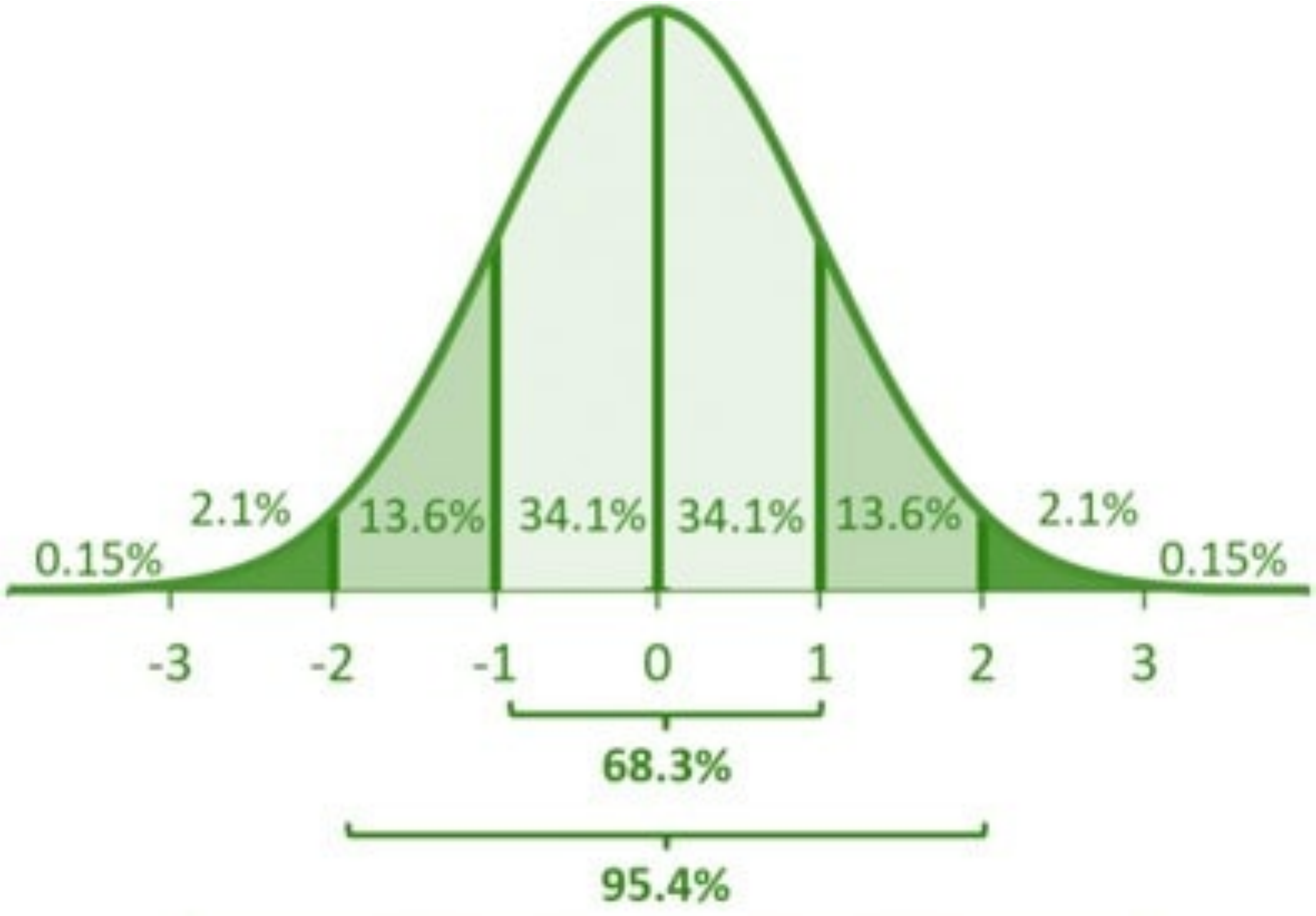
What if we want something other than 95%?

- A **level 95% confidence interval for a population mean** μ with known population standard deviation σ is given by:

$$\left(\bar{x} - \boxed{1.96}\frac{\sigma}{\sqrt{n}}, \bar{x} + \boxed{1.96}\frac{\sigma}{\sqrt{n}}\right)$$

<u>Where did 1.96 come from?</u>

- We used the empirical rule and then I vaguely said, “to be more exact…”



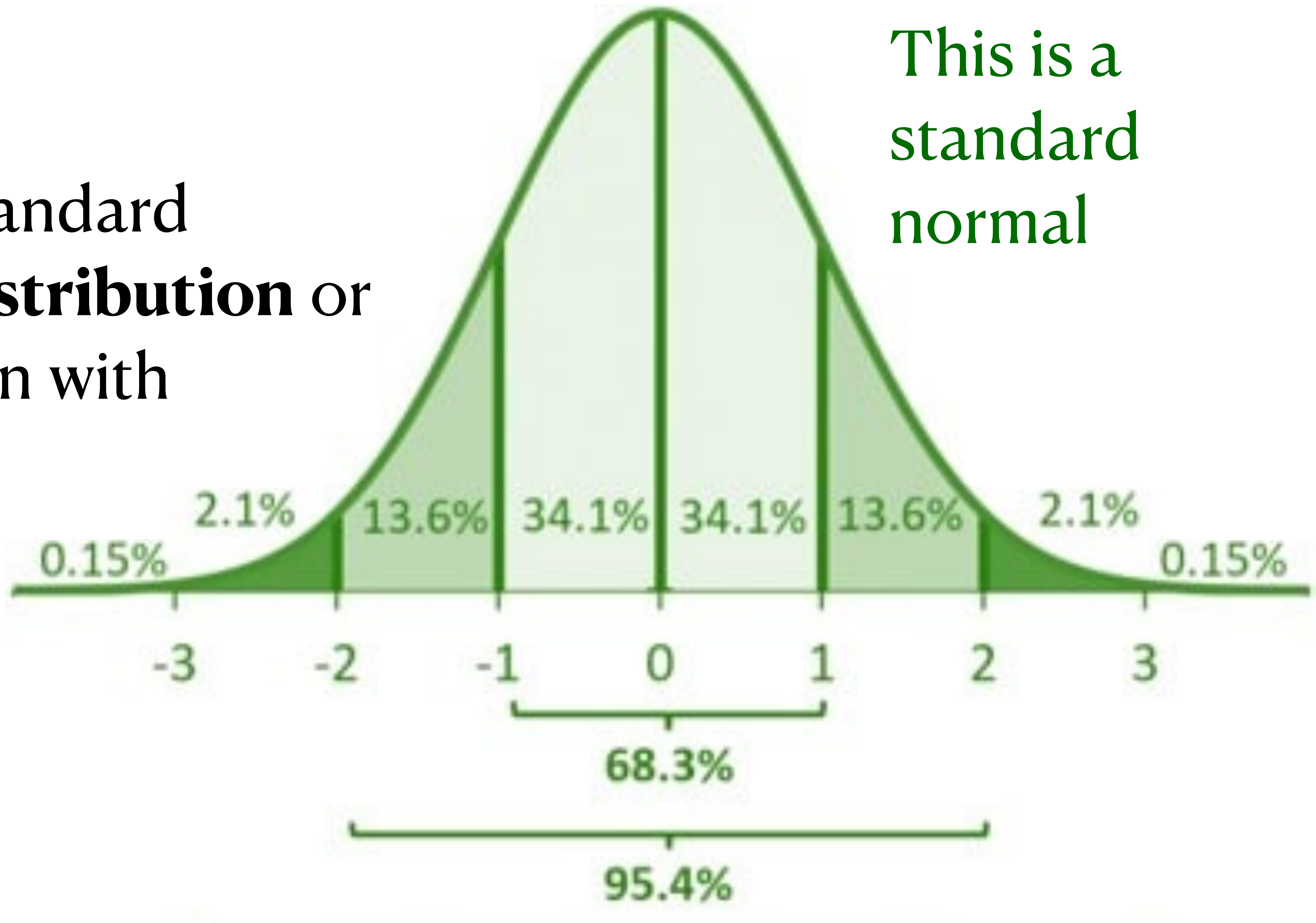
Confidence Level

A property of the normal distribution (location and scale)

- In general, if $X \sim N(\mu, \sigma^2)$, then:

$$Z = \frac{X - \mu}{\sigma} \sim N(0,\ 1)$$

if you subtract the mean and divide by the standard deviation, you obtain a **standard normal distribution** or **z-distribution**, which is a normal distribution with $\mu = 0$ and $\sigma = 1$.



This is a standard normal

Confidence Level

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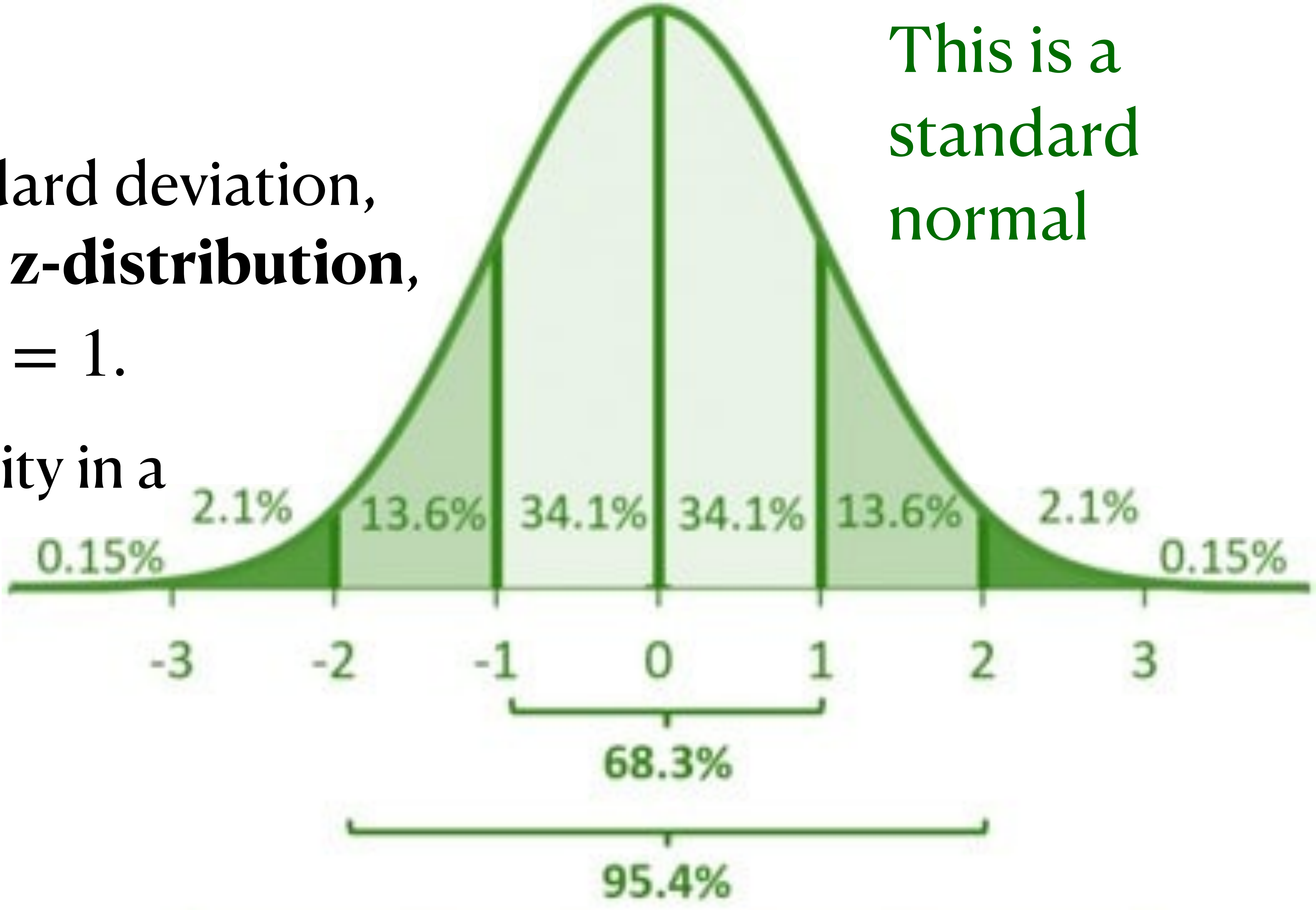
$$Z = \frac{X - \mu}{\sigma} \sim N(0, 1)$$

if you subtract the mean and divide by the standard deviation, you obtain a **standard normal distribution** or **z-distribution**, which is a normal distribution with $\mu = 0$ and $\sigma = 1$.

This is where 1.96 came from: 95% of the probability in a z-distribution is between -1.96 and 1.96

So, what do we do if we want something different from 95%?



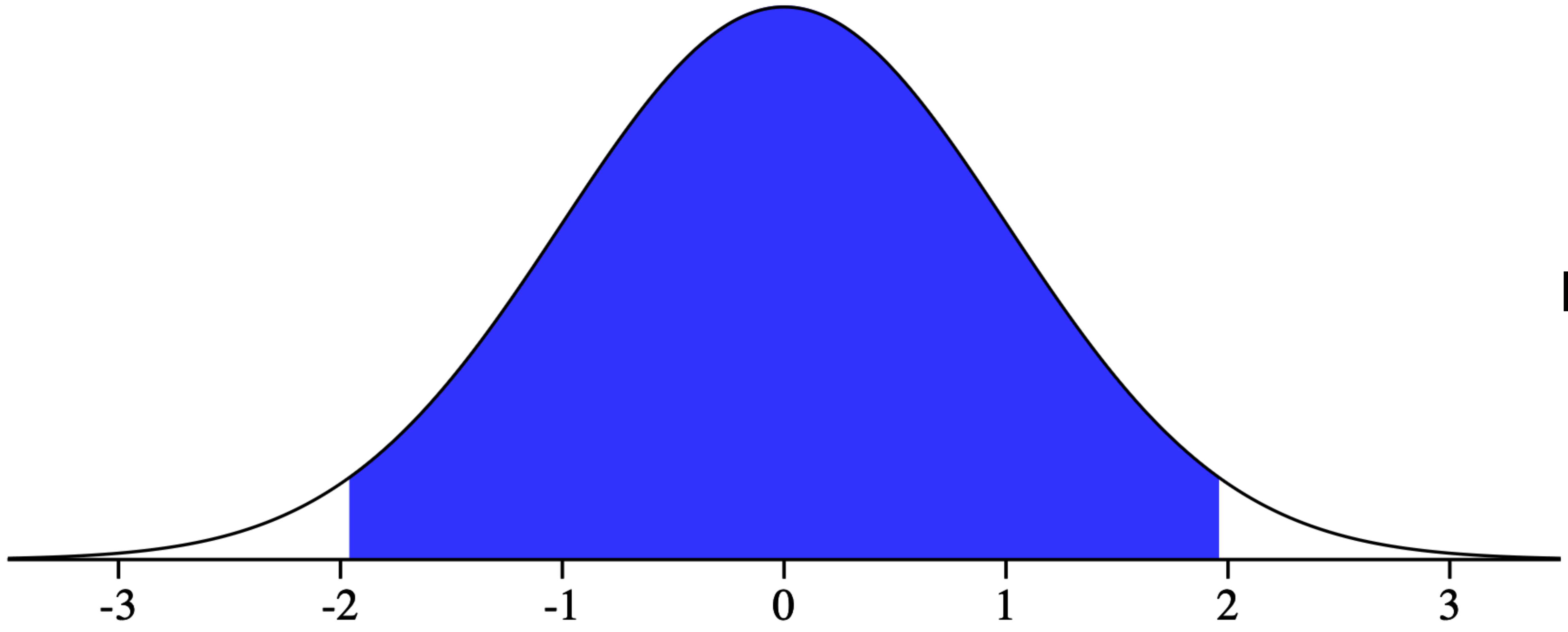
This is a standard normal

Confidence Level

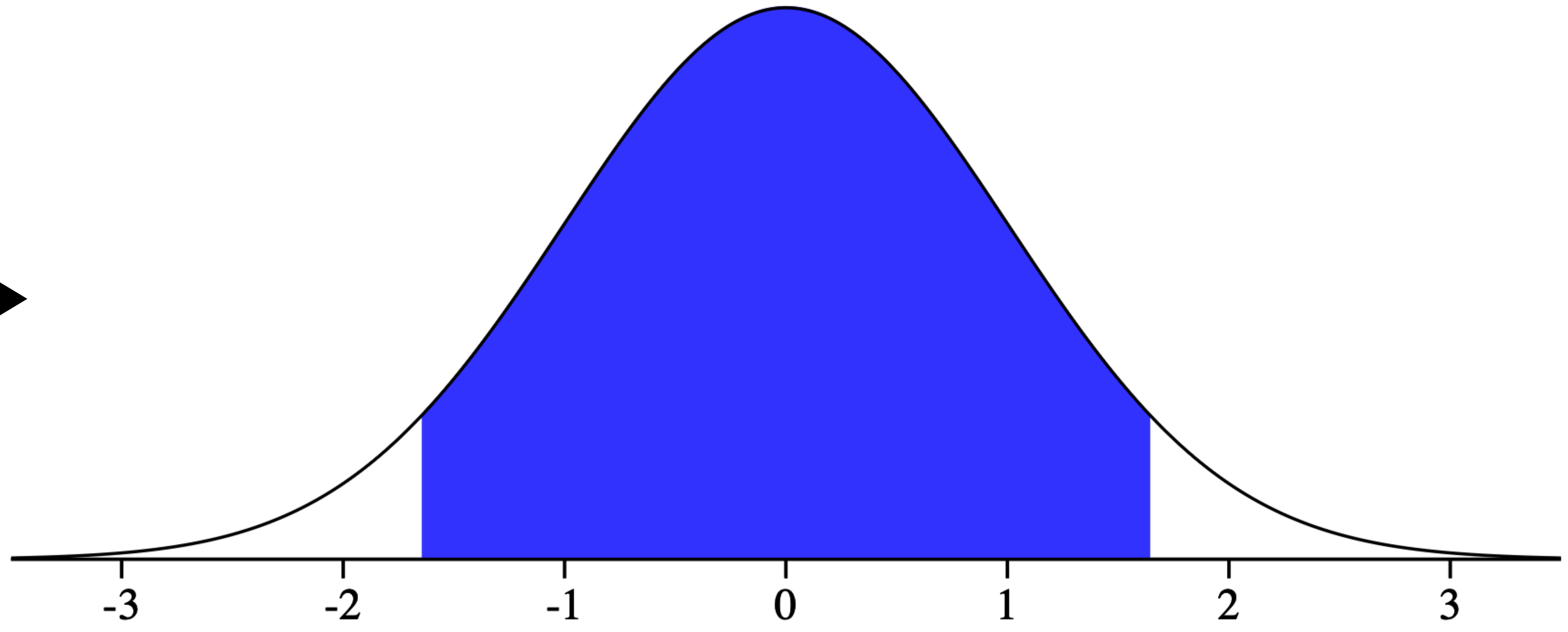
- Now, rather than a 95% interval, suppose we want a 90% interval.

$$\left(\bar{x} - \boxed{(??)}\frac{\sigma}{\sqrt{n}}, \bar{x} + \boxed{(??)}\frac{\sigma}{\sqrt{n}}\right)$$

We need to figure out what these are
for a 90% interval instead of 95%



$P(-1.96 \leq Z \leq 1.96) = 0.95$



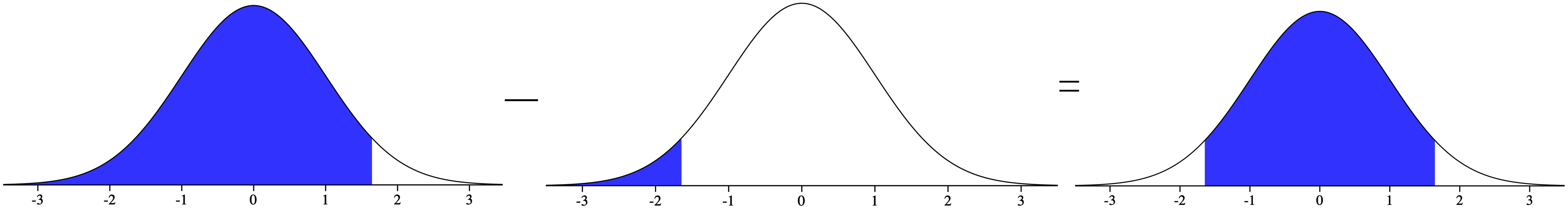
$P(?? \leq Z \leq ??) = 0.90$

Confidence Level

- Now, rather than a 95% interval, suppose we want a 90% interval.

$$\left(\bar{x} - z_1 \frac{\sigma}{\sqrt{n}}, \bar{x} + z_2 \frac{\sigma}{\sqrt{n}}\right)$$

- Do you agree that:



$P(Z \leq z_2)$ $-$ $P(Z \leq z_1)$ $=$ $P(z_1 \leq Z \leq z_2) = 0.90$

Can use quantiles of the normal cdf to find z_1 and z_2

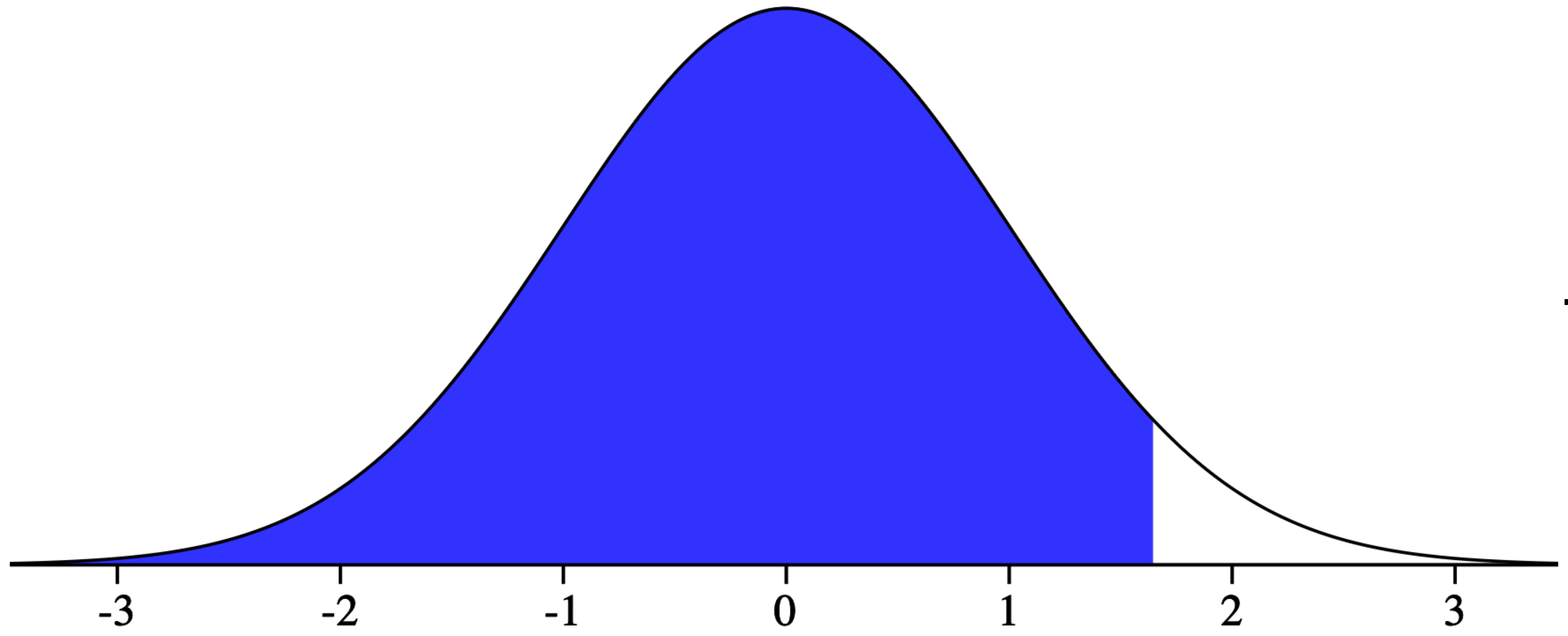
But, to do so, we need to know the probabilities...

Confidence Level

- Now, rather than a 95% interval, suppose we want a 90% interval.

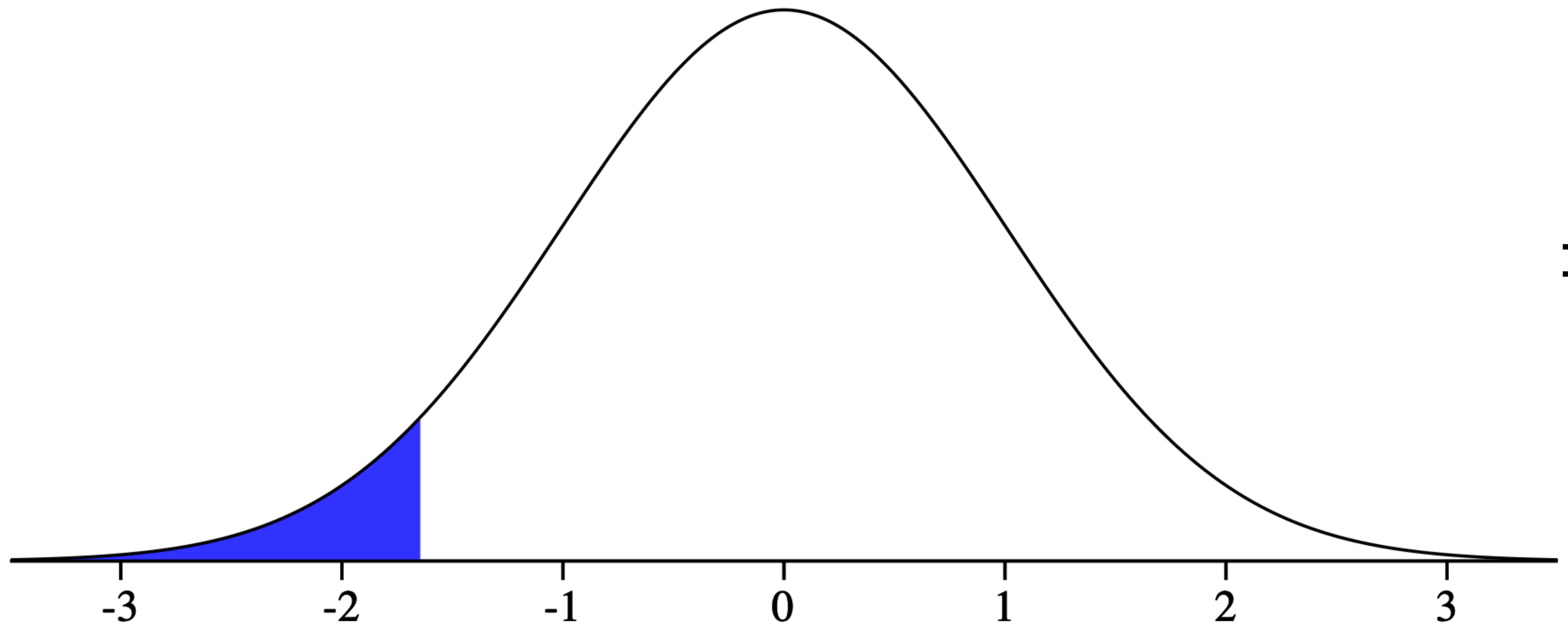
$$\left(\bar{x} - \boxed{z_1}\frac{\sigma}{\sqrt{n}},\ \bar{x} + \boxed{z_2}\frac{\sigma}{\sqrt{n}}\right)$$

By symmetry, each of these tails contains 5% probability



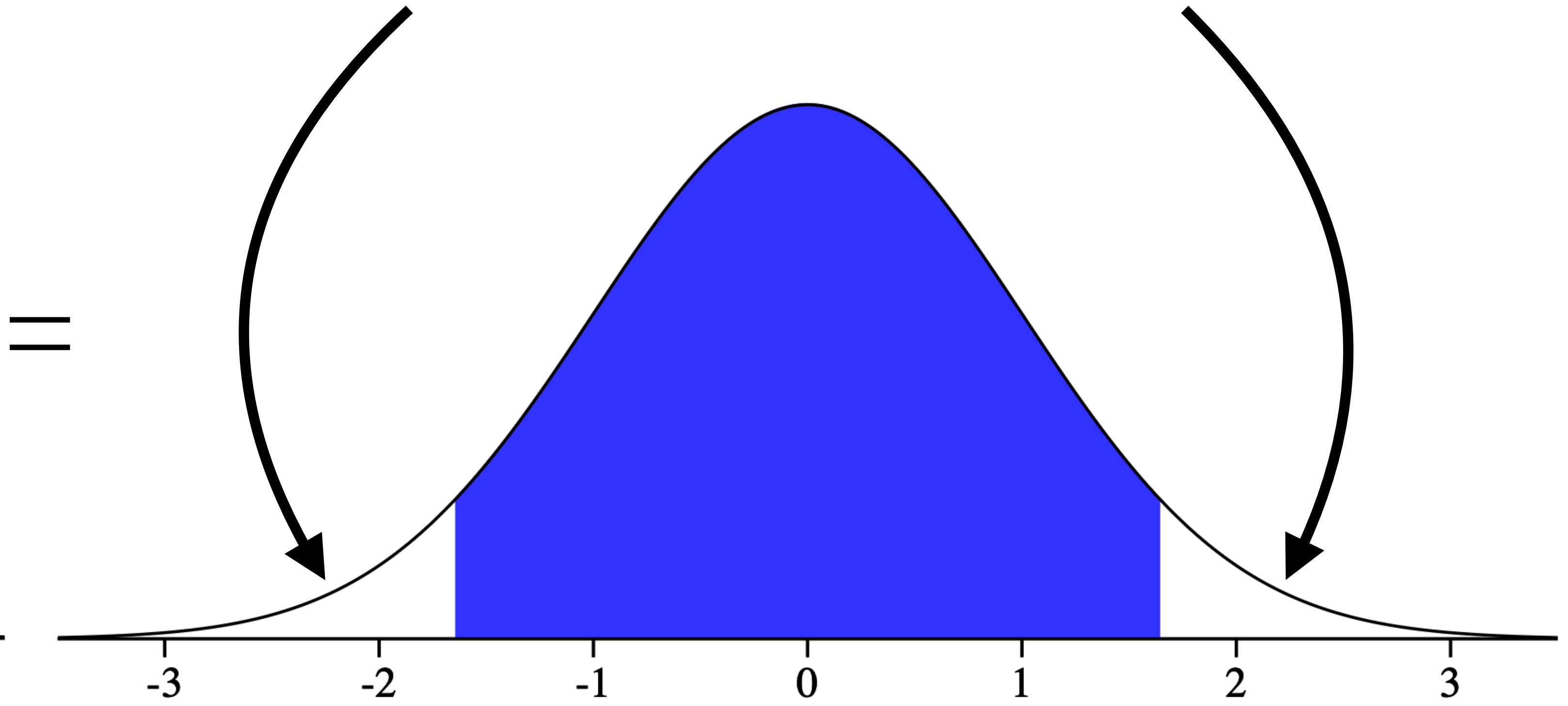
$P(Z \leq z_2)$

$-$



$P(Z \leq z_1)$

$=$



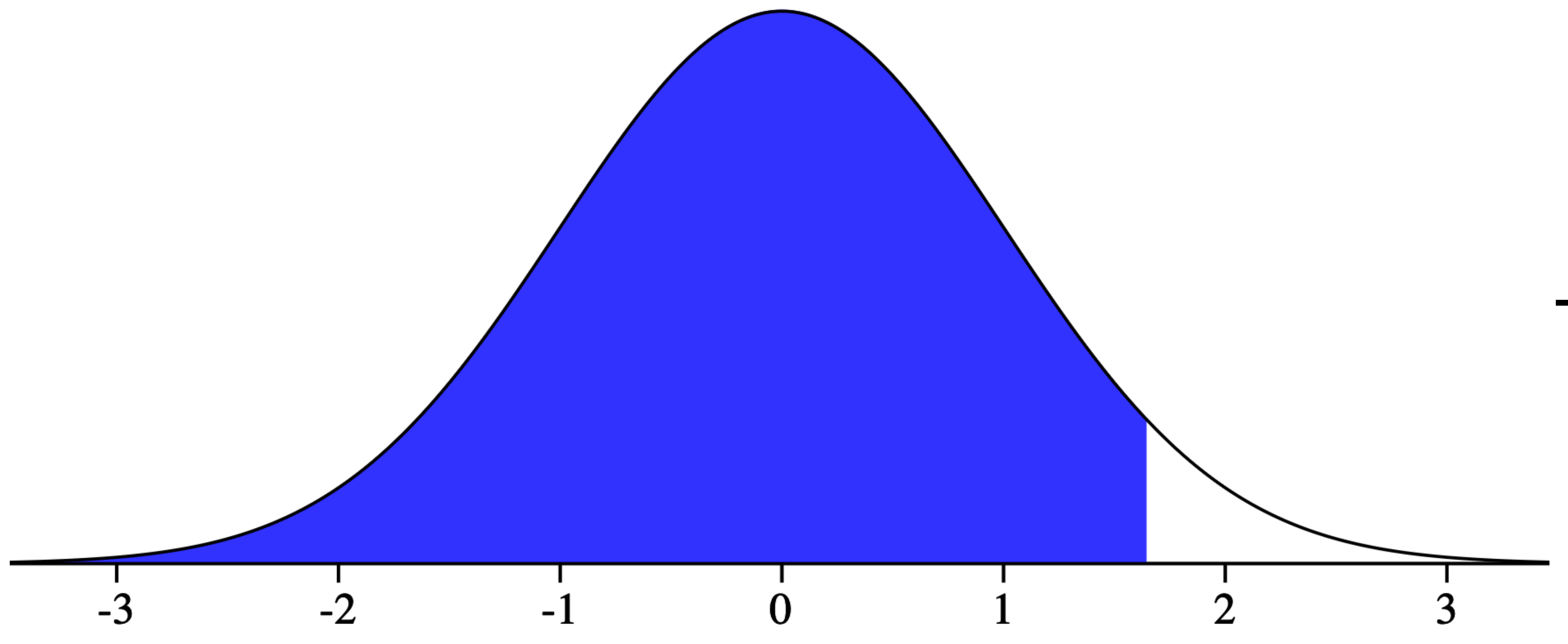
$P(z_1 \leq Z \leq z_2) = 0.90$

Confidence Level

- Now, rather than a 95% interval, suppose we want a 90% interval.

$$\left(\bar{x} - z_1 \frac{\sigma}{\sqrt{n}}, \bar{x} + z_2 \frac{\sigma}{\sqrt{n}}\right)$$

By symmetry, each of these tails contains 5% probability

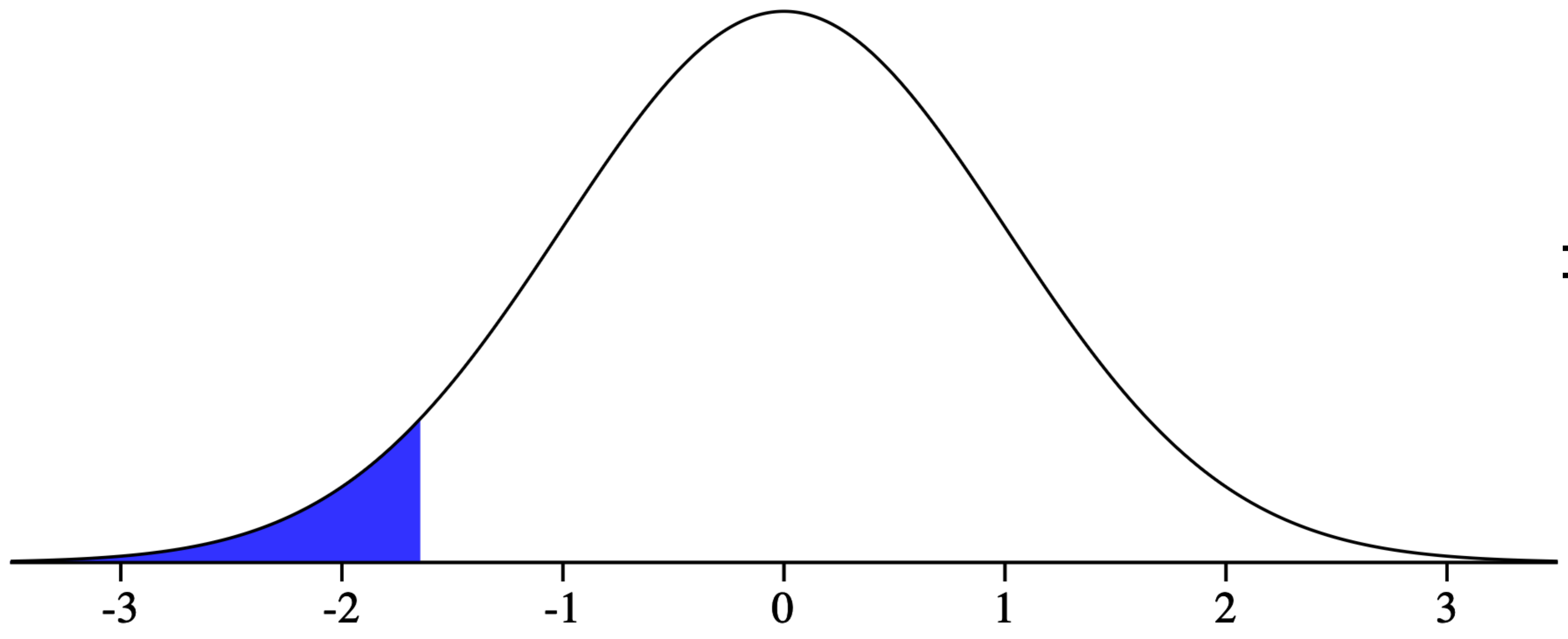


$P(Z \leq z_2)$

$= 0.95$

In R: qnorm(0.95) ≈ 1.645

$-$

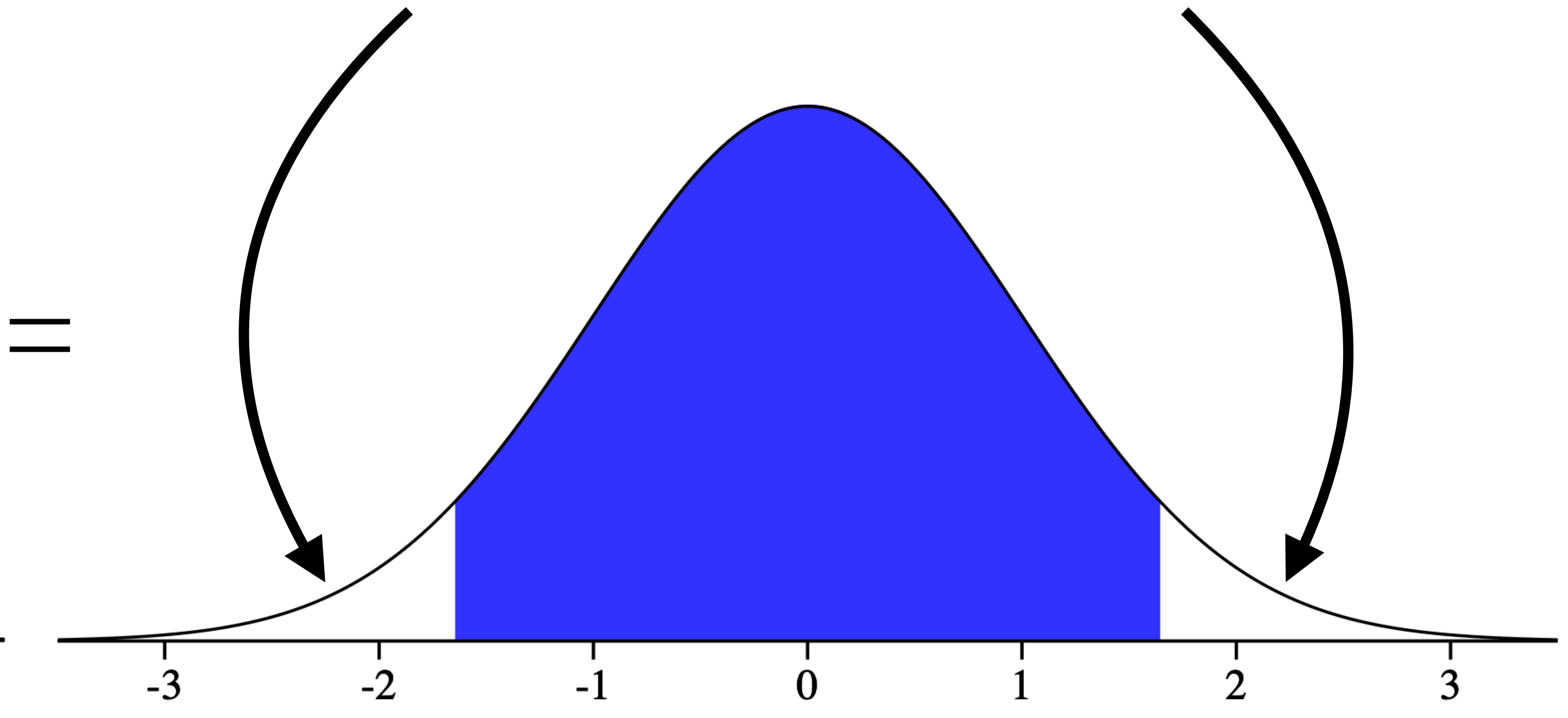


$P(Z \leq z_1)$

$= 0.05$

In R: qnorm(0.05) ≈ -1.645

$=$



$P(z_1 \leq Z \leq z_2) = 0.90$

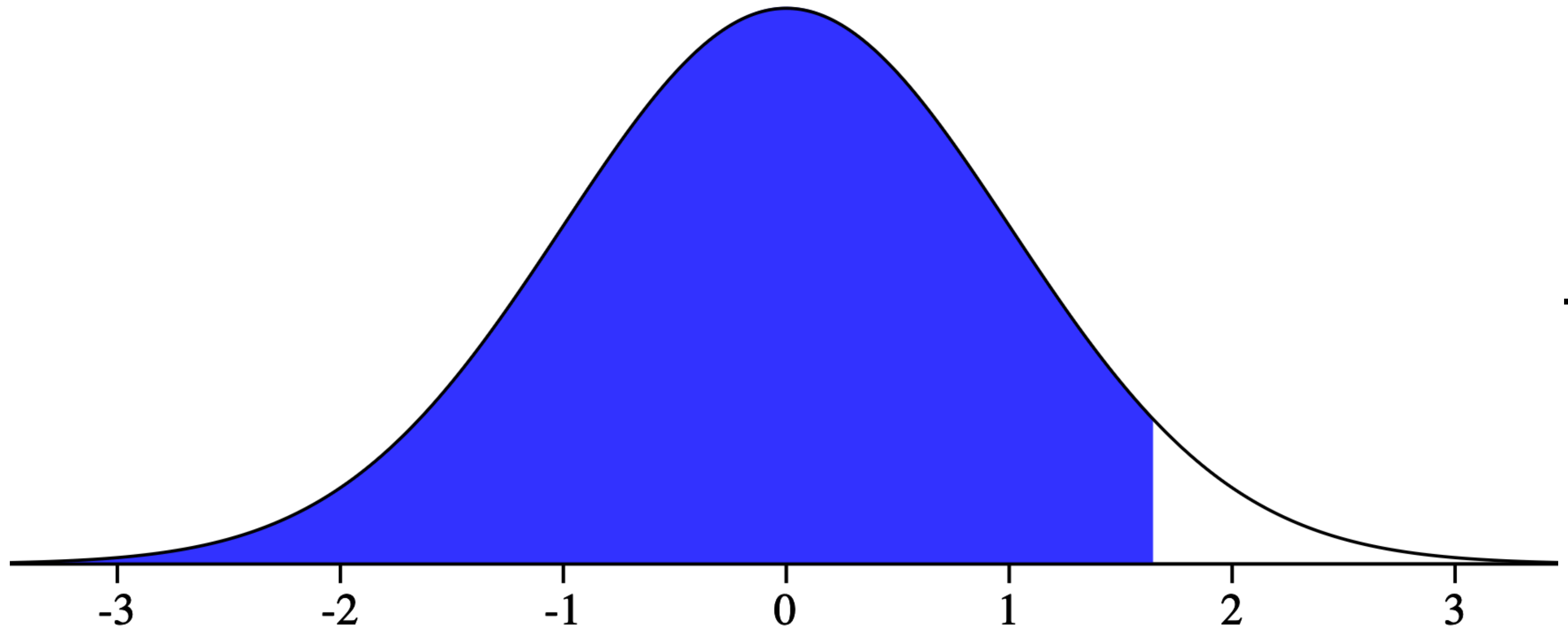
Confidence Level

- We've figured out the required multipliers for our 90% interval:

$$\left(\bar{x} - 1.645\frac{\sigma}{\sqrt{n}},\, \bar{x} + 1.645\frac{\sigma}{\sqrt{n}}\right)$$

- We denote the values below, qnorm(0.95) = $z_{0.95}$, and qnorm(0.05) = $z_{0.05}$

By symmetry, each of these tails contains 5% probability



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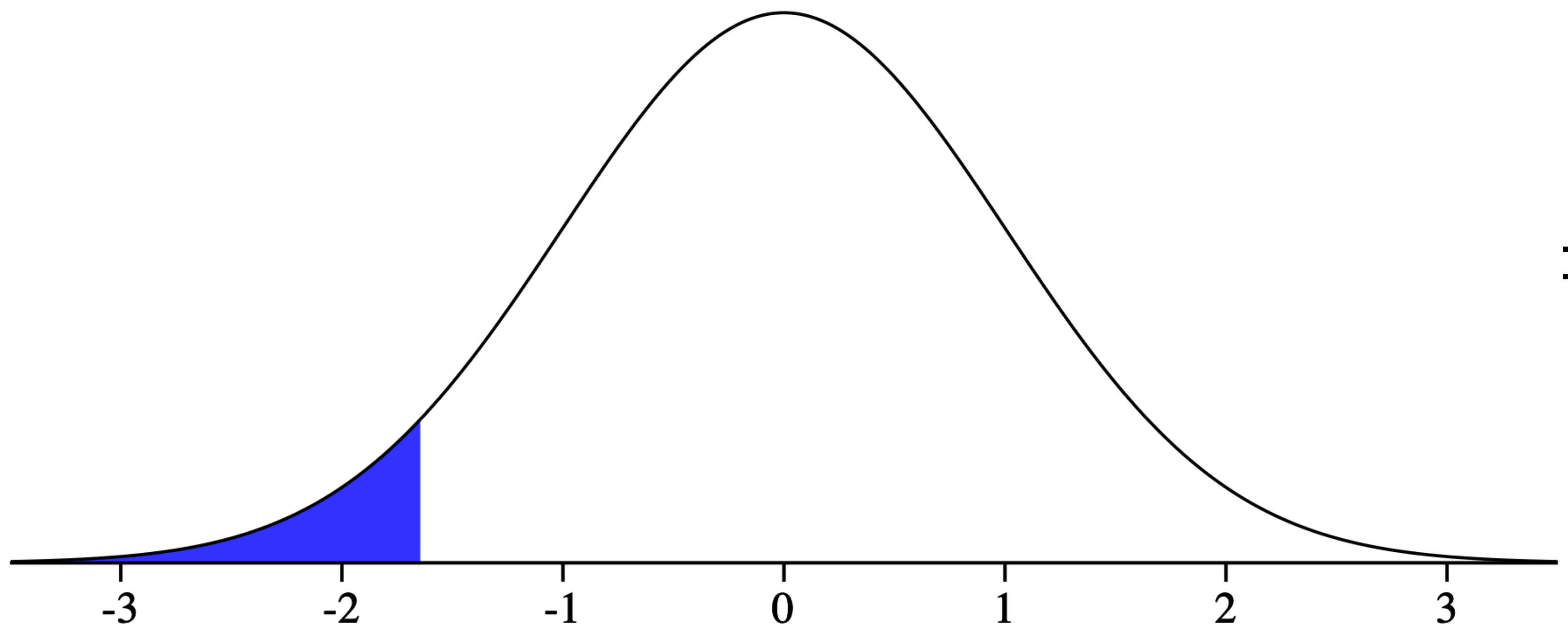

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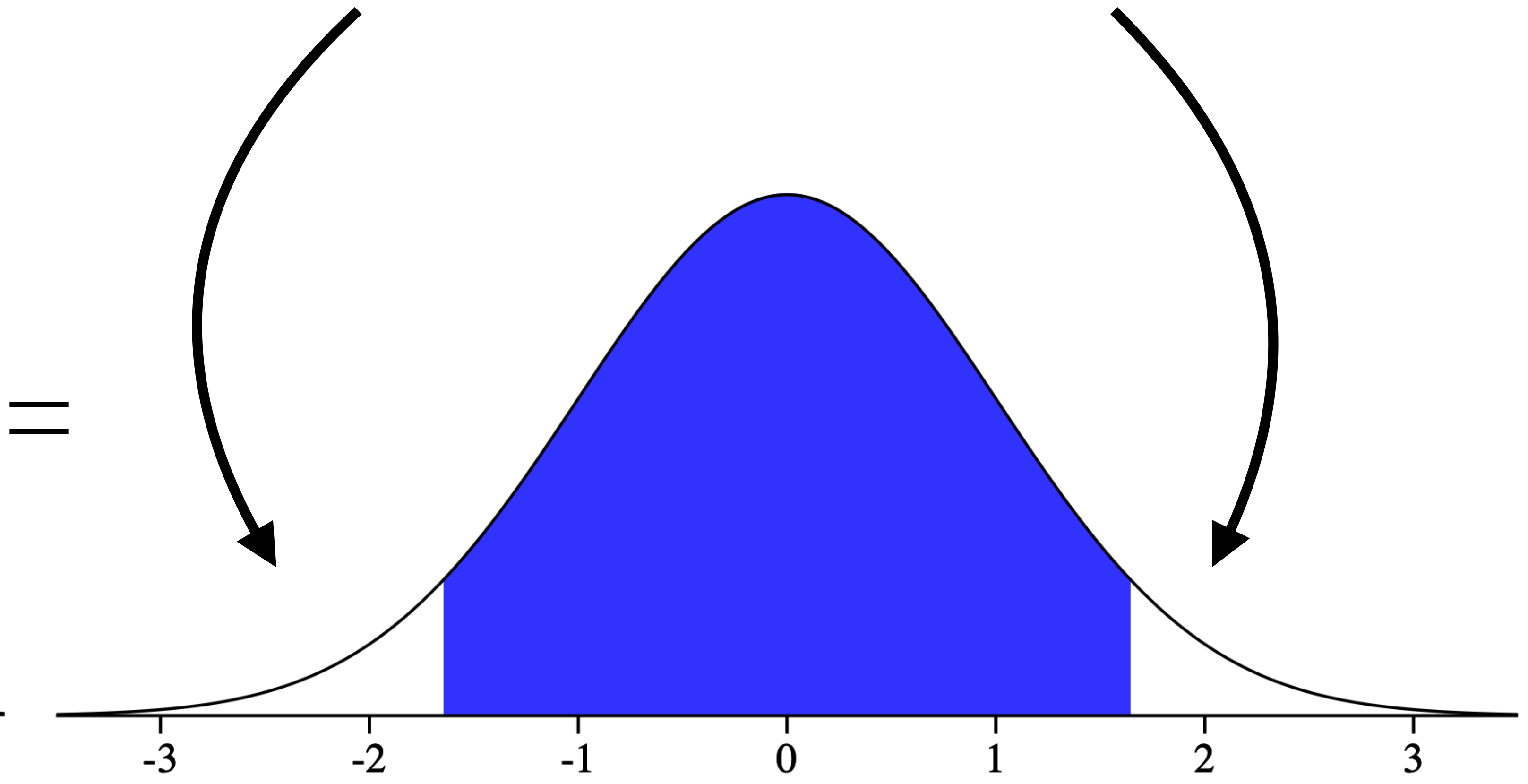


$P(Z \leq z_2)$

$= 0.95$

In R: qnorm(0.95) ≈ 1.645

$P(Z \leq z_1)$

$= 0.05$

In R: qnorm(0.05) ≈ -1.645

$P(z_1 \leq Z \leq z_2) = 0.90$

Confidence Level (General)

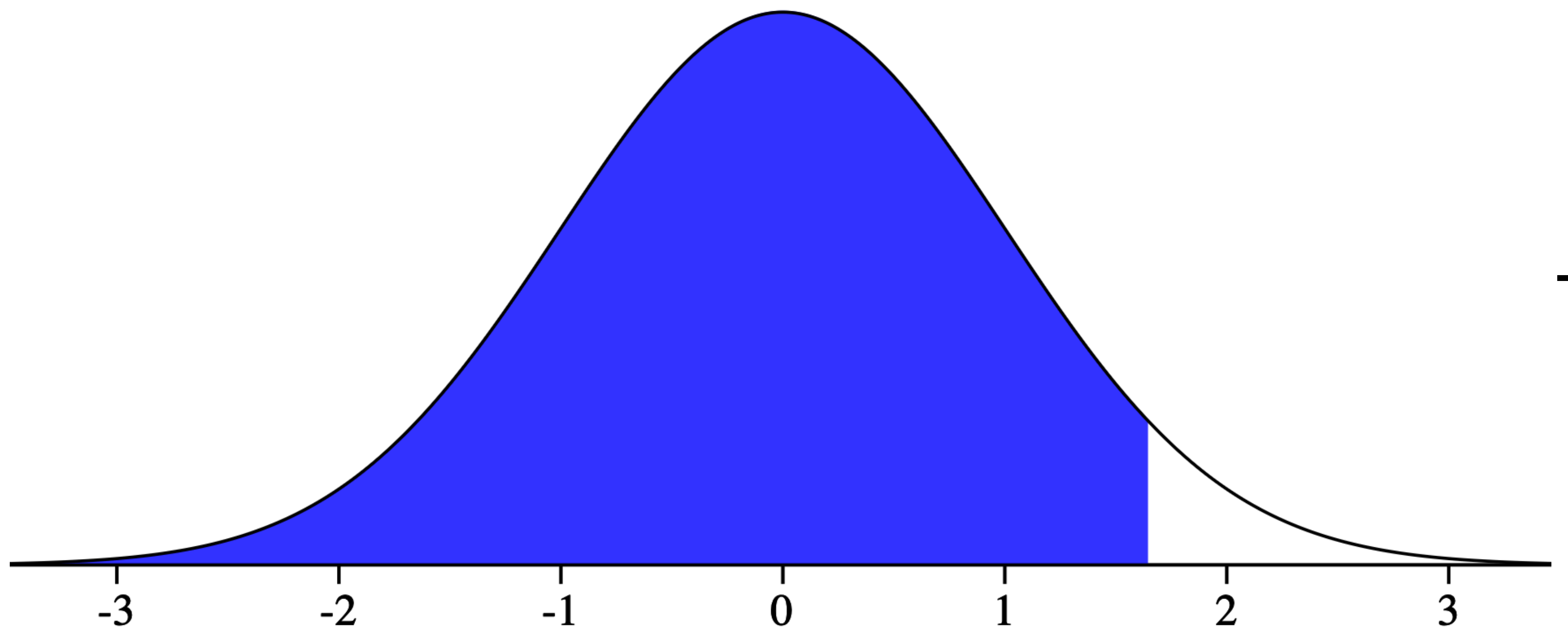
- For some α between 0 and 1, suppose you are asked for a $100 \times (1 - \alpha)\,\%$ interval:

$$\left(\bar{x} - z_{1-\alpha/2}\frac{\sigma}{\sqrt{n}}, \bar{x} + z_{1-\alpha/2}\frac{\sigma}{\sqrt{n}}\right)$$

- We denote the values below, `qnorm(1-α/2)` $= z_{1-\alpha/2}$, and

  `qnorm(α/2)` $= z_{\alpha/2}$. By symmetry, $z_{\alpha/2} = -z_{1-\alpha/2}$!

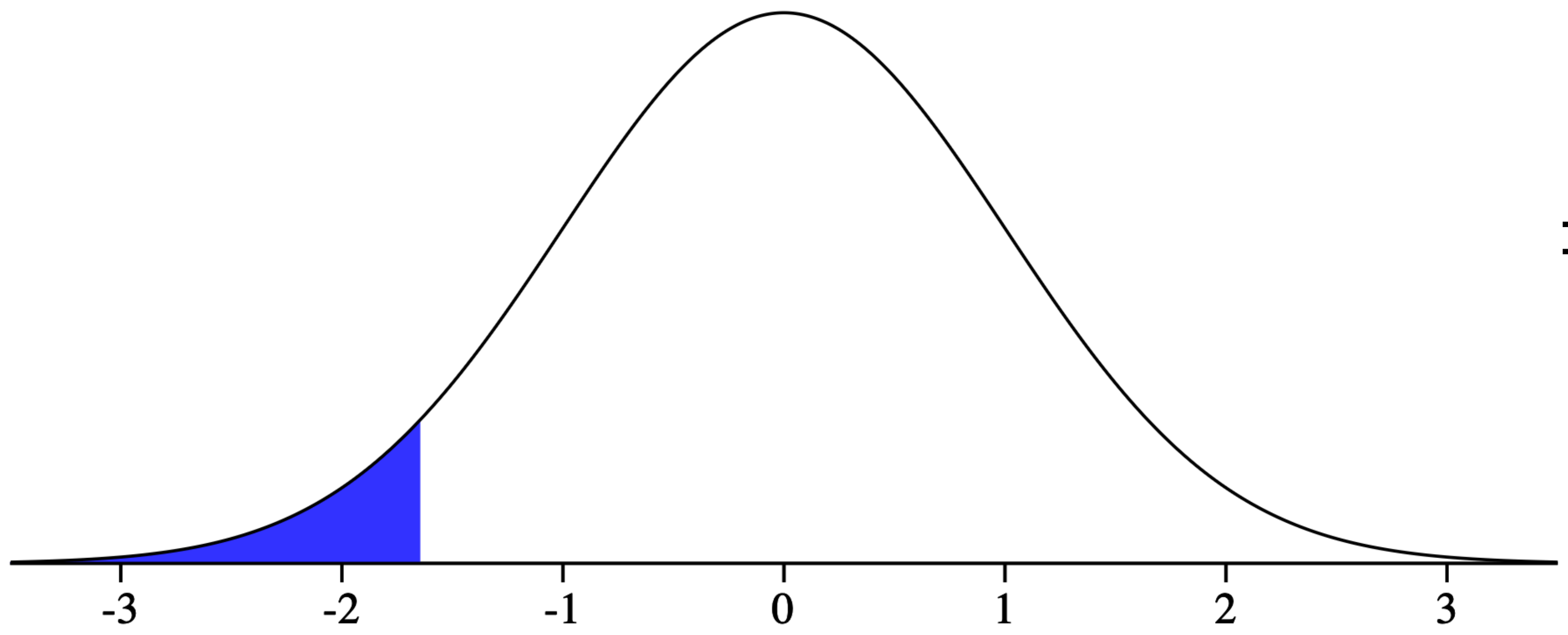
By symmetry, each of these tails contains $\alpha/2$ probability



$P(Z \leq z_{1-\alpha/2})$

$= 1 - \alpha/2$

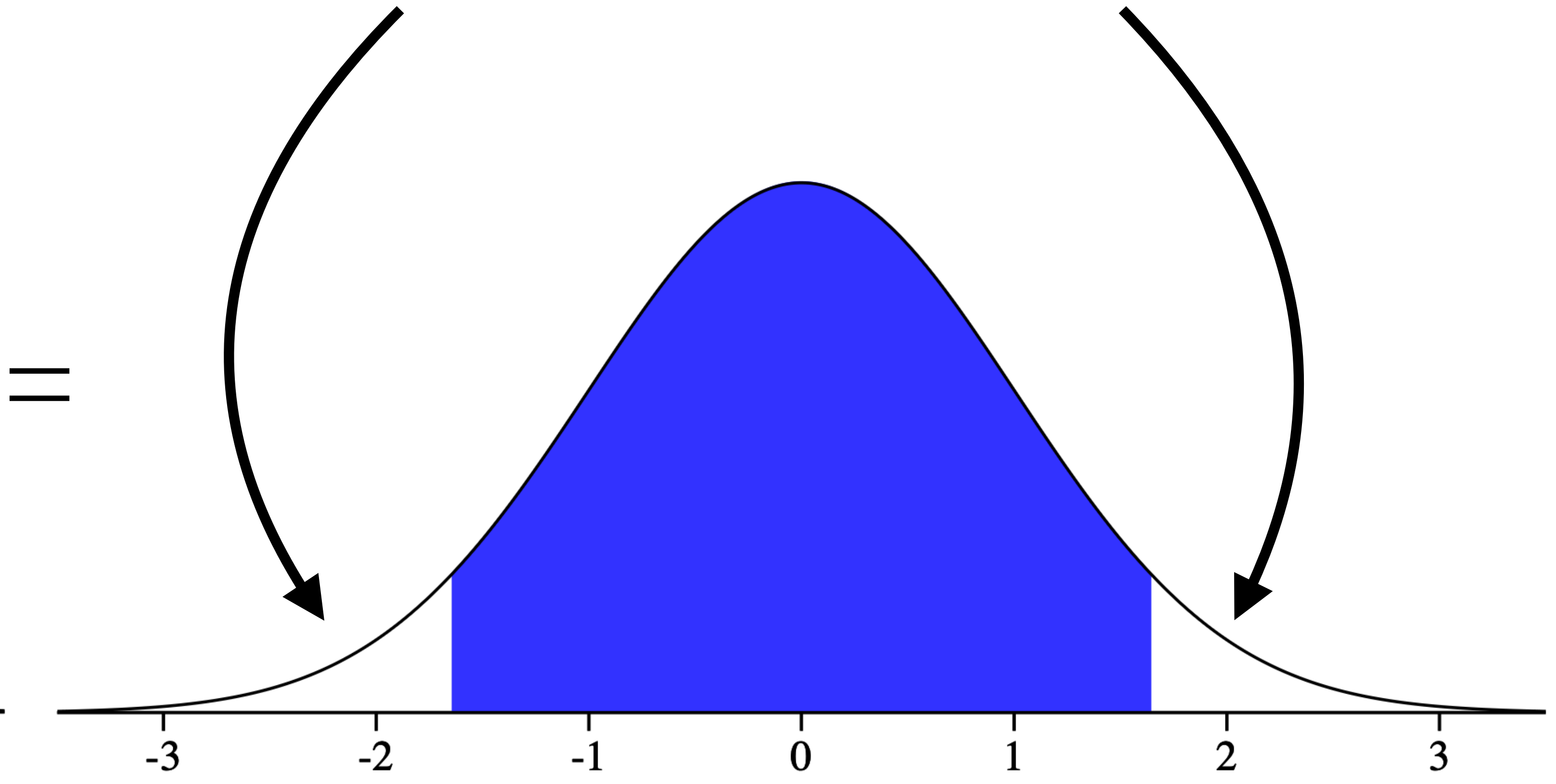
$-$



$P(Z \leq z_{\alpha/2})$

$= \alpha/2$

$=$



$P(z_{\alpha/2} \leq Z \leq z_{1-\alpha/2}) = 1 - \alpha$

Confidence Level (General)

- For some α between 0 and 1, suppose you are asked for a $100 \times (1 - \alpha)$ % interval:

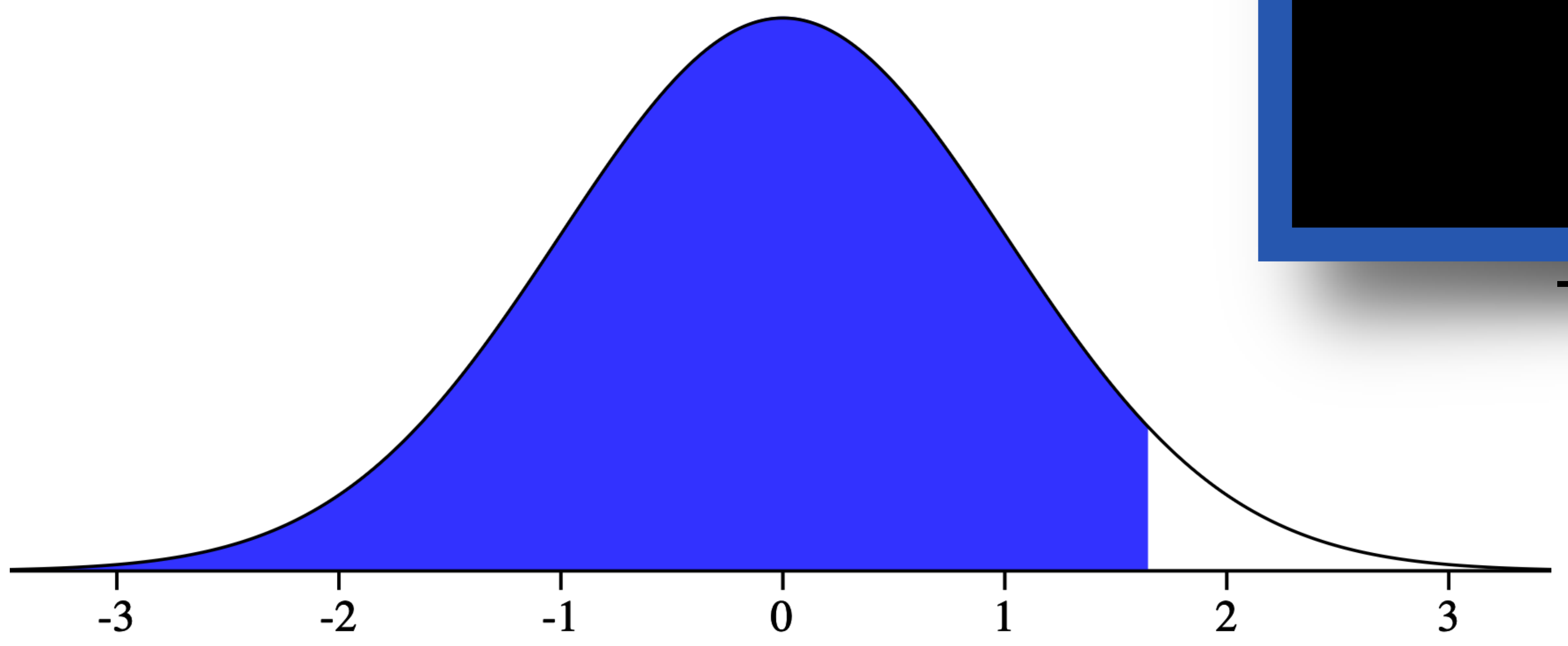
$$\left(\bar{\quad} \quad \sigma \quad \bar{\quad} \quad \sigma \right)$$

For homework/classwork, you can use R to compute $z_{\alpha/2}$ and $z_{1-\alpha/2}$. On an exam, these values would be given to you.

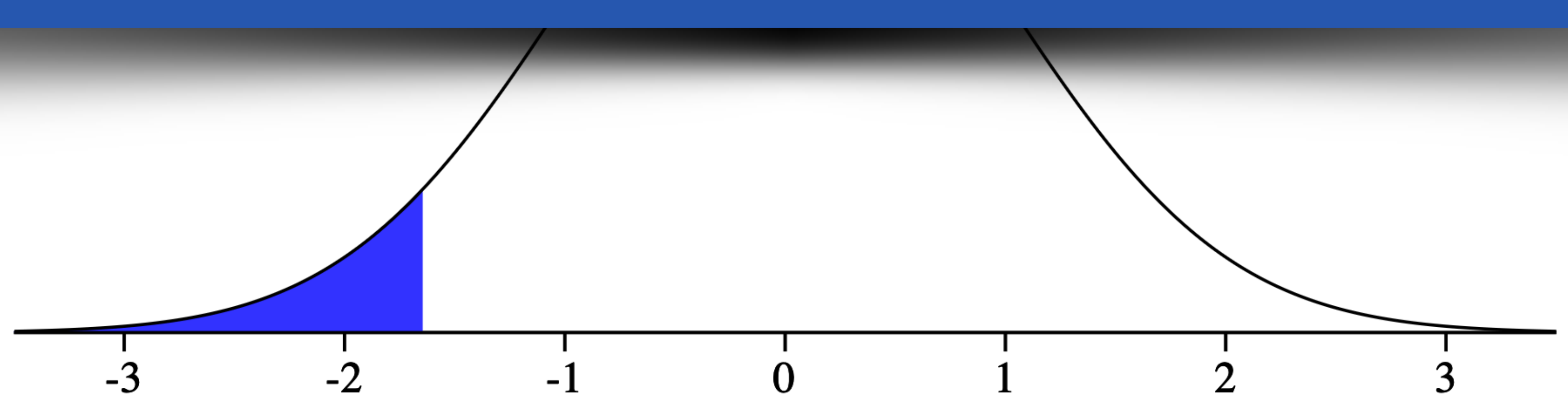
- We denote the valu

symmetry, each of
se tails contains $\alpha/2$
bability

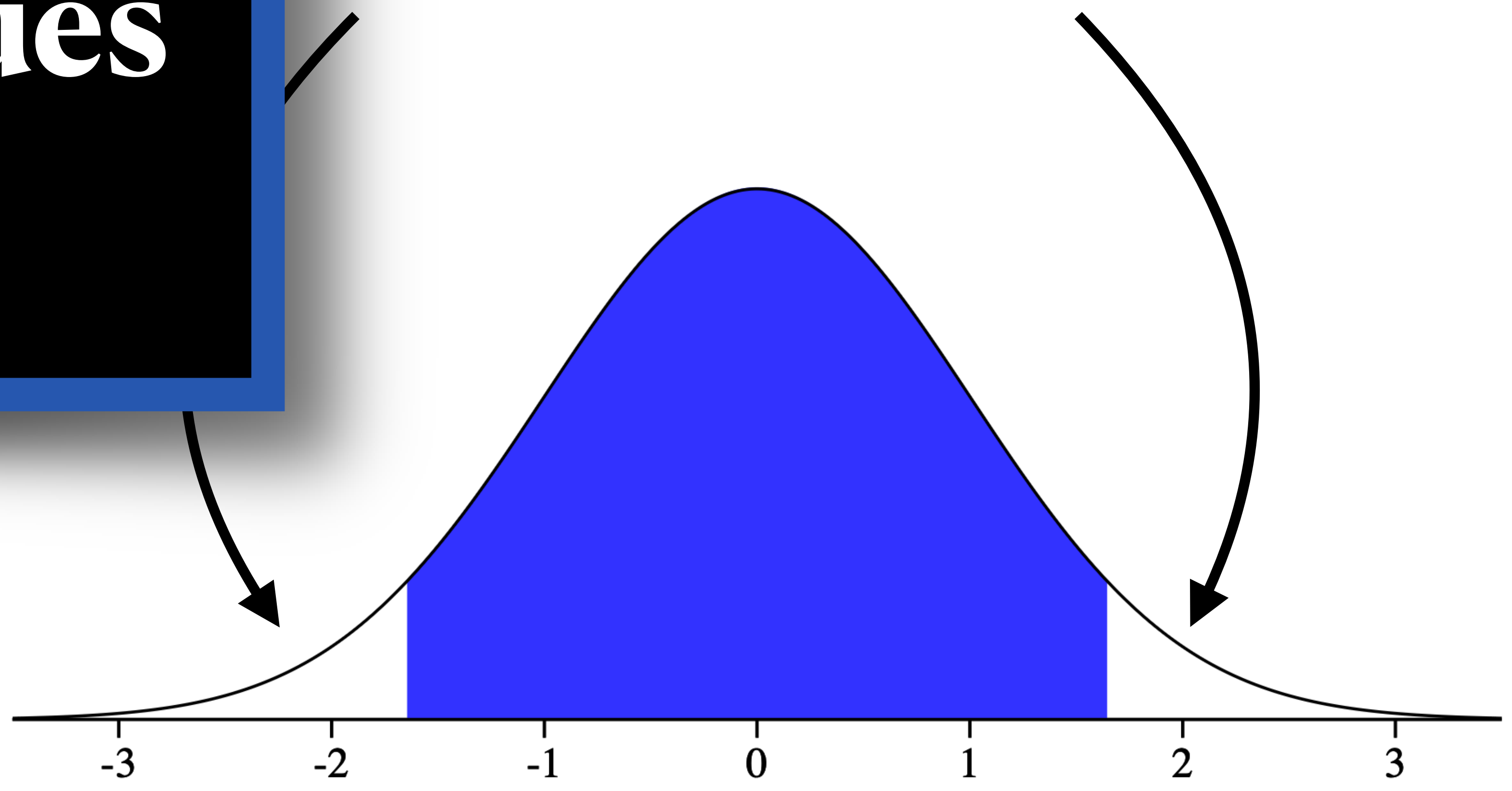
qnorm($\alpha/2$) = $z_{\alpha/}$



$P(Z \leq z_{1-\alpha/2})$
$= 1 - \alpha/2$



$P(Z \leq z_{\alpha/2})$
$= \alpha/2$



$P(z_{\alpha/2} \leq Z \leq z_{1-\alpha/2}) = 1 - \alpha$

Confidence Level

Example

- For some α between 0 and 1, suppose you are asked for a $100 \times (1 - \alpha)\,\%$ interval:

$$\left(\bar{x} - z_{1-\alpha/2}\frac{\sigma}{\sqrt{n}}, \bar{x} + z_{1-\alpha/2}\frac{\sigma}{\sqrt{n}}\right)$$

- We are interested in the mean full-term birth weight of U.S.-born babies, and we know the population standard deviation of U.S. birth weights is 1.1 lbs. We take a sample of 36 U.S. babies, and find that in the sample the average birth weight is 6.5 lbs.

- Find an 80% confidence interval for the population mean birth weight of U.S.-born babies.

Confidence Level

Example

- For some α between 0 and 1, suppose you are asked for a $100 \times (1-\alpha)\,\%$ interval:

$$\left(\bar{x} - z_{1-\alpha/2}\frac{\sigma}{\sqrt{n}},\ \bar{x} + z_{1-\alpha/2}\frac{\sigma}{\sqrt{n}}\right) = \left(6.5 - z_{1-\alpha/2}\frac{1.1}{\sqrt{36}},\ 6.5 + z_{1-\alpha/2}\frac{1.1}{\sqrt{36}}\right)$$

μ

- We are interested in the mean full-term birth weight of U.S.-born babies, and we σ know the population standard deviation of U.S. birth weights is 1.1 lbs. We take a sample of 36 U.S. babies, and find that in the sample the average birth weight is 6.5 $\bar{x}$ lbs. n

μ

- Find an 80% confidence interval for the population mean birth weight of U.S.-born babies.

Confidence Level

Example

- For some α between 0 and 1, suppose you are asked for a $100 \times (1 - \alpha)\,\%$ interval:

$$\left(\bar{x} - z_{1-\alpha/2}\frac{\sigma}{\sqrt{n}}, \bar{x} + z_{1-\alpha/2}\frac{\sigma}{\sqrt{n}}\right) = \left(6.5 - z_{1-\alpha/2}\frac{1.1}{\sqrt{36}}, 6.5 + z_{1-\alpha/2}\frac{1.1}{\sqrt{36}}\right)$$

μ

- We are interested in the mean full-term birth weight of U.S.-born babies, and we σ know the population standard deviation of U.S. birth weights is 1.1 lbs. We take a sample of 36 U.S. babies, and find that in the sample the average birth weight is 6.5 $\bar{x}$ lbs. n

μ

- Find an 80% confidence interval for the population mean birth weight of U.S.-born babies.

$$100 \times (1 - \alpha)\,\% = 80\,\%$$

$$\alpha = 0.2$$

Confidence Level

Example

- For some α between 0 and 1, suppose you are asked for a $100 \times (1-\alpha)\,\%$ interval:

$$\left(\bar{x} - z_{1-\alpha/2}\frac{\sigma}{\sqrt{n}}, \bar{x} + z_{1-\alpha/2}\frac{\sigma}{\sqrt{n}}\right) = \left(6.5 - 1.28\frac{1.1}{\sqrt{36}}, 6.5 + 1.28\frac{1.1}{\sqrt{36}}\right)$$

- We are interested in the μ mean full-term birth weight of U.S.-born babies, and we σ know the population standard deviation of U.S. birth weights is 1.1 lbs. We take a sample of 36 U.S. babies, and find that in the sample the average birth weight is 6.5 $\bar{x}$ lbs. n

- Find an 80% confidence interval for the μ population mean birth weight of U.S.-born babies.

$$100 \times (1-\alpha)\,\% = 80\,\%$$

$$\alpha = 0.2$$

$$z_{1-\alpha/2} = z_{0.9} = \texttt{qnorm(0.9)} \approx 1.28$$

Confidence Level

Example

- For some α between 0 and 1, suppose you are asked for a $100 \times (1 - \alpha)\,\%$ interval:

$$\left(\bar{x} - z_{1-\alpha/2}\frac{\sigma}{\sqrt{n}}, \bar{x} + z_{1-\alpha/2}\frac{\sigma}{\sqrt{n}}\right) = \left(6.5 - 1.28\frac{1.1}{\sqrt{36}}, 6.5 + 1.28\frac{1.1}{\sqrt{36}}\right) = (6.265, 6.735)$$

- We are interested in the mean full-term birth weight of U.S.-born babies, and we know the population standard deviation of U.S. birth weights is 1.1 lbs. We take a sample of 36 U.S. babies, and find that in the sample the average birth weight is 6.5 lbs.

- An 80% confidence interval for the population mean birth weight of U.S.-born babies is (6.265, 6.735).

Confidence Level

Example

- For some α between 0 and 1, suppose you are asked for a $100 \times (1 - \alpha)\%$ interval:

$$\left(\bar{x} - z_{1-\alpha/2}\frac{\sigma}{\sqrt{n}}, \bar{x} + z_{1-\alpha/2}\frac{\sigma}{\sqrt{n}}\right) = \left(6.5 - 1.28\frac{1.1}{\sqrt{36}}, 6.5 + 1.28\frac{1.1}{\sqrt{36}}\right) = (6.265, 6.735)$$

- We are interested in the mean full-term birth weight of U.S.-born babies, and we know the population standard deviation of U.S. birth weights is 1.1 lbs. We take a sample of 36 U.S. babies, and find that in the sample the average birth weight is 6.5 lbs.

- An 80% confidence interval for the population mean birth weight of U.S.-born babies is $(6.265, 6.735)$.

- PollEv.com/stats8: Write a sentence interpreting this interval in context.

Confidence Level

Example

- For some α between 0 and 1, suppose you are asked for a $100 \times (1 - \alpha)\,\%$ interval:

$$\left(\bar{x} - z_{1-\alpha/2}\frac{\sigma}{\sqrt{n}},\ \bar{x} + z_{1-\alpha/2}\frac{\sigma}{\sqrt{n}}\right) = \left(6.5 - 1.28\frac{1.1}{\sqrt{36}},\ 6.5 + 1.28\frac{1.1}{\sqrt{36}}\right) = (6.265,\ 6.735)$$

- We are interested in the mean full-term birth weight of U.S.-born babies, and we know the population standard deviation of U.S. birth weights is 1.1 lbs. We take a sample of 36 U.S. babies, and find that in the sample the average birth weight is 6.5 lbs.

- An 80% confidence interval for the population mean birth weight of U.S.-born babies is $(6.265,\ 6.735)$.

- We are 80% confident that the population mean birth weight of U.S.-born babies is between 6.265 and 6.735 lbs.

Standard Error

- For a $100 \times (1 - \alpha)\,\%$ interval:

$$\left(\bar{x} - z_{1-\alpha/2}\frac{\sigma}{\sqrt{n}}, \bar{x} + z_{1-\alpha/2}\frac{\sigma}{\sqrt{n}}\right)$$

- Can you think of any situation where we would want to estimate the population mean μ but we would know the population standard deviation σ?
  - *Maybe* manufacturing quality control? *Maybe* measurement error with a lab instrument?
    - These are a stretch. It's unrealistic to ever assume we know σ when we don't know μ. So what do we do?

Standard Error

- For a $100 \times (1-\alpha)\,\%$ interval:

$$\left(\bar{x} - z_{1-\alpha/2}\frac{\sigma}{\sqrt{n}}, \bar{x} + z_{1-\alpha/2}\frac{\sigma}{\sqrt{n}}\right)$$

- Can you think of any situation where we would want to estimate the population mean μ but we would know the population standard deviation σ?
  - *Maybe* manufacturing quality control? *Maybe* measurement error with a lab instrument?
    - These are a stretch. It's unrealistic to ever assume we know σ when we don't know μ. So what do we do? → **estimate it using the sample standard deviation**

Standard Error

- For a $100 \times (1 - \alpha)\,\%$ interval:

$$\left(\bar{x} - z_{1-\alpha/2} \frac{\sigma}{\sqrt{n}}, \bar{x} + z_{1-\alpha/2} \frac{\sigma}{\sqrt{n}} \right)$$

- We denote the population standard deviation using σ

- We denote the sample standard deviation using s

- We will use s to estimate σ:

$$\left(\bar{x} - z_{1-\alpha/2} \frac{s}{\sqrt{n}}, \bar{x} + z_{1-\alpha/2} \frac{s}{\sqrt{n}} \right)$$

Standard Error

- For a $100 \times (1 - \alpha)\,\%$ interval:

$$\left(\bar{x} - z_{1-\alpha/2}\frac{\sigma}{\sqrt{n}},\ \bar{x} + z_{1-\alpha/2}\frac{\sigma}{\sqrt{n}}\right)$$

This is the **standard deviation** of the sampling distribution of the sample mean

- We denote the population standard deviation using σ (this is a **p**opulation **p**arameter)

- We denote the sample standard deviation using s (this is a **s**ample **s**tatistic)

- We will use s to estimate σ:

$$\left(\bar{x} - z_{1-\alpha/2}\frac{s}{\sqrt{n}},\ \bar{x} + z_{1-\alpha/2}\frac{s}{\sqrt{n}}\right)$$

This is the version of the above using s to estimate σ, we call this the **standard error** of the sample mean

Standard Error

- For a $100 \times (1 - \alpha)\,\%$ interval:

$$\left(\bar{x} - z_{1-\alpha/2} \frac{\sigma}{\sqrt{n}},\ \bar{x} + z_{1-\alpha/2} \frac{\sigma}{\sqrt{n}} \right)$$

- We denote the population standard deviation using σ
- We denote the sample standard deviation using s
- We will use s to estimate σ:

$$\left(\bar{x} - z_{1-\alpha/2} \frac{s}{\sqrt{n}},\ \bar{x} + z_{1-\alpha/2} \frac{s}{\sqrt{n}} \right)$$

But wait — s does not always equal exactly σ. There is some uncertainty when estimating σ. Don't we have to account for that in our confidence interval?

Standard Error

- Remember, CLT says:

$$\bar{X} \sim N(\mu, \sigma^2/n)$$

- NOT:

$$\bar{X} \sim N(\mu, s^2/n)$$

- So… can't use z anymore??

$$\left(\bar{x} - (??)\frac{s}{\sqrt{n}}, \bar{x} + (??)\frac{s}{\sqrt{n}}\right)$$

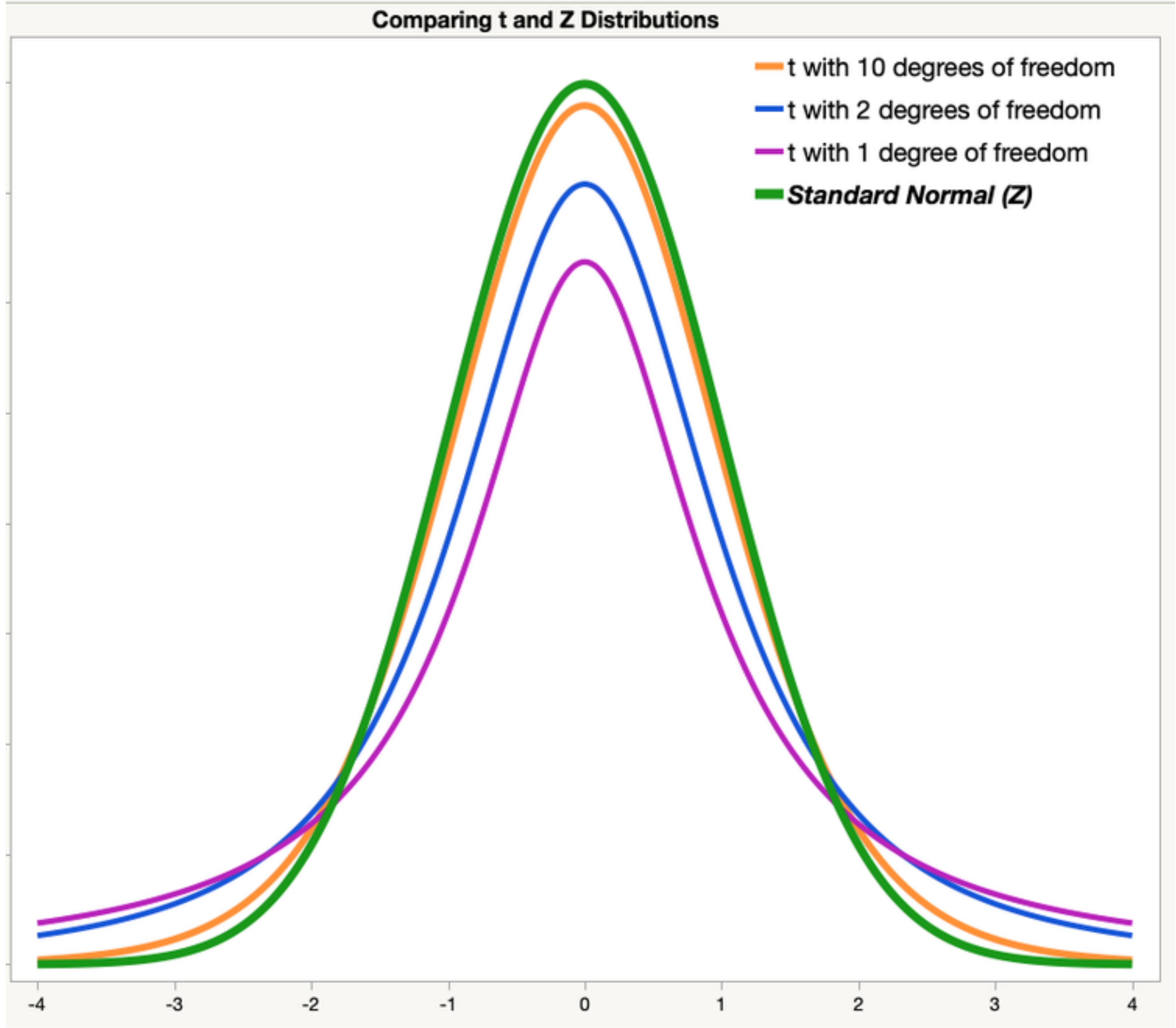
But wait — s does not always equal exactly σ. There is some uncertainty when estimating σ. Don't we have to account for that in our confidence interval?

The *t*-distribution

- Can't use z anymore??

$$\left(\bar{x} - t_{1-\alpha/2,\, n-1}\frac{s}{\sqrt{n}},\ \bar{x} + t_{1-\alpha/2,\, n-1}\frac{s}{\sqrt{n}}\right)$$

- We use t instead of z to account for the extra uncertainty in estimating σ with s
- "Degrees of freedom" is the sample size n minus 1
- Notice what happens as n gets bigger...

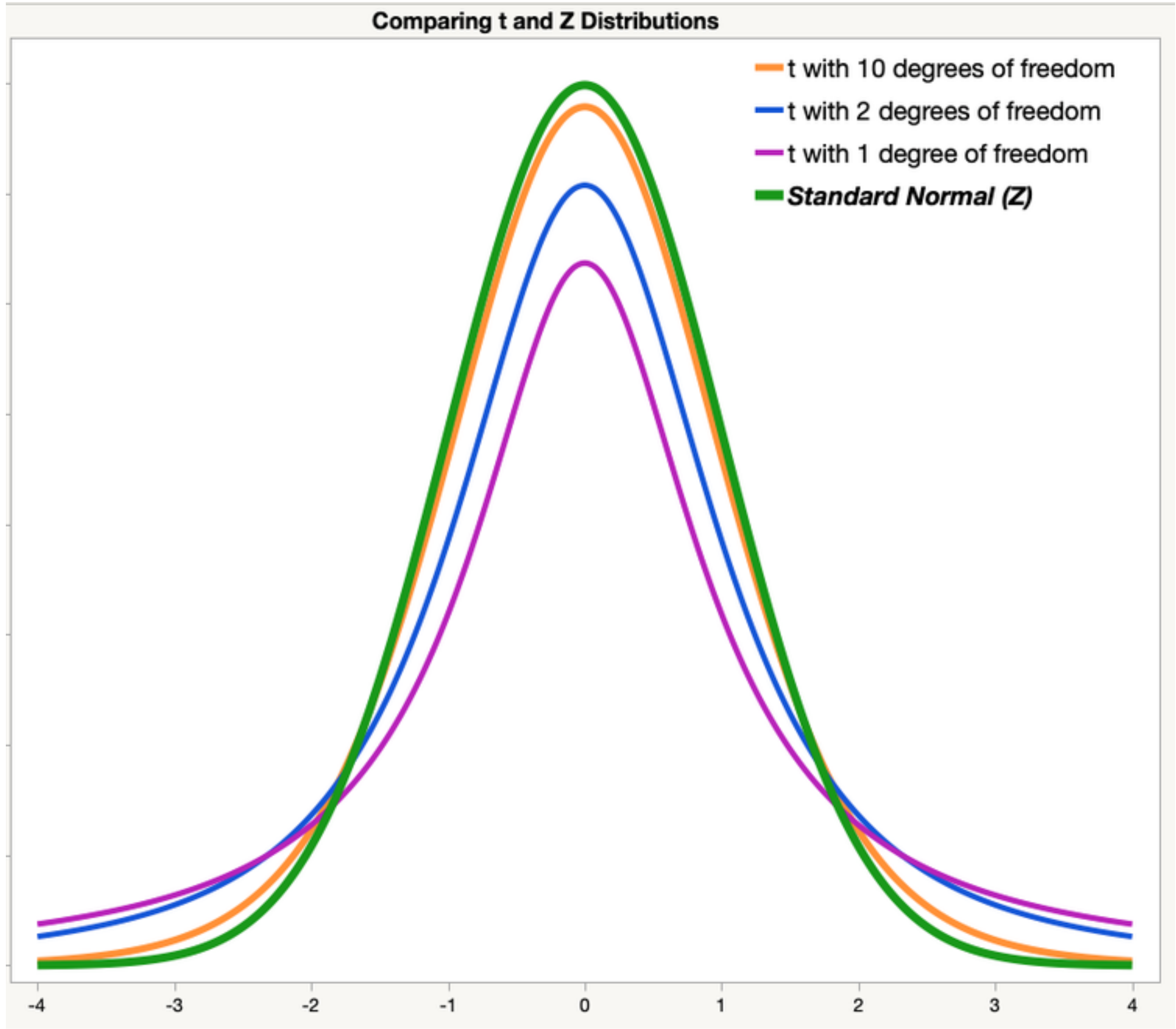


The *t*-distribution

- When n is large enough that the sampling distribution is reasonably approximated by a normal distribution (by CLT),
  - it is also large enough that the t -distribution is well-approximated by a z-distribution

★**In this class, we will just use z because our examples will use large n**

- BUT, it is good to be aware of t (used often in the literature)



Confidence Interval for a Mean

- We did it! We derived a general-use formula for a CI for a mean.

★Don't forget: there are conditions to be able to use this (conditions for CLT to apply)

  - **i.i.d. sample**: ok if we have a randomly selected sample from a much larger population
  - **sample size large enough**: rules of thumb we learned last lecture (20-30 minimum, usually)

- For a $100 \times (1 - \alpha)$ % interval:

$$\left(\bar{x} - z_{1-\alpha/2}\frac{s}{\sqrt{n}}, \bar{x} + z_{1-\alpha/2}\frac{s}{\sqrt{n}}\right)$$

Confidence Interval for a Mean

Example

- For a $100 \times (1 - \alpha)\,\%$ interval:

$$\left(\bar{x} - z_{1-\alpha/2}\frac{s}{\sqrt{n}}, \bar{x} + z_{1-\alpha/2}\frac{s}{\sqrt{n}}\right)$$

The authors of a 1992 JAMA study (source) sampled 130 healthy people aged 18-40, interested in estimating the population average body temperature in that age group. We will assume that the sample was i.i.d. from the population of interest. They found that in this sample, the average body temperature was 98.25 degrees Fahrenheit, and the standard deviation was 0.73 degrees. **Find a 90% confidence interval for the population mean body temperature of healthy people the age group 18-40.**

Confidence Interval for a Mean

PollEv.com/stats8

- For a $100 \times (1-\alpha)\,\%$ interval:

$$\left(\bar{x} - z_{1-\alpha/2}\frac{s}{\sqrt{n}},\ \bar{x} + z_{1-\alpha/2}\frac{s}{\sqrt{n}}\right)$$

- What is $\bar{x}$?

The authors of a 1992 JAMA study (source) sampled 130 healthy people aged 18-40, interested in estimating the population average body temperature in that age group. We will assume that the sample was i.i.d. from the population of interest. They found that in this sample, the average body temperature was 98.25 degrees Fahrenheit, and the standard deviation was 0.73 degrees. **Find a 90% confidence interval for the population mean body temperature of healthy people the age group 18-40.**

Confidence Interval for a Mean

PollEv.com/stats8

- For a $100 \times (1-\alpha)\,\%$ interval:

$$\left(98.25 - z_{1-\alpha/2}\frac{s}{\sqrt{n}},\ 98.25 + z_{1-\alpha/2}\frac{s}{\sqrt{n}}\right)$$

- What is $\bar{x}$?

$\bar{x} = 98.25$

The authors of a 1992 JAMA study (source) sampled 130 healthy people aged 18-40, interested in estimating the population average body temperature in that age group. We will assume that the sample was i.i.d. from the population of interest. They found that in this sample, the average body temperature was 98.25 degrees Fahrenheit, and the standard deviation was 0.73 degrees. **Find a 90% confidence interval for the population mean body temperature of healthy people the age group 18-40.**

Confidence Interval for a Mean

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- For a $100 \times (1 - \alpha)\,\%$ interval:

$$\left(98.25 - z_{1-\alpha/2}\frac{s}{\sqrt{n}},\ 98.25 + z_{1-\alpha/2}\frac{s}{\sqrt{n}}\right)$$

- What is s?

The authors of a 1992 JAMA study (source) sampled 130 healthy people aged 18-40, interested in estimating the population average body temperature in that age group. We will assume that the sample was i.i.d. from the population of interest. They found that in this sample, the average body temperature was 98.25 degrees Fahrenheit, and the standard deviation was 0.73 degrees. **Find a 90% confidence interval for the population mean body temperature of healthy people the age group 18-40.**

Confidence Interval for a Mean

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- For a $100 \times (1 - \alpha)\,\%$ interval:

$$\left(98.25 - z_{1-\alpha/2}\frac{0.73}{\sqrt{n}}, 98.25 + z_{1-\alpha/2}\frac{0.73}{\sqrt{n}}\right)$$

- What is s?

$s = 0.73$

The authors of a 1992 JAMA study (source) sampled 130 healthy people aged 18-40, interested in estimating the population average body temperature in that age group. We will assume that the sample was i.i.d. from the population of interest. They found that in this sample, the average body temperature was 98.25 degrees Fahrenheit, and the standard deviation was 0.73 degrees. **Find a 90% confidence interval for the population mean body temperature of healthy people the age group 18-40.**

Confidence Interval for a Mean

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- For a $100 \times (1 - \alpha)\,\%$ interval:

$$\left(98.25 - z_{1-\alpha/2}\frac{0.73}{\sqrt{n}},\ 98.25 + z_{1-\alpha/2}\frac{0.73}{\sqrt{n}}\right)$$

- What is α?

The authors of a 1992 JAMA study (source) sampled 130 healthy people aged 18-40, interested in estimating the population average body temperature in that age group. We will assume that the sample was i.i.d. from the population of interest. They found that in this sample, the average body temperature was 98.25 degrees Fahrenheit, and the standard deviation was 0.73 degrees. **Find a 90% confidence interval for the population mean body temperature of healthy people the age group 18-40.**

Confidence Interval for a Mean

PollEv.com/stats8

- For a $100 \times (1 - 0.1)\,\%$ interval:

$$\left(98.25 - z_{1-0.1/2}\frac{0.73}{\sqrt{n}},\ 98.25 + z_{1-0.1/2}\frac{0.73}{\sqrt{n}}\right)$$

- What is α?

$$100(1-\alpha)\,\% = 90\,\% \rightarrow \alpha = 0.1$$

The authors of a 1992 JAMA study (source) sampled 130 healthy people aged 18-40, interested in estimating the population average body temperature in that age group. We will assume that the sample was i.i.d. from the population of interest. They found that in this sample, the average body temperature was 98.25 degrees Fahrenheit, and the standard deviation was 0.73 degrees. **Find a 90% confidence interval for the population mean body temperature of healthy people the age group 18-40.**

Confidence Interval for a Mean

PollEv.com/stats8

- For a $100 \times (1 - 0.1)\,\%$ interval:

$$\left(98.25 - z_{1-0.1/2}\frac{0.73}{\sqrt{n}},\ 98.25 + z_{1-0.1/2}\frac{0.73}{\sqrt{n}}\right)$$

- What is $z_{1-\alpha/2}$? (Use R)

The authors of a 1992 JAMA study (source) sampled 130 healthy people aged 18-40, interested in estimating the population average body temperature in that age group. We will assume that the sample was i.i.d. from the population of interest. They found that in this sample, the average body temperature was 98.25 degrees Fahrenheit, and the standard deviation was 0.73 degrees. **Find a 90% confidence interval for the population mean body temperature of healthy people the age group 18-40.**

Confidence Interval for a Mean

PollEv.com/stats8

- For a $100 \times (1 - 0.1)\,\%$ interval:

$$\left(98.25 - 1.645\frac{0.73}{\sqrt{n}},\ 98.25 + 1.645\frac{0.73}{\sqrt{n}}\right)$$

- What is $z_{1-\alpha/2}$? (Use R)

`qnorm(0.95)` ≈ 1.645

The authors of a 1992 JAMA study (source) sampled 130 healthy people aged 18-40, interested in estimating the population average body temperature in that age group. We will assume that the sample was i.i.d. from the population of interest. They found that in this sample, the average body temperature was 98.25 degrees Fahrenheit, and the standard deviation was 0.73 degrees. **Find a 90% confidence interval for the population mean body temperature of healthy people the age group 18-40.**

Confidence Interval for a Mean

PollEv.com/stats8

- For a $100 \times (1 - 0.1)\,\%$ interval:

$$\left(98.25 - 1.645\frac{0.73}{\sqrt{n}},\ 98.25 + 1.645\frac{0.73}{\sqrt{n}}\right)$$

The authors of a 1992 JAMA study (source) sampled 130 healthy people aged 18-40, interested in estimating the population average body temperature in that age group. We will assume that the sample was i.i.d. from the population of interest. They found that in this sample, the average body temperature was 98.25 degrees Fahrenheit, and the standard deviation was 0.73 degrees. **Find a 90% confidence interval for the population mean body temperature of healthy people the age group 18-40.**

Confidence Interval for a Mean

PollEv.com/stats8

- For a $100 \times (1 - 0.1)\,\%$ interval:

$$\left(98.25 - 1.645\frac{0.73}{\sqrt{n}},\ 98.25 + 1.645\frac{0.73}{\sqrt{n}}\right)$$

(98.14, 98.36)

The authors of a 1992 JAMA study (source) sampled 130 healthy people aged 18-40, interested in estimating the population average body temperature in that age group. We will assume that the sample was i.i.d. from the population of interest. They found that in this sample, the average body temperature was 98.25 degrees Fahrenheit, and the standard deviation was 0.73 degrees. A 90% confidence interval for the population mean body temperature of healthy people the age group 18-40 is (98.14, 98.36). **Write one sentence interpreting this interval.**

Confidence Interval for a Mean

PollEv.com/stats8

- For a $100 \times (1 - 0.1)\,\%$ interval:

$$\left(98.25 - 1.645\frac{0.73}{\sqrt{n}},\ 98.25 + 1.645\frac{0.73}{\sqrt{n}}\right)$$

$$(98.14,\ 98.36)$$

The authors of a 1992 JAMA study (source) sampled 130 healthy people aged 18-40, interested in estimating the population average body temperature in that age group. We will assume that the sample was i.i.d. from the population of interest. They found that in this sample, the average body temperature was 98.25 degrees Fahrenheit, and the standard deviation was 0.73 degrees. A 90% confidence interval for the population mean body temperature of healthy people the age group 18-40 is (98.14, 98.36). **Write one sentence interpreting this interval.**

Confidence Interval for a Mean

PollEv.com/stats8

- For a $100 \times (1 - 0.1)\,\%$ interval:

$$\left(98.25 - 1.645\frac{0.73}{\sqrt{n}},\ 98.25 + 1.645\frac{0.73}{\sqrt{n}}\right)$$

$$(98.14,\ 98.36)$$

We are 90% confident that the mean body temperature in the population of healthy adults aged 18-40 is between 98.14 and 98.36 degrees Fahrenheit.

The authors of a 1992 JAMA study (source) sampled 130 healthy people aged 18-40, interested in estimating the population average body temperature in that age group. We will assume that the sample was i.i.d. from the population of interest. They found that in this sample, the average body temperature was 98.25 degrees Fahrenheit, and the standard deviation was 0.73 degrees. A 90% confidence interval for the population mean body temperature of healthy people the age group 18-40 is (98.14, 98.36). **Write one sentence interpreting this interval.**

Confidence Interval for a Proportion

(It's basically the same!)

- A $100 \times (1 - \alpha)\,\%$ interval for a mean:

$$\left(\bar{x} - z_{1-\alpha/2}\frac{s}{\sqrt{n}},\, \bar{x} + z_{1-\alpha/2}\frac{s}{\sqrt{n}} \right)$$

- For Bernoulli-distributed data, the population mean is p, so we estimate it with $\hat{p}$; the population SD is $p(1 - p)$, so we estimate it with $\hat{p}(1 - \hat{p})$.

- A $100 \times (1 - \alpha)\,\%$ interval for a proportion:

$$\left(\hat{p} - z_{1-\alpha/2}\sqrt{\frac{\hat{p}(1-\hat{p})}{n}},\, \hat{p} + z_{1-\alpha/2}\sqrt{\frac{\hat{p}(1-\hat{p})}{n}} \right)$$

Anatomy of a Confidence Interval

"**Point estimate**" — the middle of the CI

"multiplier" — based on the confidence level

$$\left(\bar{x} - \boxed{z_{1-\alpha/2}\frac{s}{\sqrt{n}}},\ \bar{x} + \boxed{z_{1-\alpha/2}\frac{s}{\sqrt{n}}}\right) \qquad \left(\hat{p} - \boxed{z_{1-\alpha/2}\frac{\hat{p}(1-\hat{p})}{\sqrt{n}}},\ \hat{p} + \boxed{z_{1-\alpha/2}\frac{\hat{p}(1-\hat{p})}{\sqrt{n}}}\right)$$

"**Standard error** of the sample statistic" — estimating the standard deviation of the sampling distribution with the standard deviation from the sample

"**Margin of error**" — half the width of the CI (maximum distance between the sample statistic and the population parameter in $100 \times (1-\alpha)$ % random samples)

General form of *any* confidence interval is [point estimate] ± [margin of error]

The result of all our hard work:

<u>Means</u>

A $100(1-\alpha)\%$ CI for a mean is given by:

$$\left(\bar{x} - z_{1-\alpha/2}\frac{s}{\sqrt{n}}, \bar{x} + z_{1-\alpha/2}\frac{s}{\sqrt{n}}\right)$$

"We are $[100(1-\alpha)]\%$ confident that the population mean of [context] is between [lower bound] and [upper bound] [units]."

<u>Proportions</u>

A $100(1-\alpha)\%$ CI for a proportion is given by:

$$\left(\hat{p} - z_{1-\alpha/2}\frac{\hat{p}(1-\hat{p})}{\sqrt{n}}, \hat{p} + z_{1-\alpha/2}\frac{\hat{p}(1-\hat{p})}{\sqrt{n}}\right)$$

"We are $[100(1-\alpha)]\%$ confident that the population proportion of [context] is between [lower bound] and [upper bound]."

*<u>Note</u>: you may hear these called "one-sample z-intervals"

Confidence Intervals: How-To

- So, you're given a scenario and you want a confidence interval. What do you do?
  1. Confirm what **parameter** you want the interval for (for now, just mean or proportion).
  2. Check whether **assumptions/conditions** for the CLT (so, for the interval) are met (i.i.d. + sample size "rules of thumb").
  3. **Identify** the relevant quantities provided: $\hat{p}$ or $\bar{x}$, s; get α based on the confidence level; use R to get $z_{1-\alpha/2}$. **Plug in** to the formula.
  4. **Interpret** the interval in context.

Confidence Interval

Example

In the spring of 2020, researchers in Grand Nancy, France randomly sampled individuals from electoral lists and tested them for SARS-CoV-2 antibodies. (This is called a *seroprevalence* study). Among 2006 participants, 43 tested positive. Find a 95% confidence interval for the population proportion of people in Grand Nancy, France who had SARS-CoV-2 antibodies.

<u>Step 1</u>: What parameter are we interested in?

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<u>Step 1</u>: What parameter are we interested in?

p, the population proportion of people in France who had COVID antibodies in spring 2020

Confidence Interval

Example

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<u>Step 2</u>: Are the conditions for a valid interval met?

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Step 2: Are the conditions for a valid interval met?

Conditions for CLT:

i.i.d. sample (random sample from large population) → met because they randomly sampled 2006 people in the large population of France

Sample size: for a proportion, we want $n\hat{p} > 10$ and $n(1-\hat{p}) > 10$ → met because $n\hat{p} = 43$ and $n(1-\hat{p}) = 1963$

Confidence Interval

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Step 3: Identify relevant values; plug in to get the CI.

Confidence Interval

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Step 3: Identify relevant values; plug in to get the CI.

$\hat{p}$ = the proportion in the sample of people who tested positive $= 43/2006 \approx 0.0214$

$\alpha = 1 - 0.95 = 0.05$

$z_{1-0.05/2} = z_{0.975} = \texttt{qnorm(0.975)} \approx 1.96$

Confidence Interval

Example

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<u>Step 3</u>: Identify relevant values; plug in to get the CI.

$\hat{p} = 43/2006 \approx 0.0214$; $\alpha = \ = 0.05$; $z_{0.975} = \text{qnorm(0.975)} \approx 1.96$

$$\left(\hat{p} - z_{1-\alpha/2}\sqrt{\frac{\hat{p}(1-\hat{p})}{n}}, \hat{p} + z_{1-\alpha/2}\sqrt{\frac{\hat{p}(1-\hat{p})}{n}}\right)$$

$$=\left(43/2006 - 1.96\sqrt{\frac{43/2006(1963/2006)}{2006}}, 43/2006 + 1.96\sqrt{\frac{43/2006(1963/2006)}{2006}}\right)$$

$$\approx (0.0151, 0.0278)$$

Confidence Interval

Example

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<u>Step 4</u>: Interpret in context.

Confidence Interval

Example

In the spring of 2020, researchers in Grand Nancy, France randomly sampled individuals from electoral lists and tested them for SARS-CoV-2 antibodies. (This is called a *seroprevalence* study). Among 2006 participants, 43 tested positive. Find a 95% confidence interval for the population proportion of people in Grand Nancy, France who had SARS-CoV-2 antibodies.

Step 4: Interpret in context. (0.0151, 0.0278)

We are 95% confident that the population proportion of people in France with SARS-CoV-2 antibodies was between 0.0151 and 0.0278 in the spring of 2020.

→ if you want, you can use percentages instead: 95% confident that the population percentage was between 1.51% and 2.78%

Confidence Interval

Checking Understanding

Suppose a study presents a 90% confidence interval.

PollEv.com/stats8: Suppose we increase the confidence level from 90% to 99%. What happens to the interval?

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Gets wider, because to be more certain we need to increase the width (the corresponding multiplier from the z-distribution gets bigger)

Confidence Interval

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Suppose a study presents a 90% confidence interval.

PollEv.com/stats8: Suppose we increase the sample size, n. What happens to the interval?

Confidence Interval

Checking Understanding

Suppose a study presents a 90% confidence interval.

PollEv.com/stats8: Suppose we increase the sample size, n. What happens to the interval?

Gets narrower, we gain more information by increasing the sample size, so we become more certain (and, the sample size is in the denominator of the margin of error, so increasing it makes the margin of error smaller)

Confidence Interval

Checking Understanding

Suppose a study presents a 90% confidence interval.

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Confidence Interval

Checking Understanding

Suppose a study presents a 90% confidence interval.

PollEv.com/stats8: Why can't we say that, after computing the interval, there is a 90% probability that the population parameter is in the interval?

Because once the interval is computed, the bounds are fixed numbers. The population parameter is also a fixed number, though it is unknown. We can't speak in probabilities about anything here, because nothing in random, though some things are unknown.

Hypothesis Testing

Statistical Inference

- We will learn two types of statistical inference:

✅ **Confidence intervals** — when you want to estimate the numeric value of a parameter, with some uncertainty around it. *What values of the parameter are plausible?*

⭐ **Hypothesis testing** — when you want to provide evidence for or against a claim about a parameter. *Is this one claimed value plausible?*

★Remember: the idea behind inference is that there is a population we are interested in...

- but we cannot collect data on the whole population...
- so we have to rely on a sample from that population to learn about it.

- The uncertainty comes entirely from the fact that we are taking a random sample from a larger population rather than getting information from the whole population.

Hypothesis Testing Concepts

Body Temp Example

The authors of a 1992 JAMA study (source) sampled 130 healthy people aged 18-40, interested in estimating the population average body temperature in that age group. We will assume that the sample was i.i.d. from the population of interest.

- The reason for this study was to try to see whether the old commonly-stated average of 98.6 was actually true

So, they want to test: **is the population mean body temperature, μ, equal to 98.6?**

- PollEv.com/stats8: One idea would be to look at the probability that $\mu = 98.6$, and if it's low, say that the hypothesis is unlikely. Do you agree with this? Why or why not?

Hypothesis Testing Concepts

Body Temp Example

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So, they want to test: **is the population mean body temperature, μ, equal to 98.6?**

- PollEv.com/stats8: One idea would be to look at the probability that $\mu = 98.6$, and if it's low, say that the hypothesis is unlikely. Do you agree with this? Why or why not?
- No! If we talk about something in probabilities, something has to be **random**.

Hypothesis Testing Concepts

Proof by Contradiction

- Proof by contradiction is a trick in math where you assume something is true, and if it "breaks" everything else, then you can say you've proved it's not true
- Example: I want to show that dividing by 0 is not a valid mathematical operation.
  - *Assume* it *is* a valid mathematical operation. That is, for some number, dividing by 0 produces another meaningful number.
  - Now I will operate under that assumption: suppose division by 0 is valid. For arbitrary nonzero numbers a and b where $a = b$, consider the equation $(a - b)(a + b) = b(a - b)$.
  - If dividing by 0 is valid, then I should be able to divide both sides of this equation by $a - b$, which is 0 because $a = b$. This gives us $a + b = b$. Since $a = b$, this implies $2b = b$.
  - But if b is not 0, this implies that $2 = 1$. This is obviously false, and we have reached a contradiction.
  - Therefore, our original assumption must be wrong: dividing by 0 is not valid.

Hypothesis Testing Concepts

Proof by Contradiction

- Proof by contradiction is a trick in math where you assume something is true, and if it "breaks" everything else, then you can say you've proved it's not true
- Non-math example: I want to prove that some statements are not lies.
  - *Assume* the opposite: that *all* statements are lies.
  - "All statements are lies." is itself a statement.
  - So, by our assumption that all statements are lies, "all statements are lies." must be a lie.
  - Therefore, it is not true that all statements are lies, so at least one statement is not a lie.
  - This contradicts the original assumption that all statements are lies, so we may say that the original assumption is invalid.
  - Therefore, some statements are not lies.

Hypothesis Testing Concepts

- Hypothesis testing is not a true "proof by contradiction" but it uses a similar concept
- We can't speak in probabilities about the hypothesis $\mu = 98.6$, but we *can* assume $\mu = 98.6$,
  - then look at the probability of observing the data we collect, given that $\mu = 98.6$ is true,
  - and if that probability is very very small, we may assume that our original statement that $\mu = 98.6$ is probably wrong.

Hypothesis Testing

JAMA Body Temp Example

- The researchers performing the JAMA study wanted to see whether they could find evidence that the old commonly-stated average of 98.6 degrees Fahrenheit was actually not true.
- Procedure for the hypothesis test:
  - Assume the population average *is* 98.6.
  - Collect data.
  - How likely is the data under that assumption? If unlikely, then they have evidence against the assumption that the population average is 98.6.
- So how do we actually do this?

Hypothesis Testing

JAMA Body Temp Example

- The researchers performing the JAMA study wanted to see whether they could find evidence that the old commonly-stated average of 98.6 degrees Fahrenheit was actually not true.
- *Assume* that $\mu = 98.6$. This is called the **null hypothesis**, and is usually written as H_0,

$$H_0 : \mu = 98.6$$

- Everything that is *not* the null hypothesis is called the **alternative hypothesis**, and it is usually written as H_a,

$$H_a : \mu \neq 98.6$$

Hypothesis Testing

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$$H_a : \mu \neq 98.6$$

🚨The textbook readings do not necessarily define the alternative hypothesis as being complementary to the null. We will define it that way, so that the null and alternative encompass all possible values of a parameter. 🚨

Null and Alternative Hypotheses

PollEv.com/stats8

- Suppose your colleague is performing a hypothesis test on a mean. They write the null and alternative hypotheses:

$$H_0 : \bar{x} = 10;\ H_a : \bar{x} \neq 10$$

Explain to your colleague why this does not make sense.

Null and Alternative Hypotheses

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$$H_0 : \bar{x} = 10; \; H_a : \bar{x} \neq 10$$

Explain to your colleague why this does not make sense.

The idea behind a hypothesis test is that you are testing hypotheses about the *population* — about the population parameter, μ. You do not need to test hypotheses about $\bar{x}$, because you will know it after you collect the sample.

Hypothesis Testing

JAMA Body Temp Example

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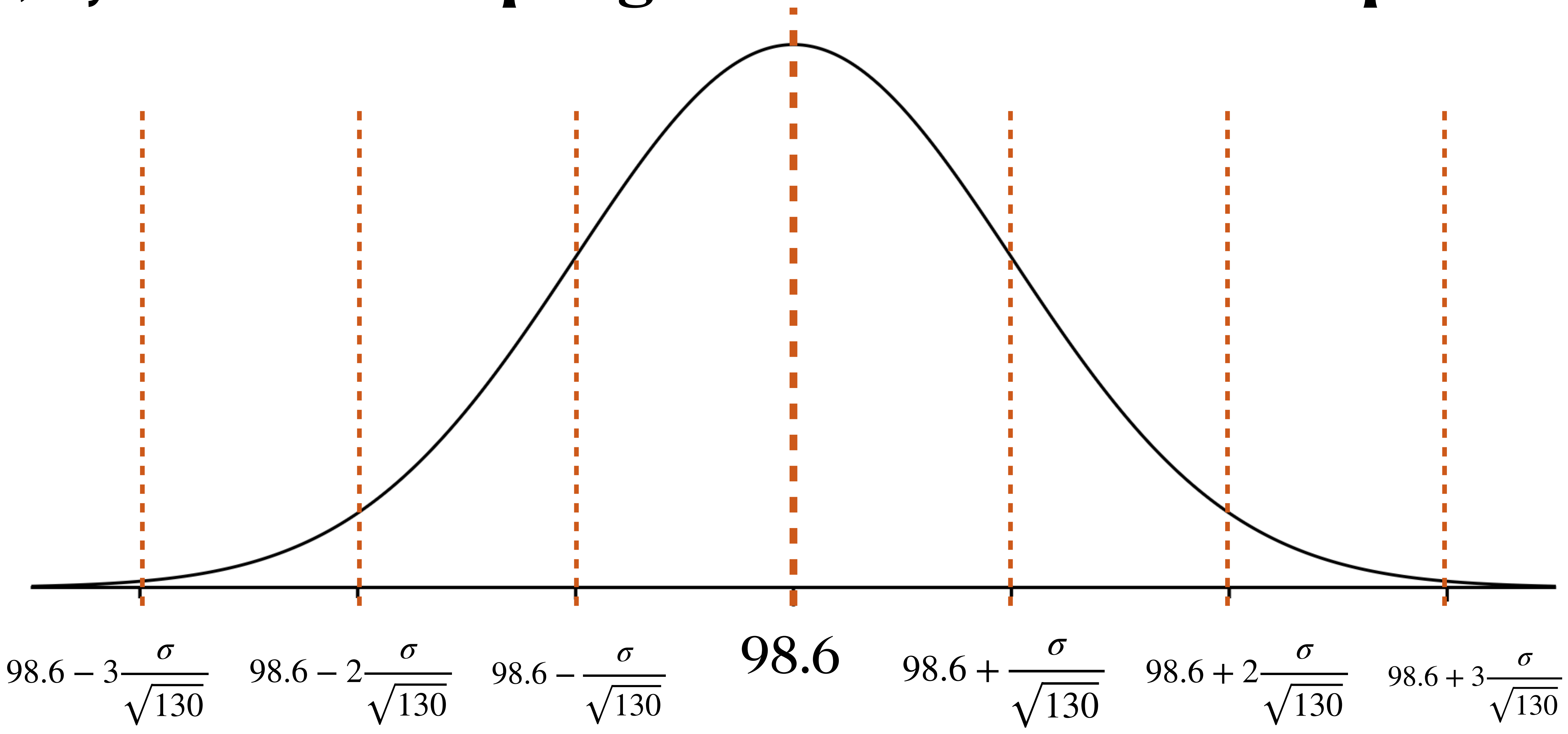
- Null hypothesis, $H_0 : \mu = 98.6$; alternative hypothesis, $H_a : \mu \neq 98.6$.
- Under the null, (that is, *assuming the null is true*), if we take a random sample of size 130 from this population, by CLT the **sampling distribution of the sample mean** should look like this:



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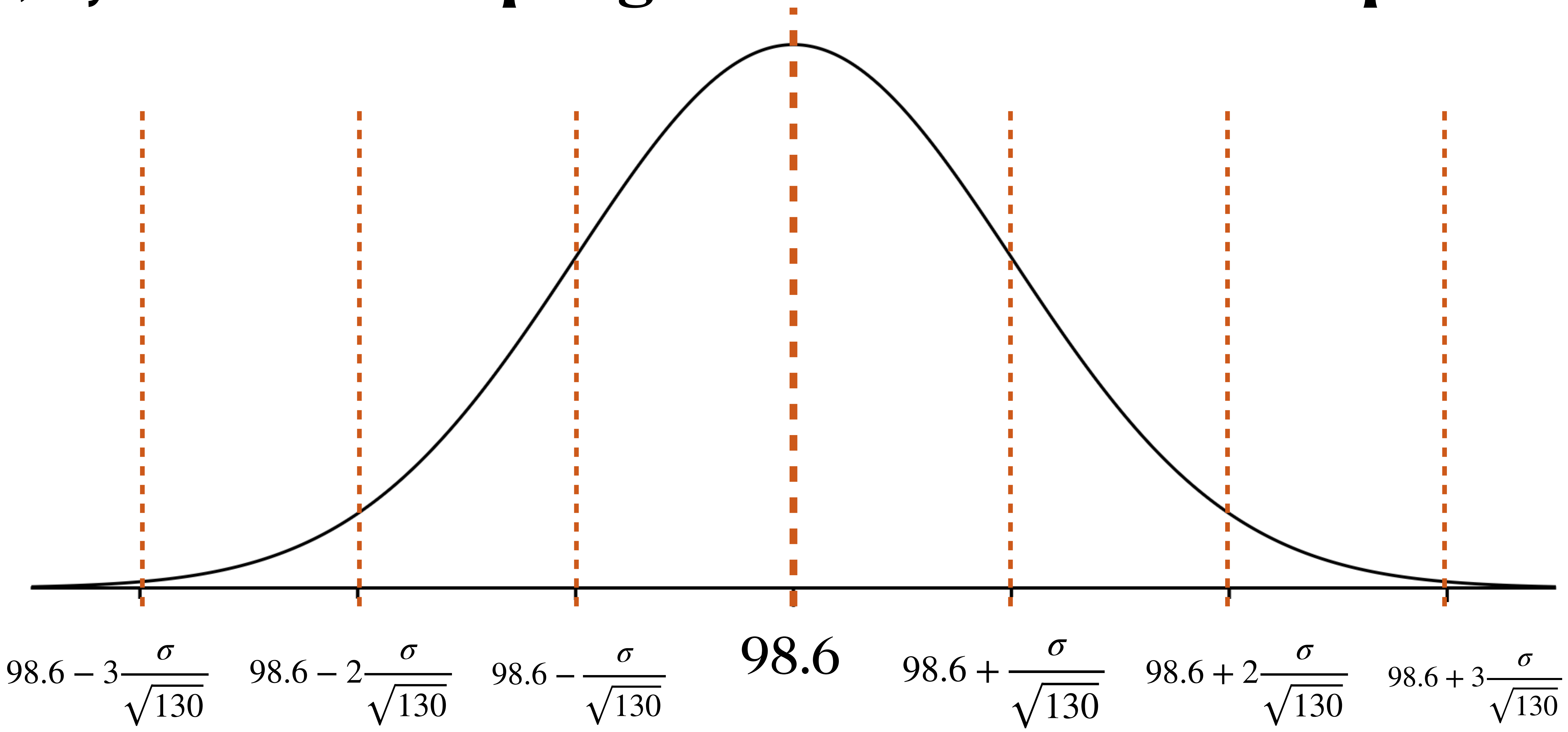

Remember that this is the theoretical distribution of sample means in many samples of 130

Hypothesis Testing

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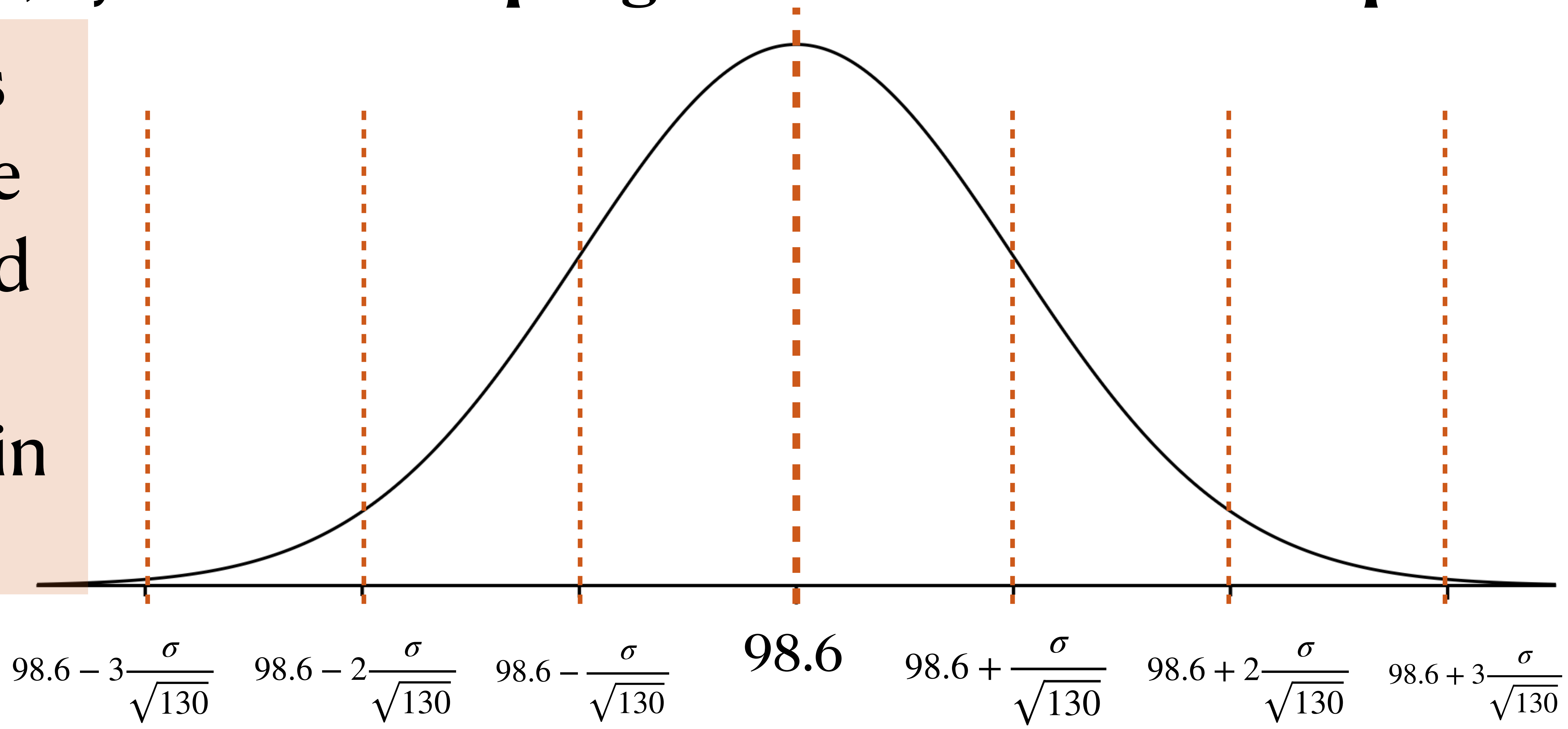
- Under the null, (that is, *assuming the null is true*), if we take a random sample of size 130 from this population, by CLT the **sampling distribution of the sample mean** should look like this:

These researchers observed a sample mean of **98.25** and a standard deviation of **0.73** in their data



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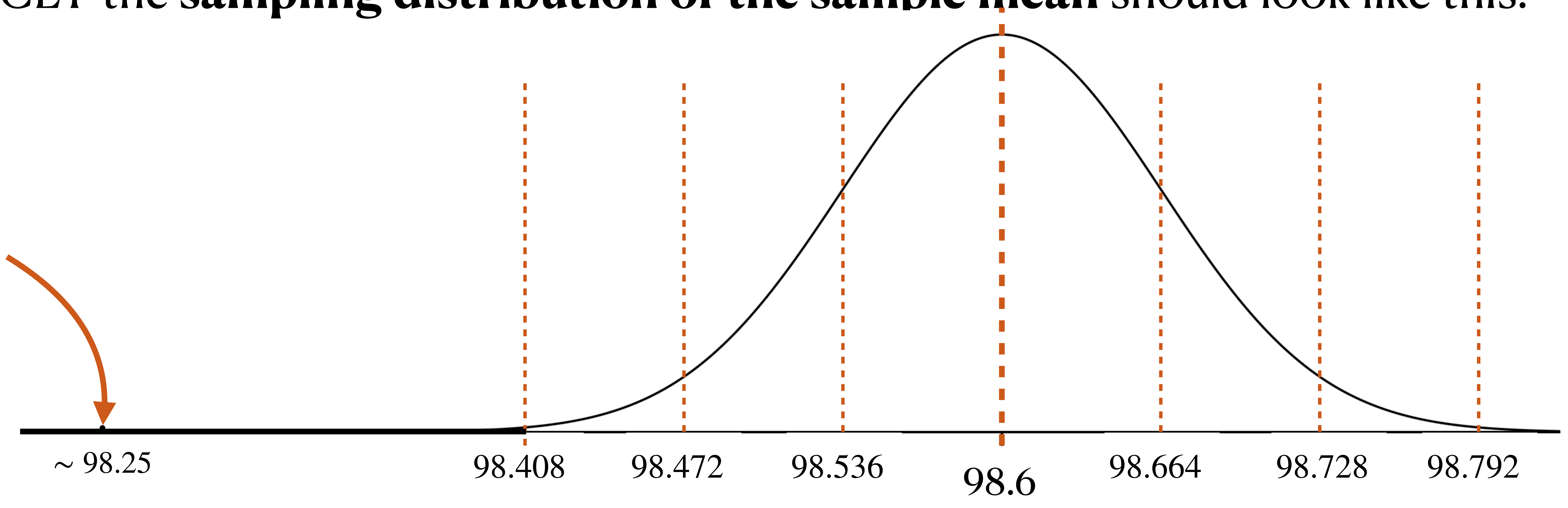
If we assume the null is true and approximate σ with s, the mean that the researchers found is somewhere over here



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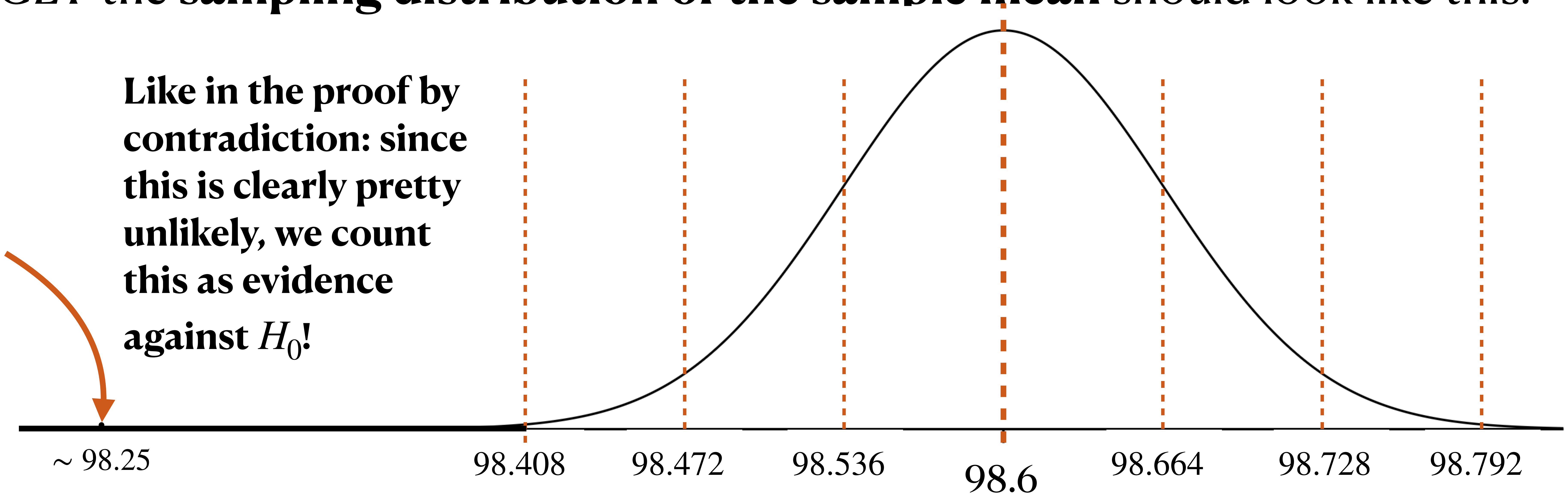
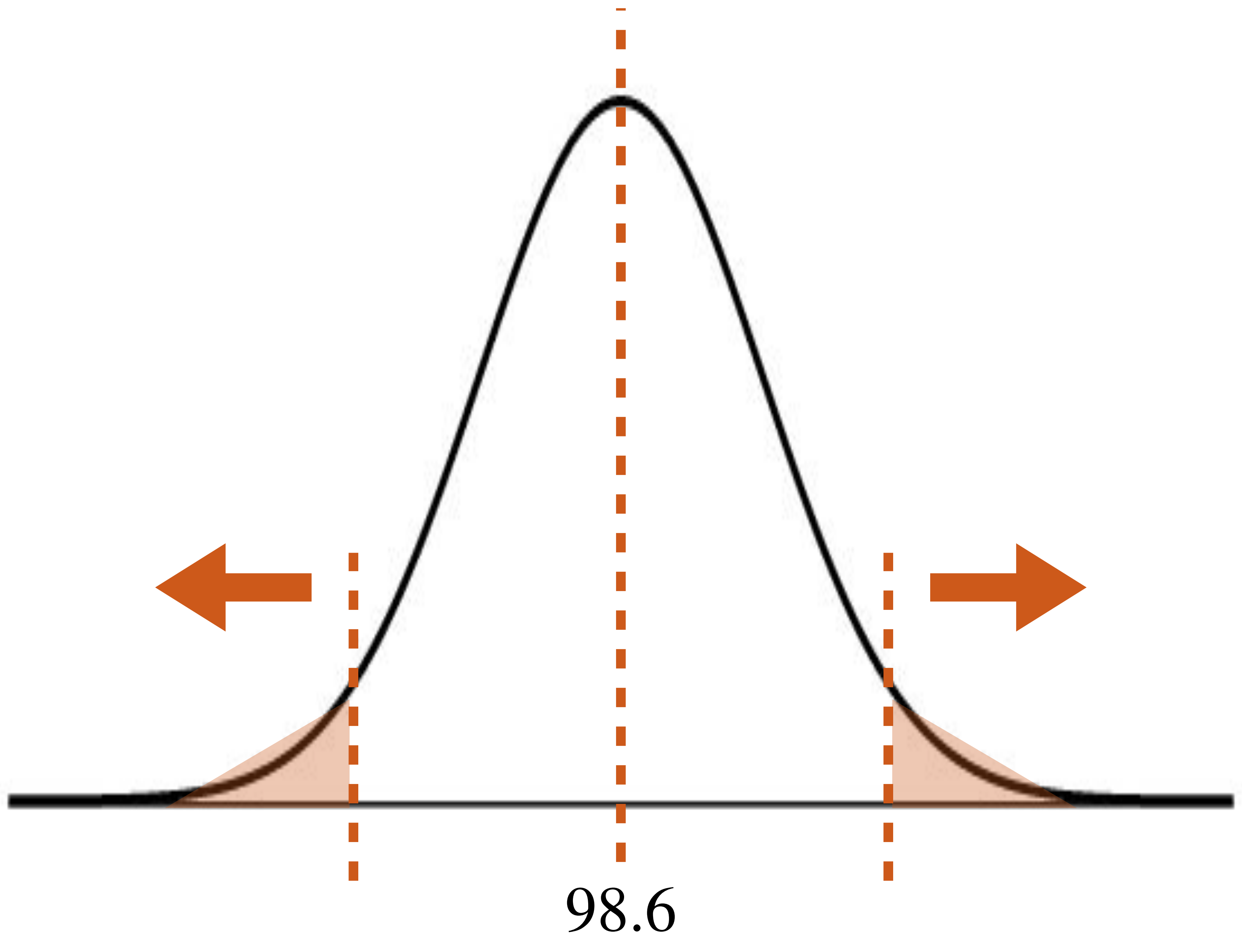
Like in the proof by contradiction: since this is clearly pretty unlikely, we count this as evidence against H_0!



Hypothesis Testing

Formalizing the Procedure

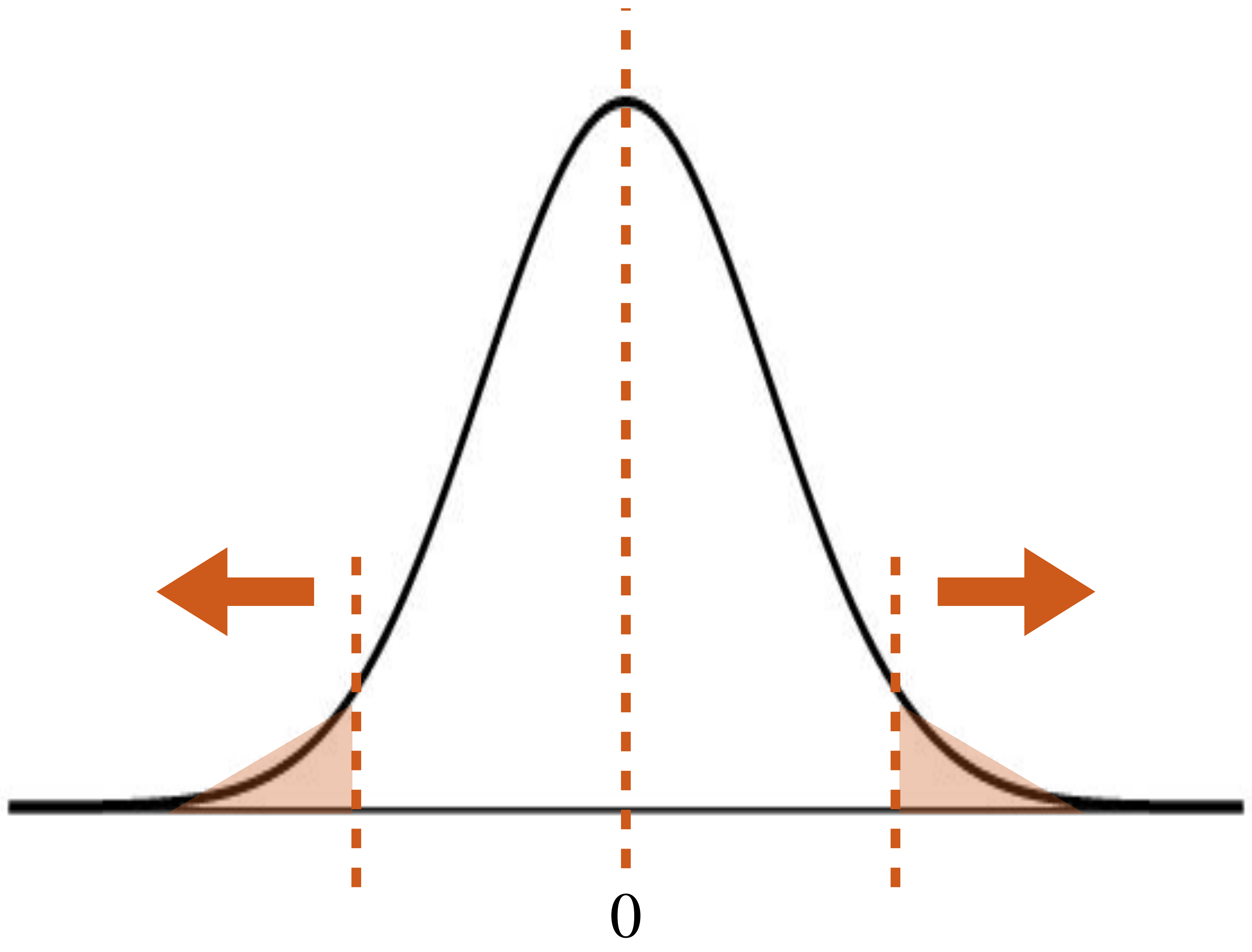
- First, we determine our hypotheses,
  - null hypothesis, $H_0 : \mu = 98.6$; alternative hypothesis, $H_a : \mu \neq 98.6$.
- We assume the null hypothesis H_0 is true, and figure out what our approximate sampling distribution would be under that assumption.
- We collect our sample and compute sample statistics $(\bar{x}, s)$.
- For a formal test, we wouldn't just plot the distribution and qualitatively assess the sample:
  - We compute the probability of observing our data or "more extreme" (that is, more unlikely under the null) under the assumption of the null hypothesis



Hypothesis Testing

Test Statistics

- We compute the probability of observing our data or "more extreme" (that is, more unlikely under the null) under the assumption of the null hypothesis
- It is a convention that we first compute a **test statistic**
  - A **test statistic** in general is the data summary used to evaluate the null and alternative hypotheses
  - Usually a "standardized" version of the sample statistic we are interested in
- Z-score!



Hypothesis Testing

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- Null hypothesis, $H_0 : \mu = 98.6$; alternative hypothesis, $H_a : \mu \neq 98.6$.
- Z-score is $\dfrac{\bar{X} - \mu}{s/\sqrt{n}}$; since $\bar{X}$ is approximately normal (by CLT), we know that $Z = \dfrac{\bar{X} - \mu}{s/\sqrt{n}} \sim N(0,1)$
- Assuming the null hypothesis is true, the z-score for the researchers' observation is:

$$z = \frac{98.25 - 98.6}{0.73/\sqrt{130}} \approx -5.467$$

Hypothesis Testing

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Test statistic

- We compute the probability of seeing this test statistic or one further away from the null hypothesis ("more extreme")

Hypothesis Testing

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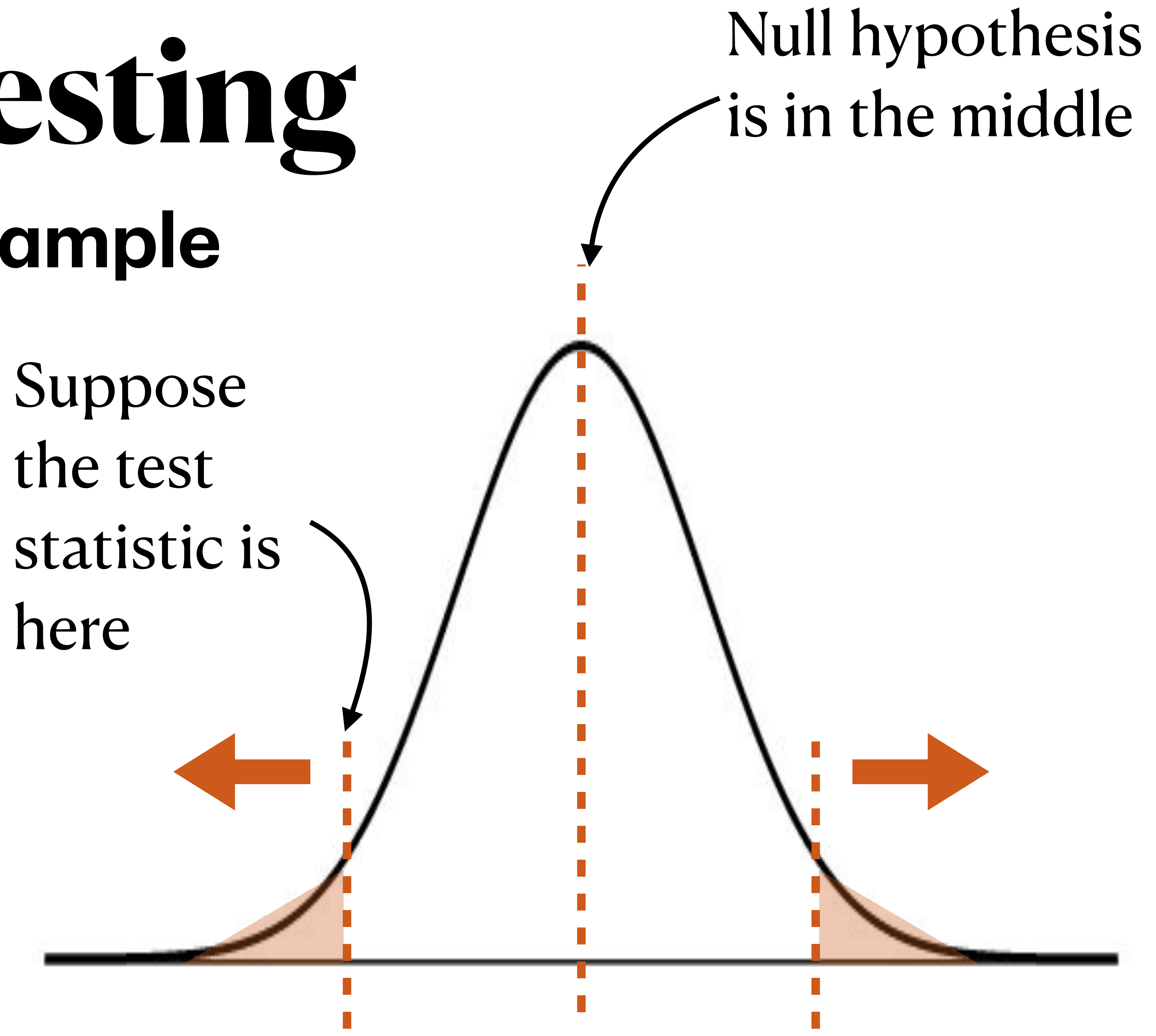

★We want the probability of the data *or* *anything* that would be less likely under the null. That means *both tails!*

$$P(Z \leq -5.467 \text{ or } Z \geq 5.467)$$

Hypothesis Testing

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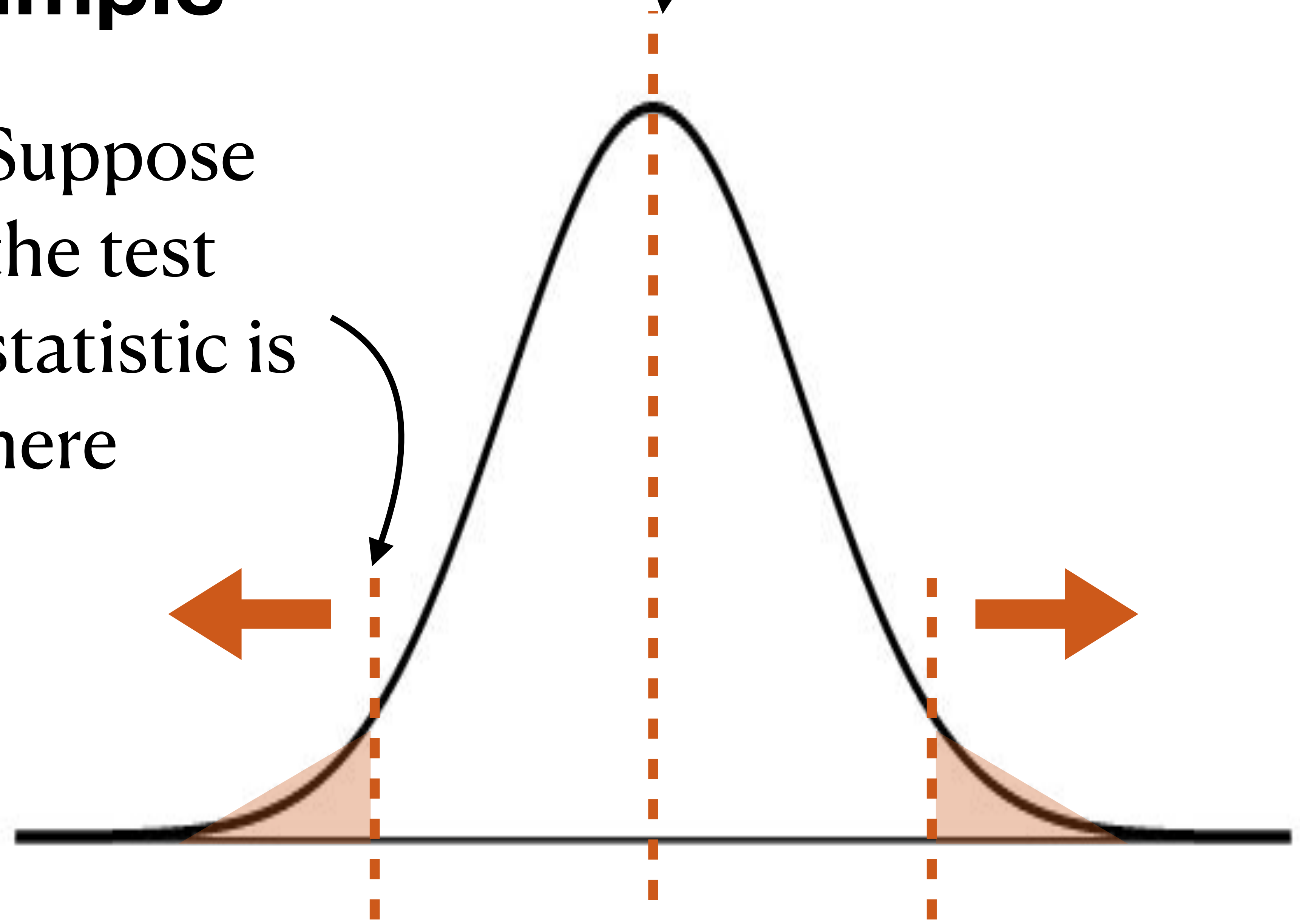


★We want the probability of the data *or* <u>*anything*</u> that would be less likely under the null. That means *both tails!*

$$P(Z \leq -5.467 \text{ or } Z \geq 5.467) \approx 4.6 \times 10^{-8}$$

This is called a **p-value**

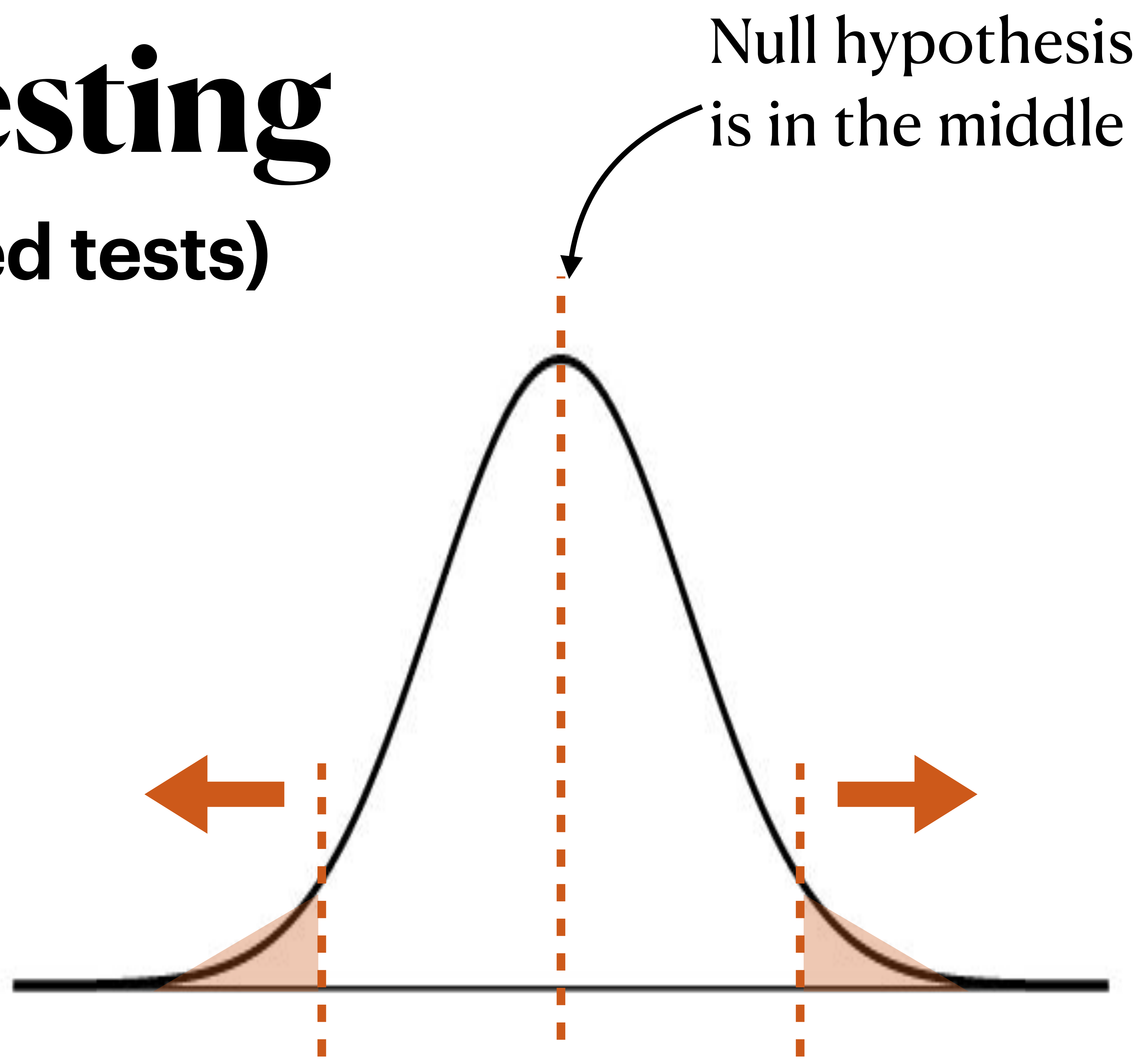
Hypothesis Testing

Procedure (for two-sided tests)

1. Write down the null and alternative hypotheses (statements about the **population parameter**)
2. Assuming the null hypothesis is true, think about what the sampling distribution of the statistic is.

   ➡ Are the conditions for CLT met? If so, we can say it is approximately normal.

3. Compute the test statistic. For this class, that means the z-score.



4. Knowing that if the statistic is approximately normal, then the z-score is $N(0,1)$, compute the p-value: $2P(Z \leq -|\text{z-score}|)$.
5. Determine whether the p-value is small enough to provide evidence against the null or not.

   ↑ This begs the question… how small is small?

Significance Level α

- The **significance level** is a pre-defined cutoff value used in hypothesis testing to decide whether a p-value is small enough to reject the null hypothesis
  - When the p-value is less than or equal to α, we reject the null hypothesis (i.e., say that we have evidence against it). When the p-value is larger than α, we cannot reject the null hypothesis.
- We usually choose this before data collection based on how willing we are to incorrectly reject the null hypothesis
  - **Ex 1**: A pharmaceutical company tests whether a new cancer drug works better than standard treatment. It would be pretty bad to find evidence that it is better, when it is not, because cancer patients may not see improvement/may have worse side effects.

→ make α small, perhaps 0.01 (1%) or less

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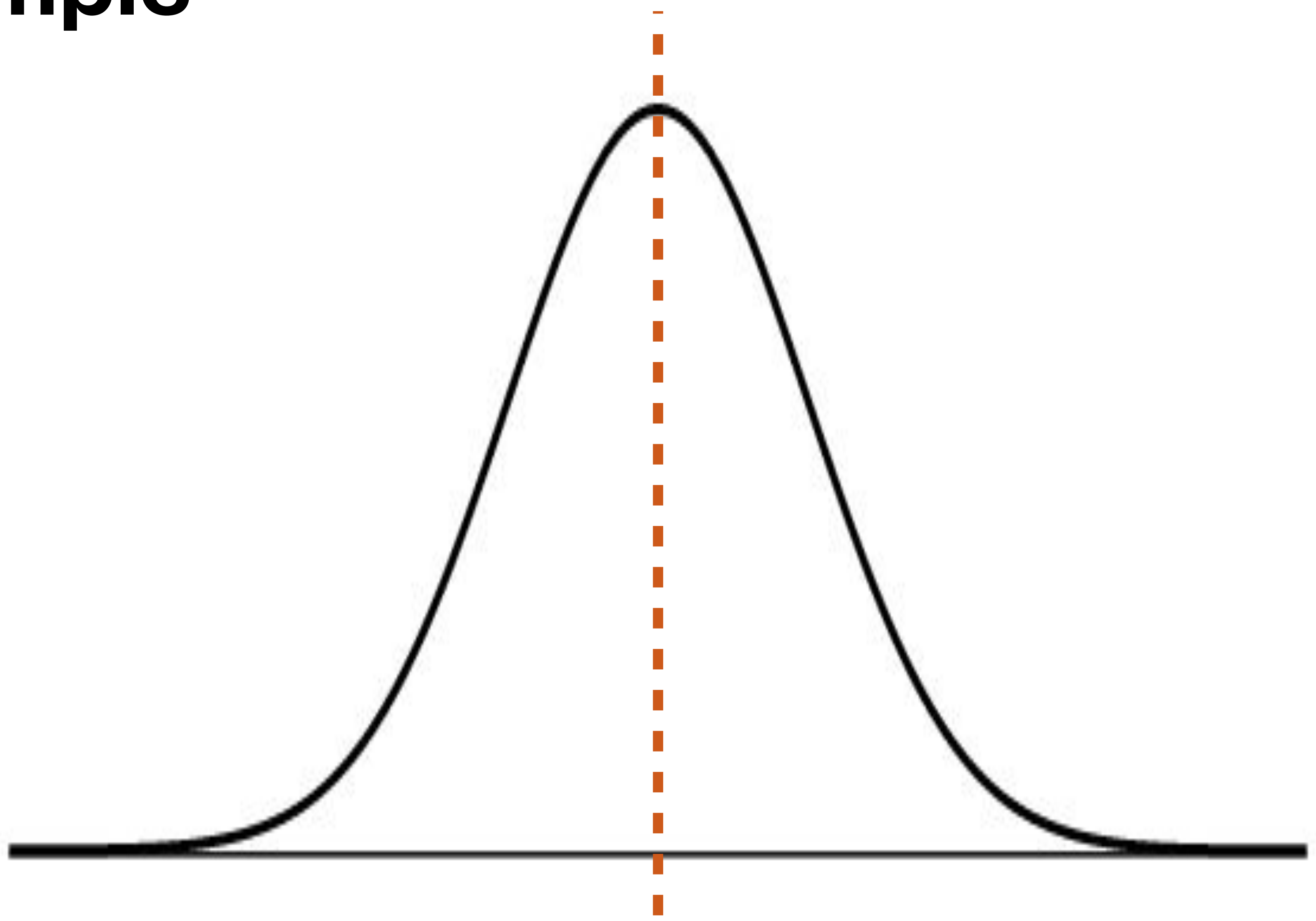
- **Ex 2**: A company tests whether a green “Buy Now” button increases clicks over a blue button. The consequences of incorrectly finding evidence that that’s the case are low (almost no cost to the company, just no benefit).

→ make α larger, perhaps 0.10 (10%)

Hypothesis Testing

Lunch Nutrients Example



- The recommended amount of folate intake at lunch is 100 micrograms (NIH). A group of researchers in Ontario wanted to see if healthy children aged 7-10 are eating that amount of folate at lunch, on average (source).
- They sampled 321 individuals from that age group (we may assume this was a random sample from the population).
- What are the null and alternative hypotheses?

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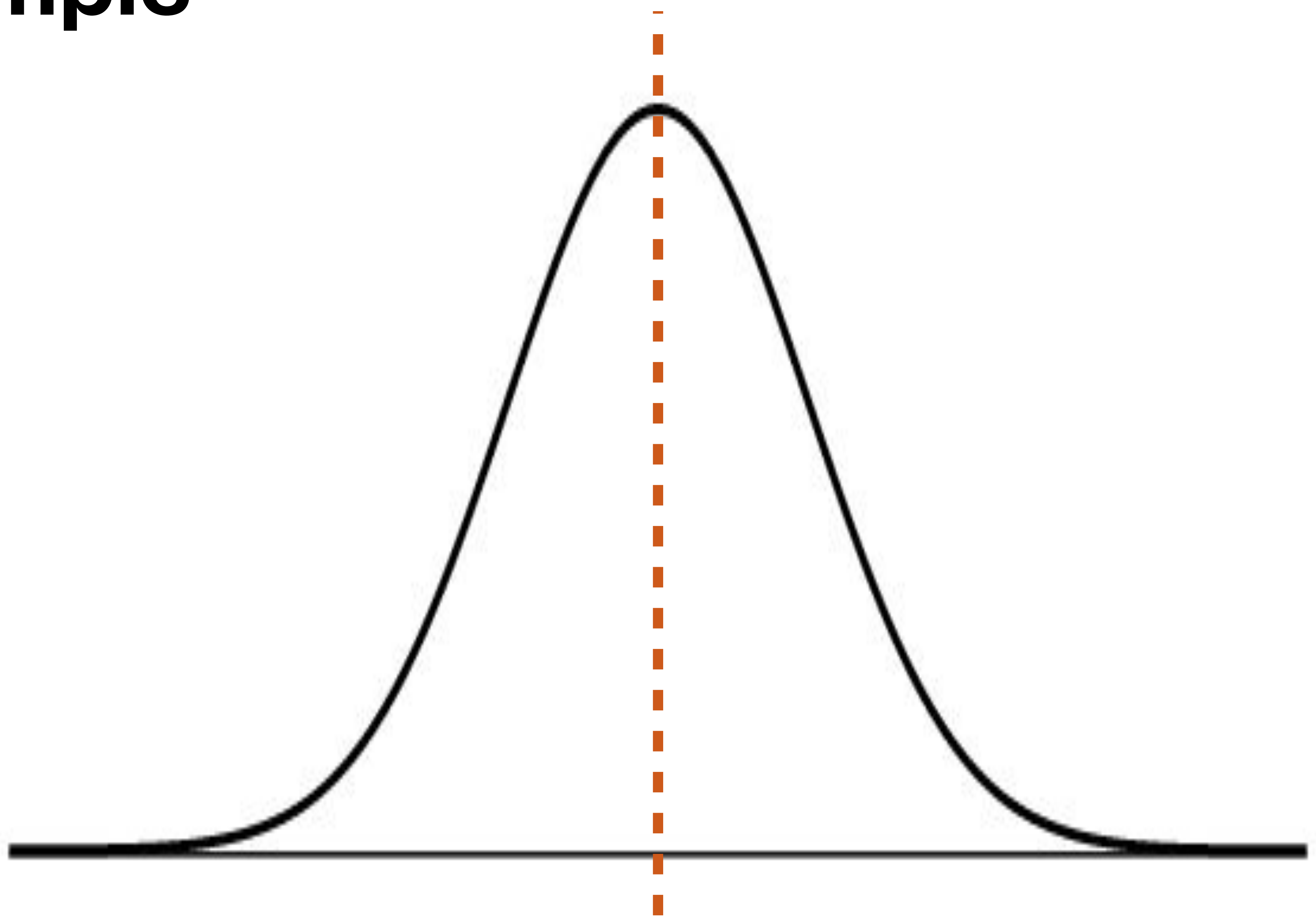

$$H_0 : \mu = 100; \; H_a : \mu \neq 100$$

Note: it can sometimes be confusing to figure out which is which. Use these two guidelines:

1. The null hypothesis must include a statement of equality (we have to have something to assume!)
2. There are only two possible results of the test: finding evidence against the null, or **not** finding evidence against the null. Select the null based on what you want to learn.

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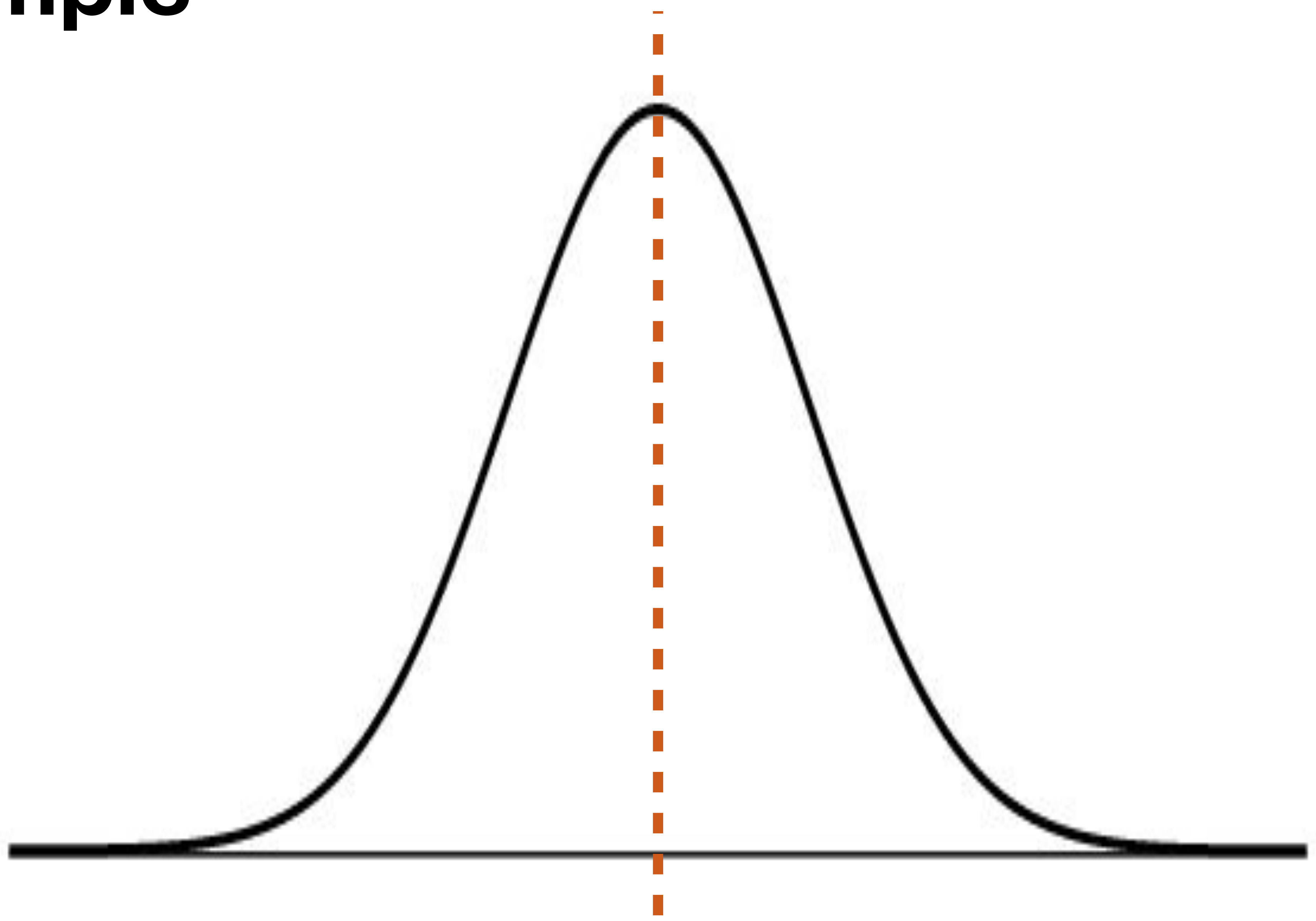

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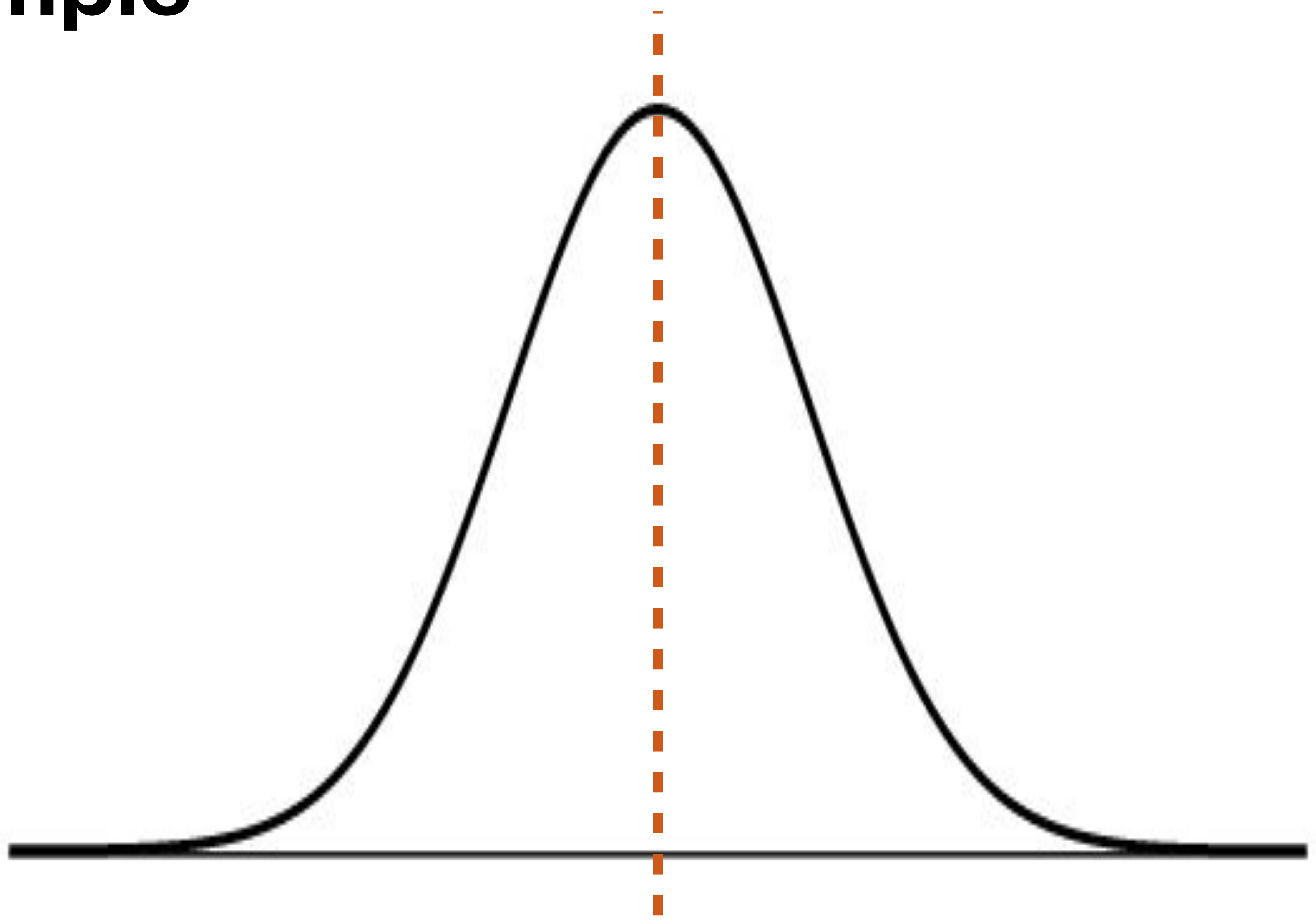

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- They sampled 321 individuals from that age group (we may assume this was a random sample from the population).
- To think about the distribution of the sample mean, we need to use CLT. Are the conditions for CLT met?
  - i.i.d. → met because we are assuming random sample from large population
  - Sample size → met because 321 is much larger than the rule-of-thumb 30

Hypothesis Testing

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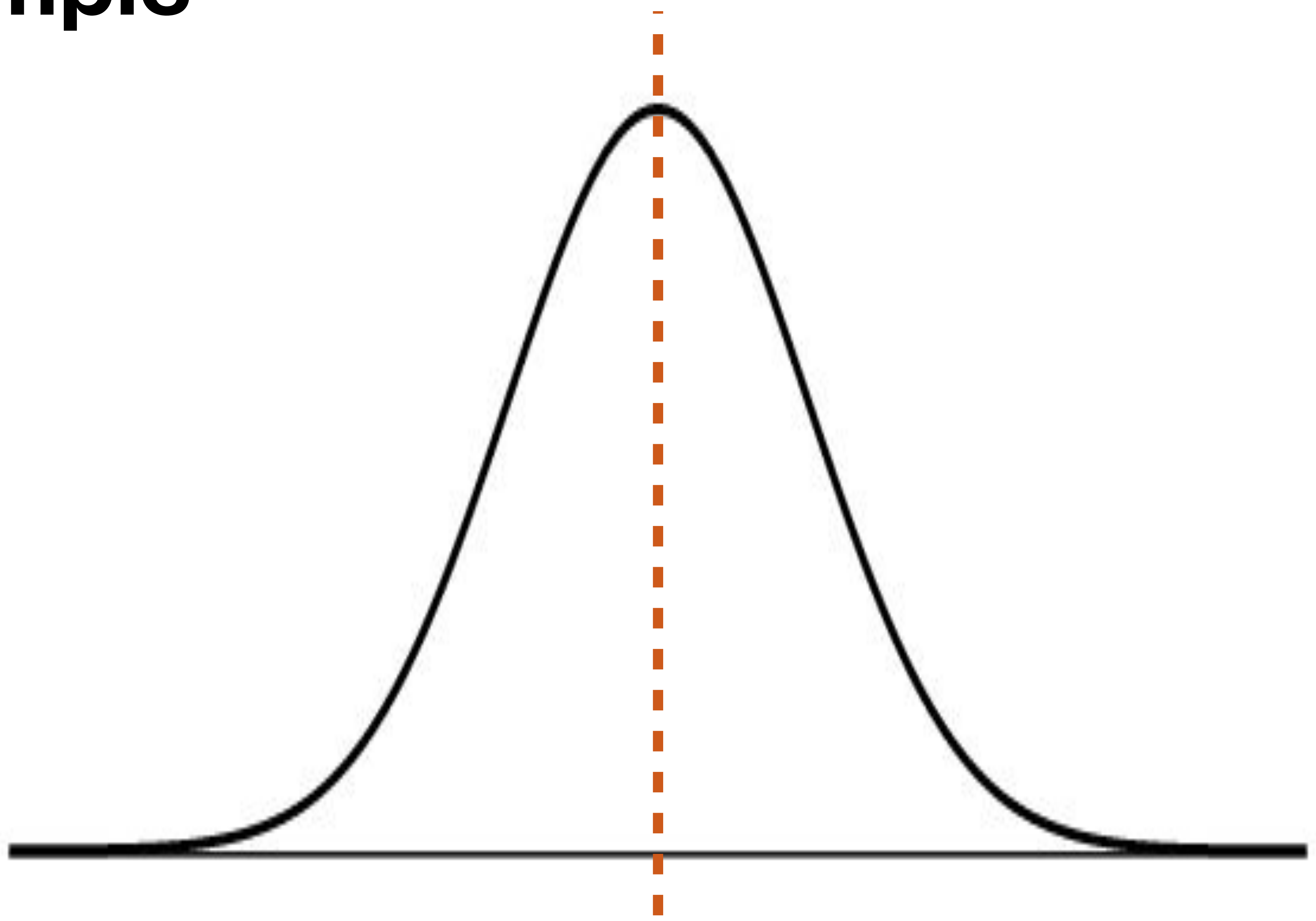

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- What is the test statistic?

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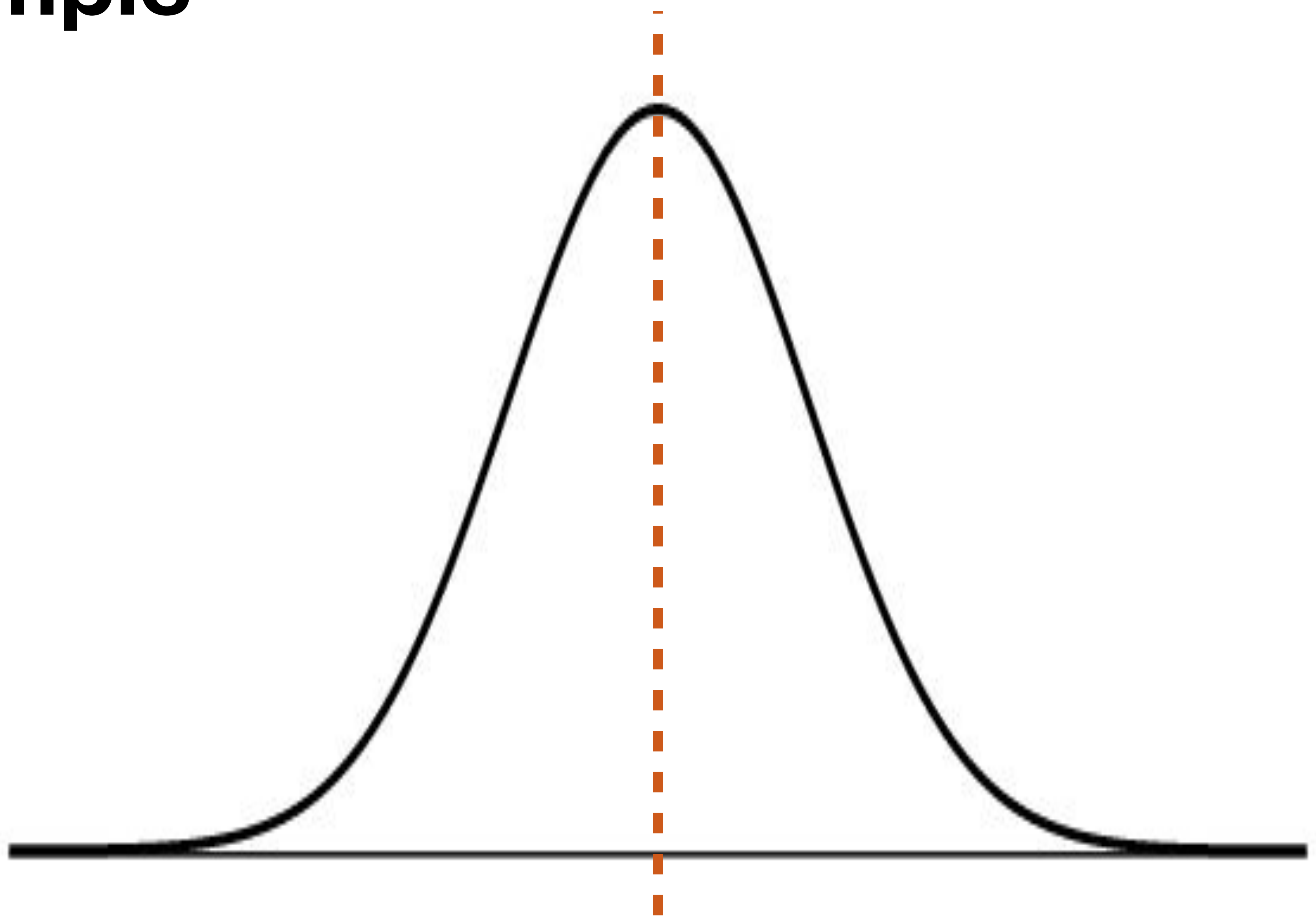
$$z = \frac{81.15 - 100}{69.57/\sqrt{321}} \approx -4.85$$



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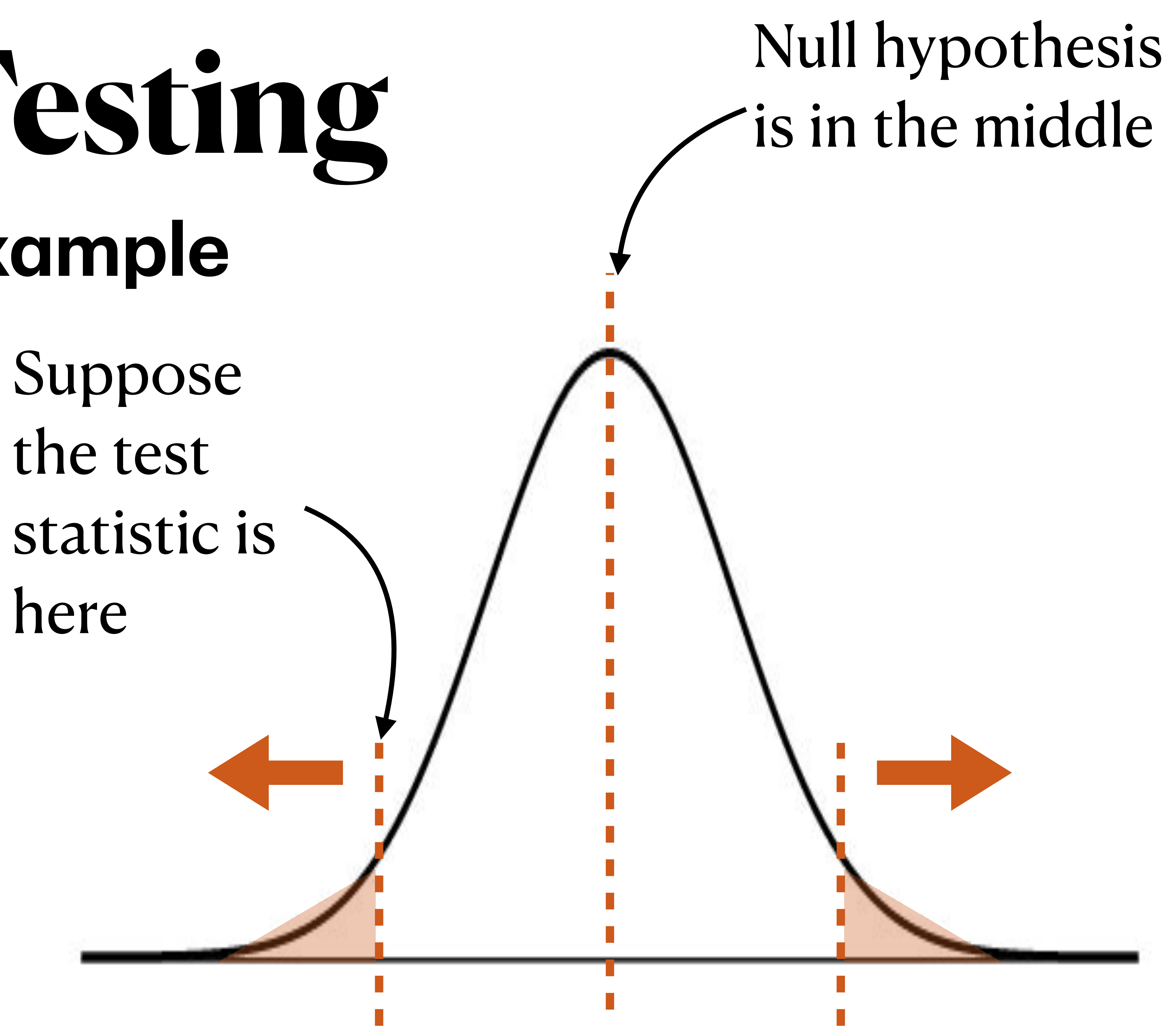

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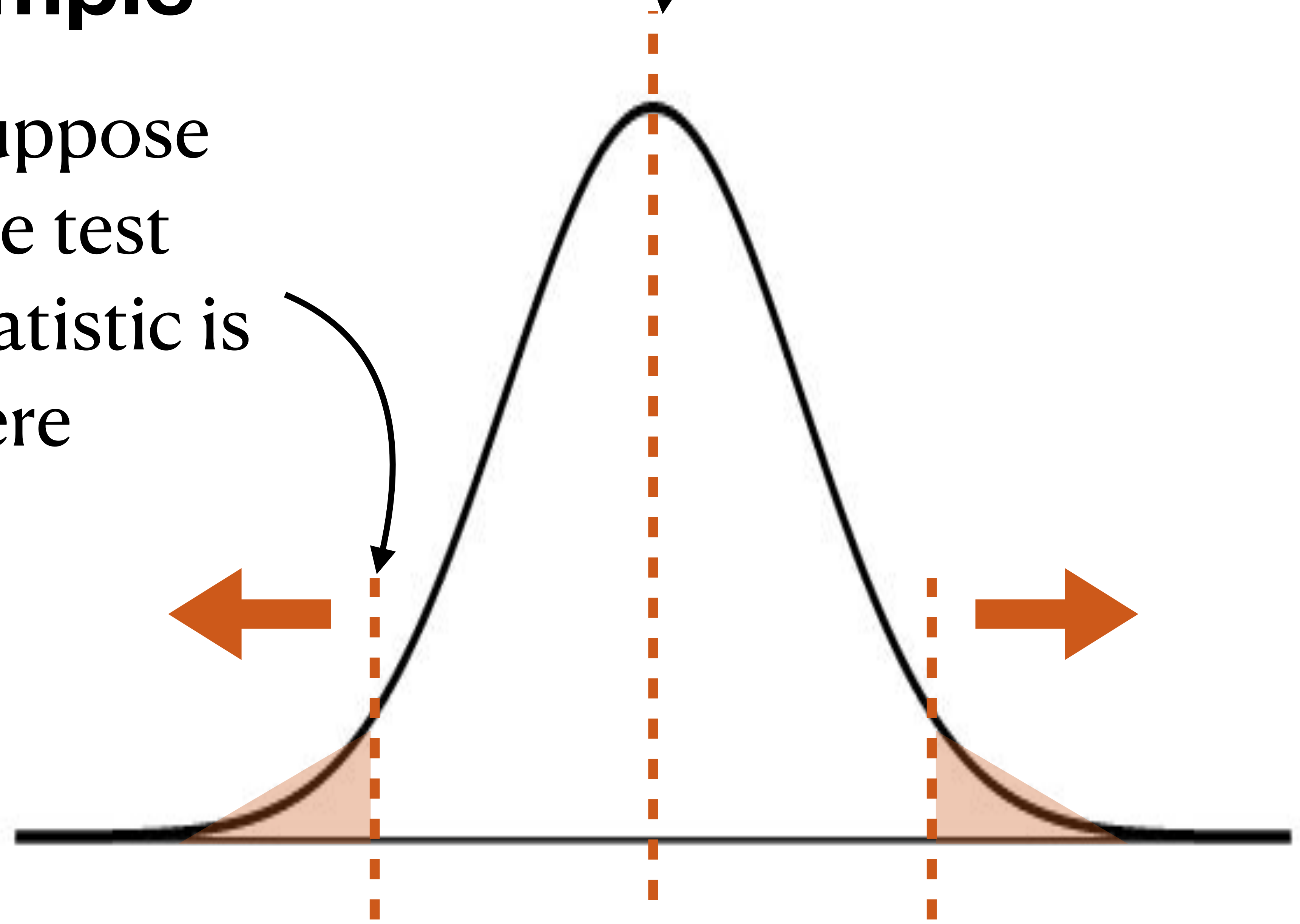

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$$2P(Z \leq -4.85) = \texttt{2*pnorm(-4.85)} \approx 1.2 \times 10^{-6}$$

Hypothesis Testing

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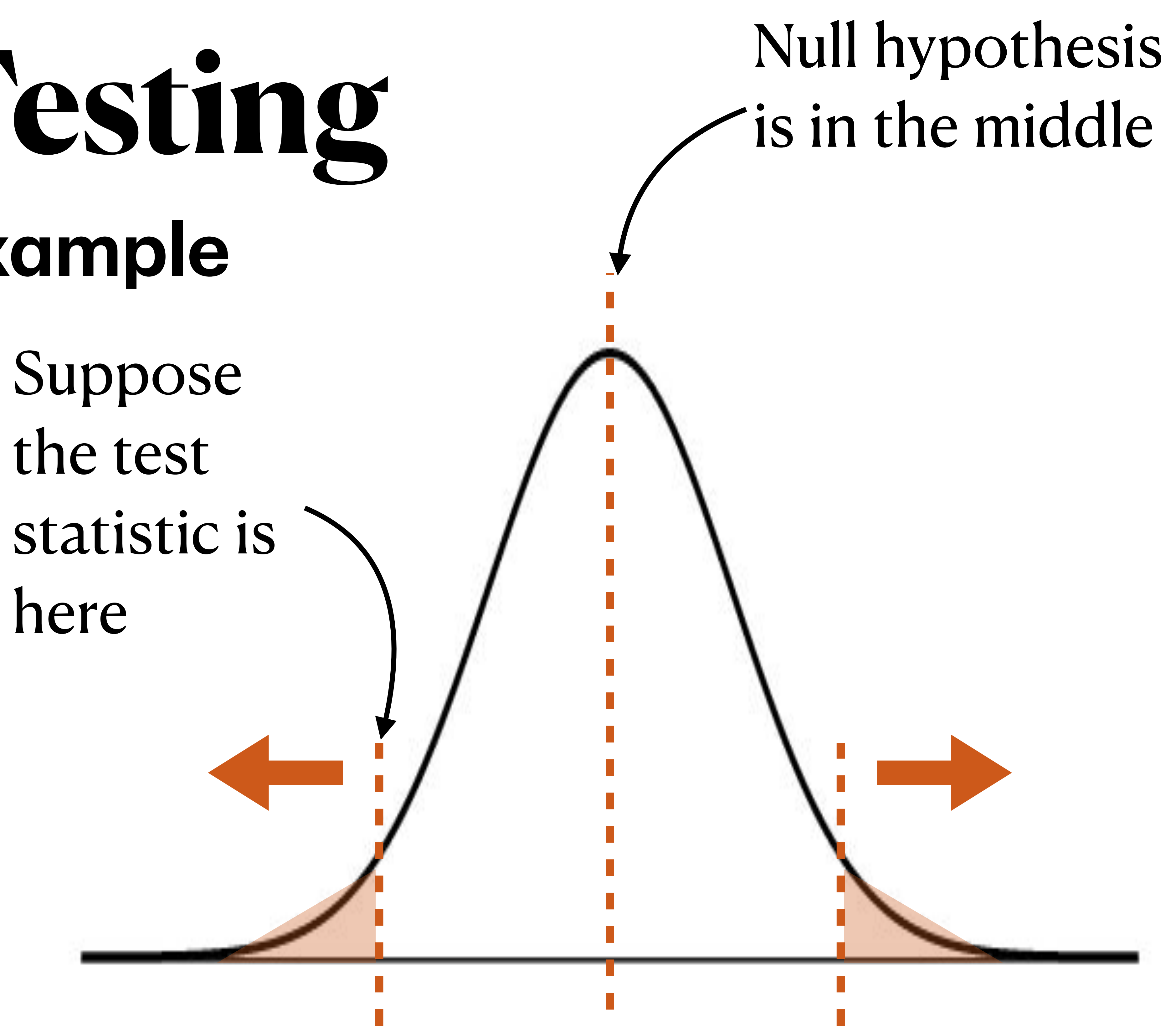
- At a significance level of $\alpha = 0.05$, do we have evidence to reject the null hypothesis?



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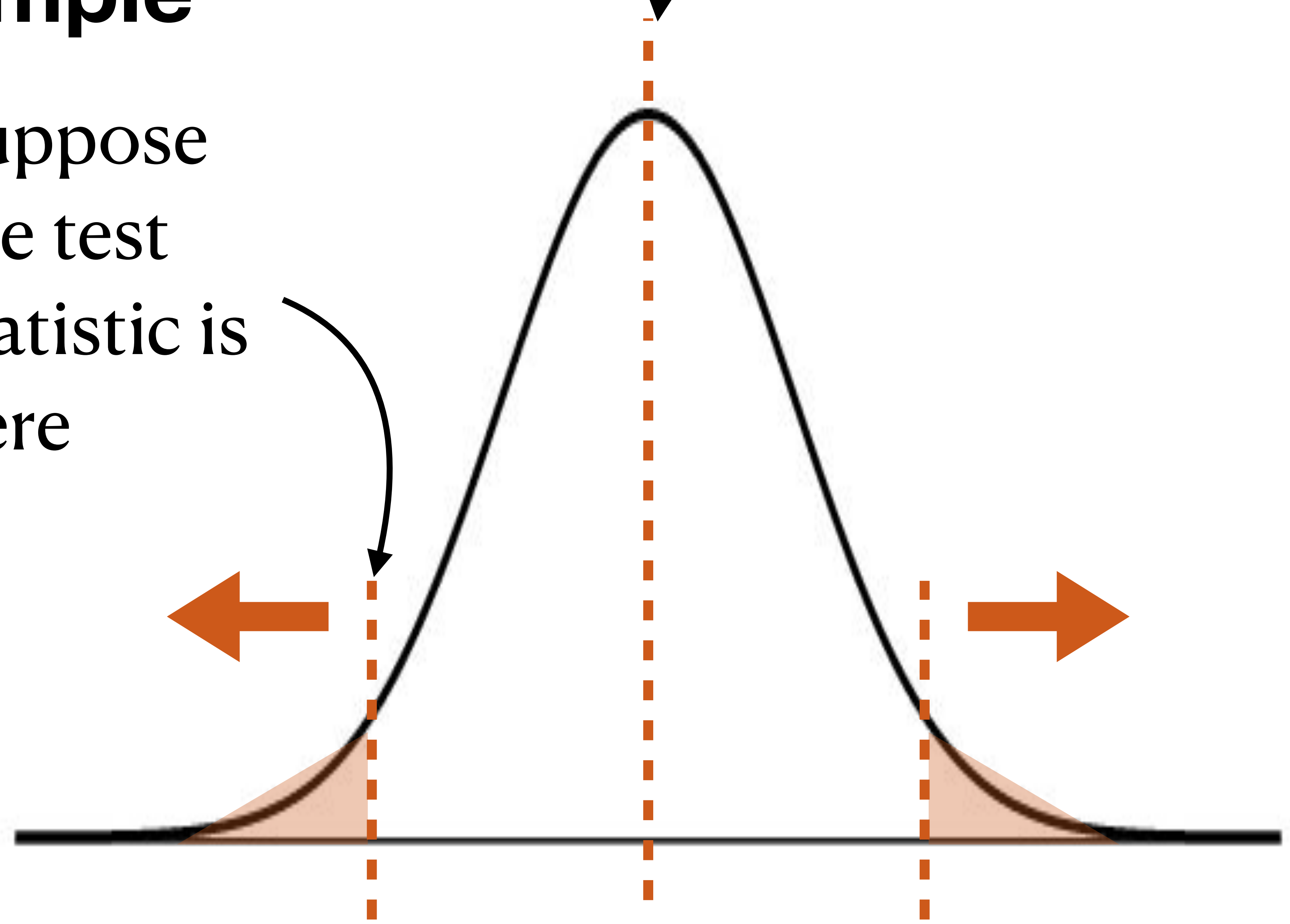

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p-value $\approx 0.0000012 \rightarrow$ p-value $< 0.05 \rightarrow$ reject the null

Hypothesis Testing

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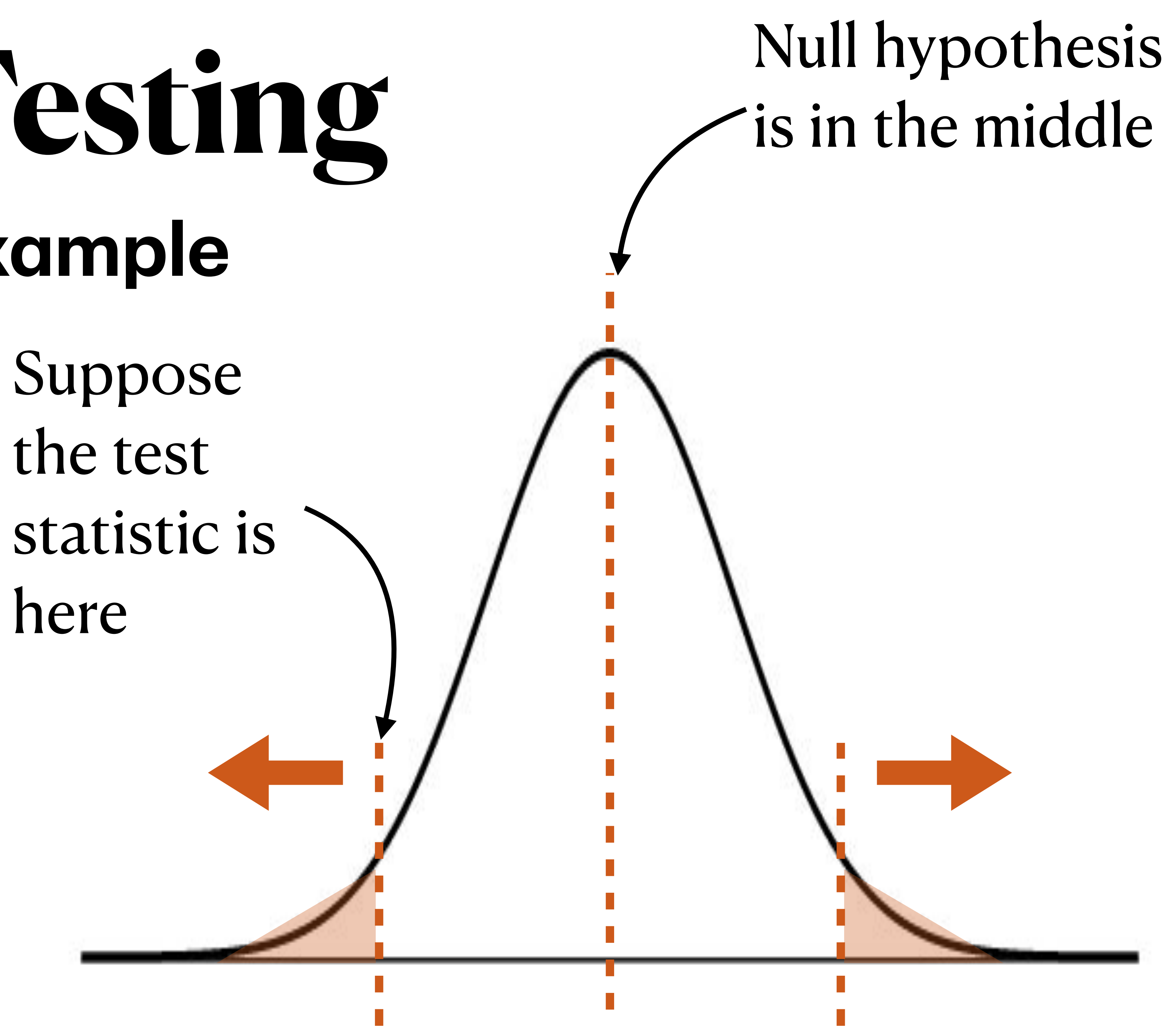

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- They sampled 321 individuals from that age group (we may assume this was a random sample from the population). They find that the sample mean folate intake is 81.15 micrograms with a standard deviation of 69.57 micrograms.
- Rejected H_0 at a significance level of 0.05. What does this mean?
  - At a significance level of 5%, we reject the null hypothesis. We have evidence to suggest that children are not eating 100 micrograms of folate at lunch, on average.

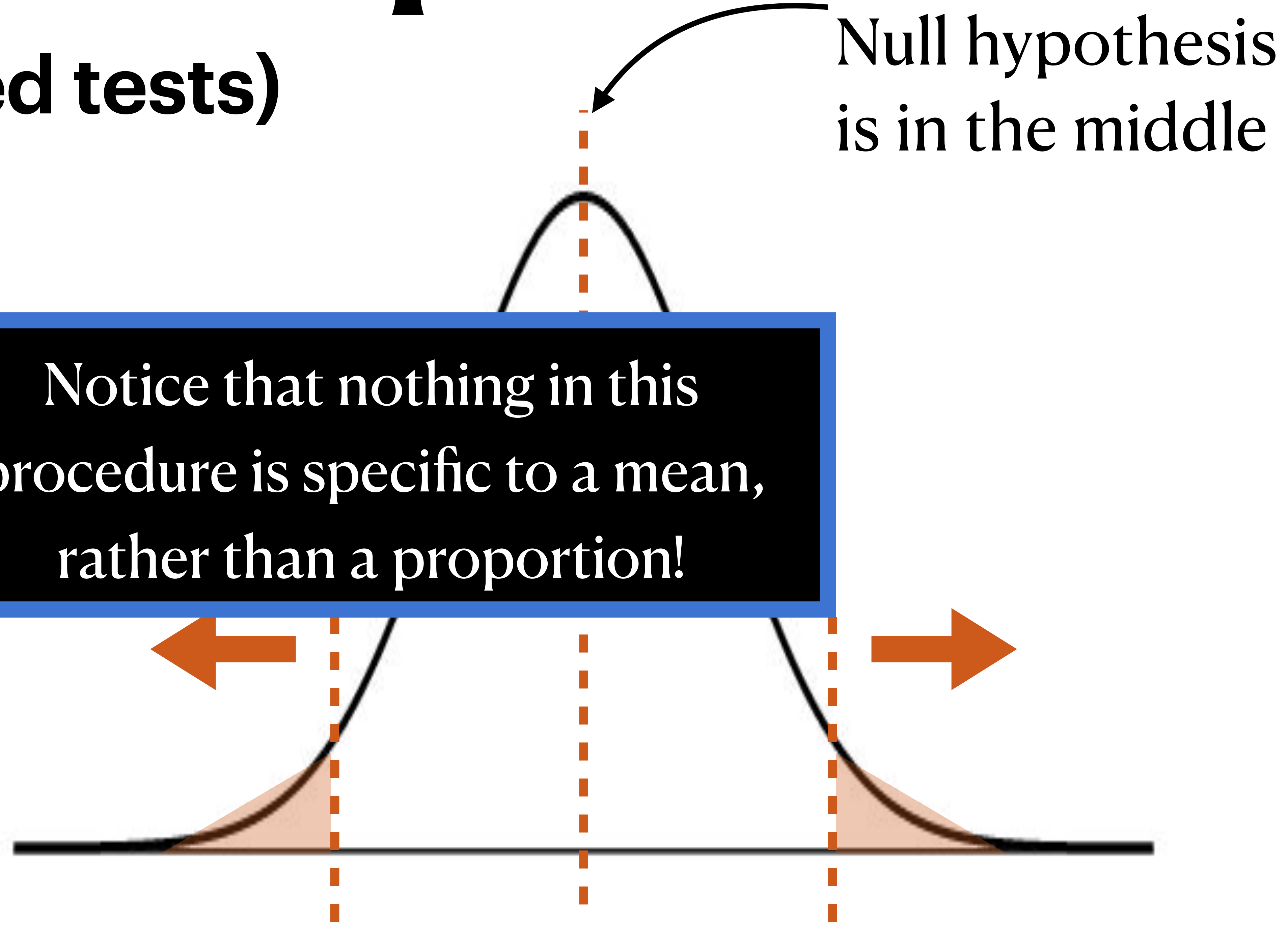
Hypothesis Testing for a Proportion

Procedure (for two-sided tests)

1. Write down the null and alternative hypotheses (statements about the **population parameter**)

2. Assuming the null hypothesis is true, think about wha the sampling distribution of the statistic is.

   ➡ Are the conditions for CLT met? If so, we can say it is approximately normal.



Notice that nothing in this procedure is specific to a mean, rather than a proportion!

3. Compute the test statistic. For this class, that means the z-score.

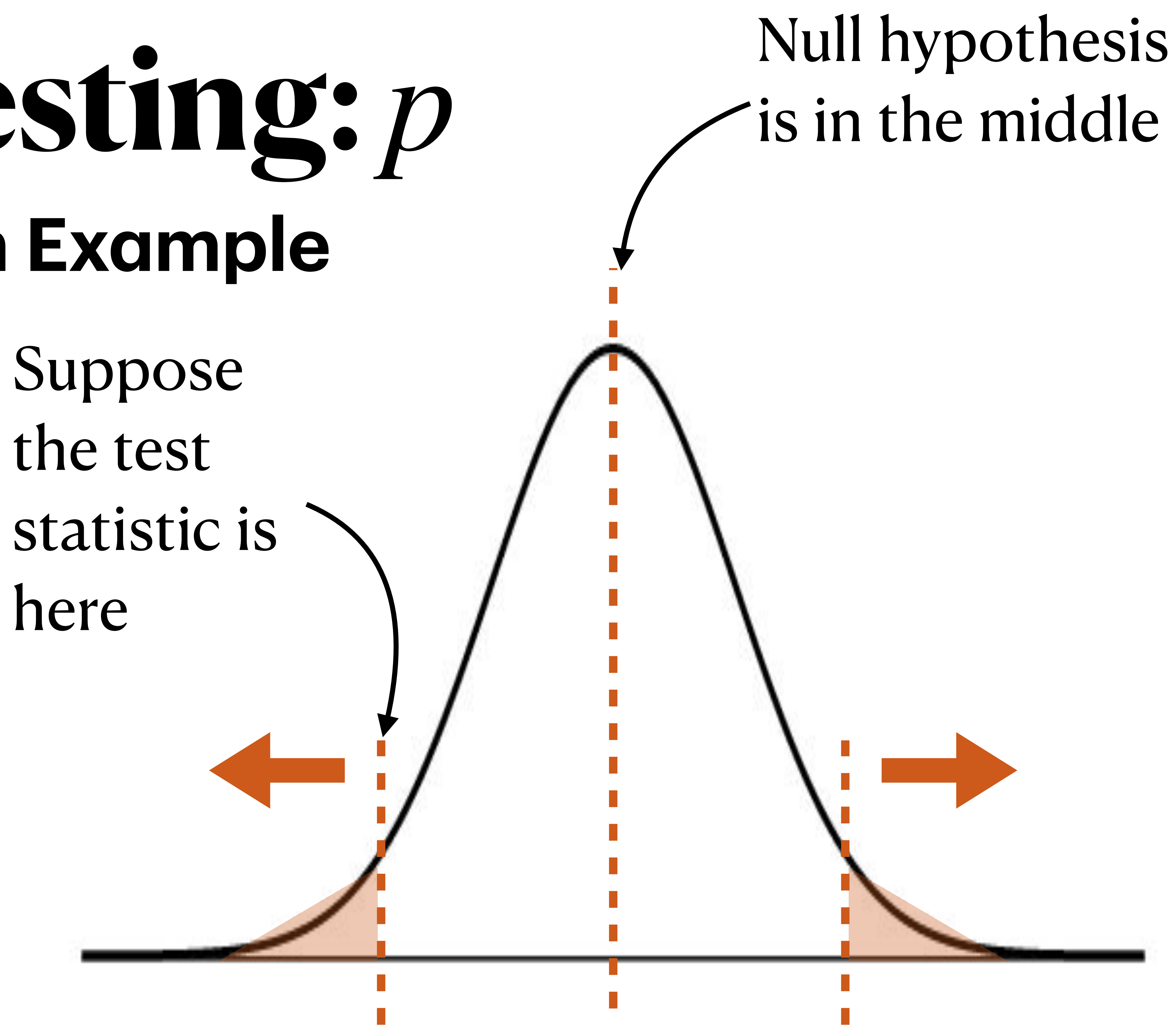
4. Knowing that if the statistic is approximately normal, then the z-score is $N(0,1)$, compute the p-value: $2P(Z \leq -|\text{z-score}|)$.

5. Determine whether the p-value is small enough to provide evidence against the null or not.

Hypothesis Testing: p

Genetic Transmission Example

- Suppose researchers are studying an allele that may be associated with a disease. A simple genetic model says that a parent who carries one copy of this allele has a 50% chance of passing it on to a child.
- Researchers randomly sampled 150 parent-child pairs in which the parent carried one copy of the allele and the child had the disease. The allele was transmitted to the child in 65 of the 150 pairs.



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Hypothesis Testing: *p*

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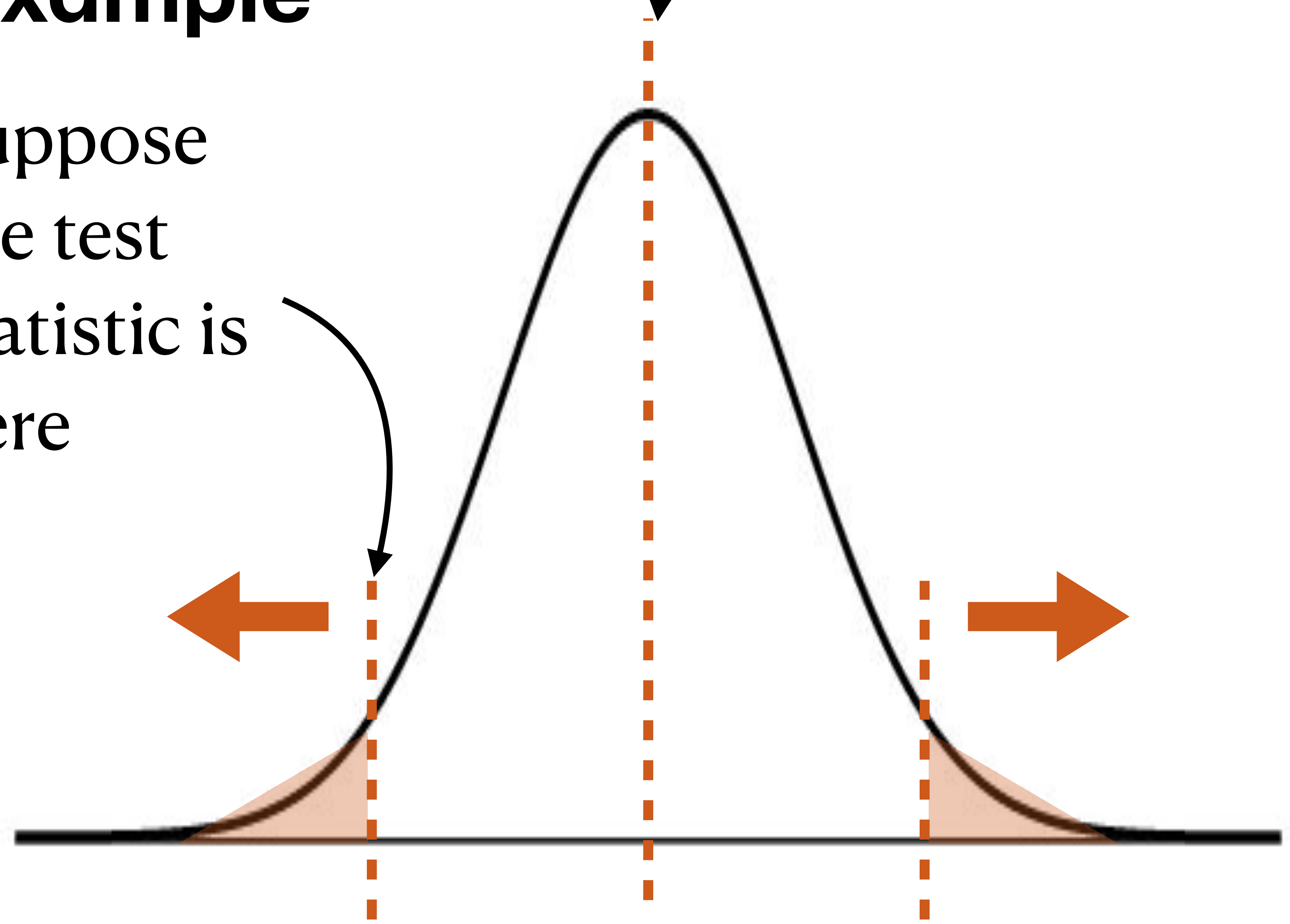

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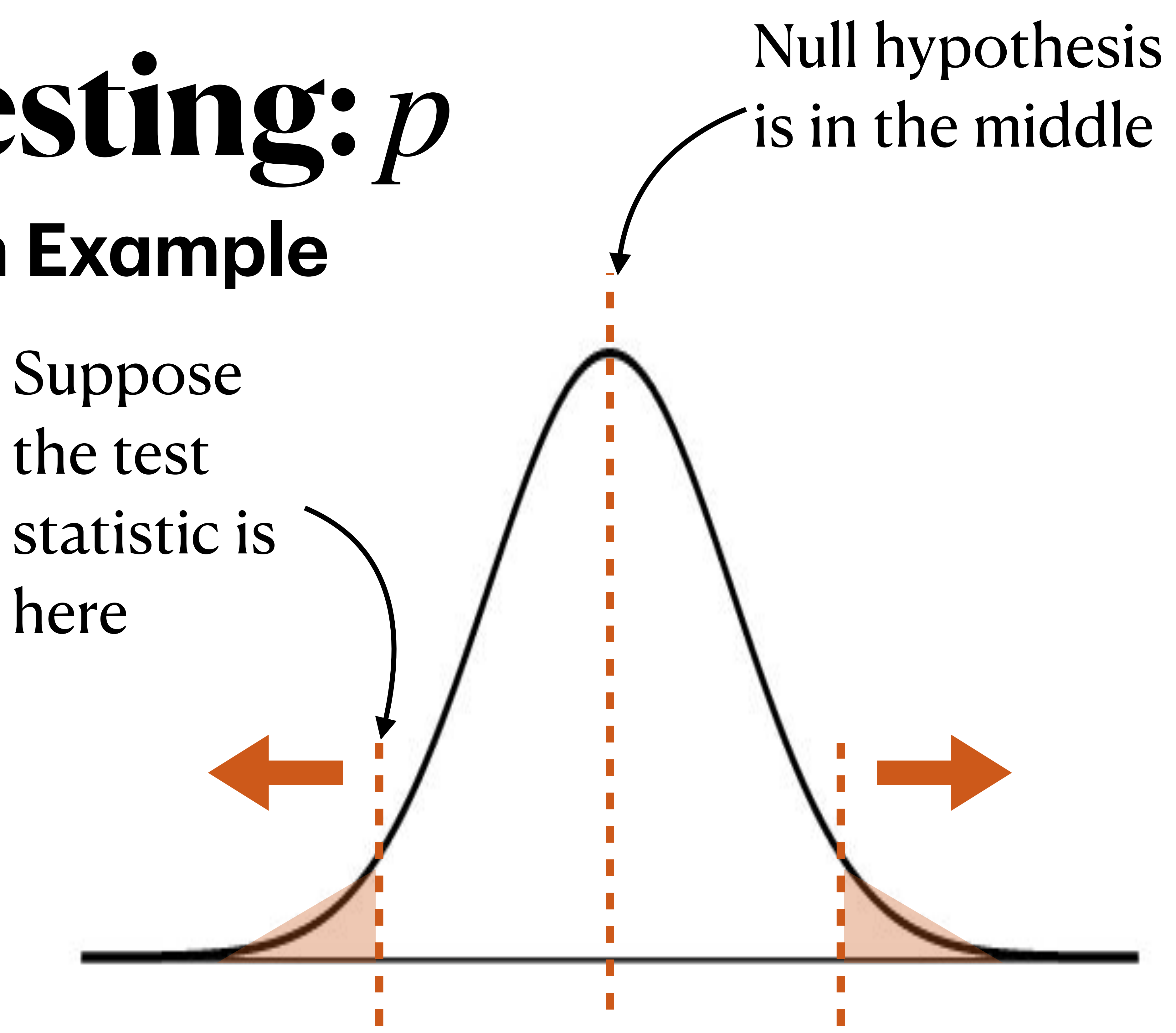

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$$H_0 : p = 0.5; H_a : p \neq 0.5$$

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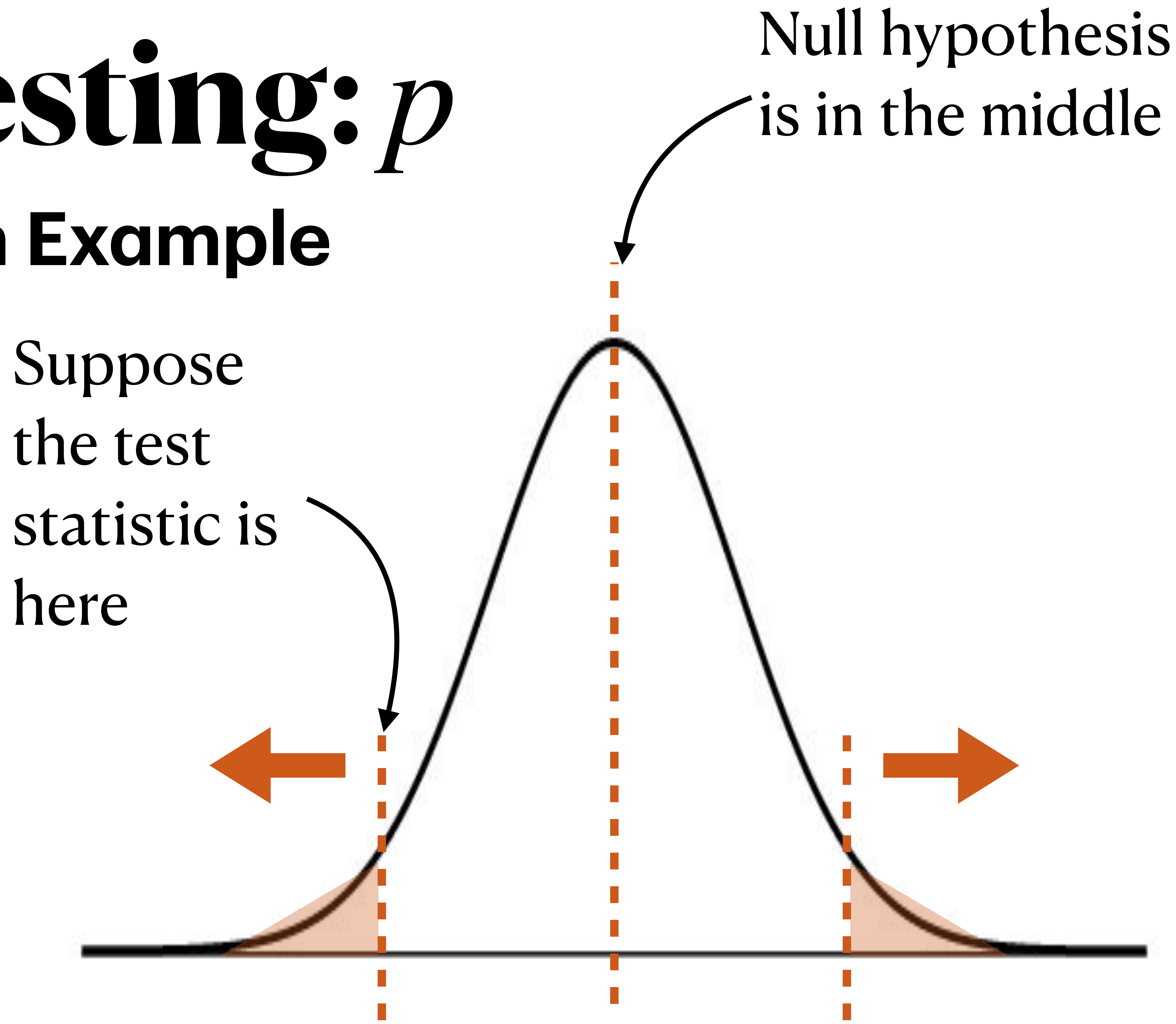

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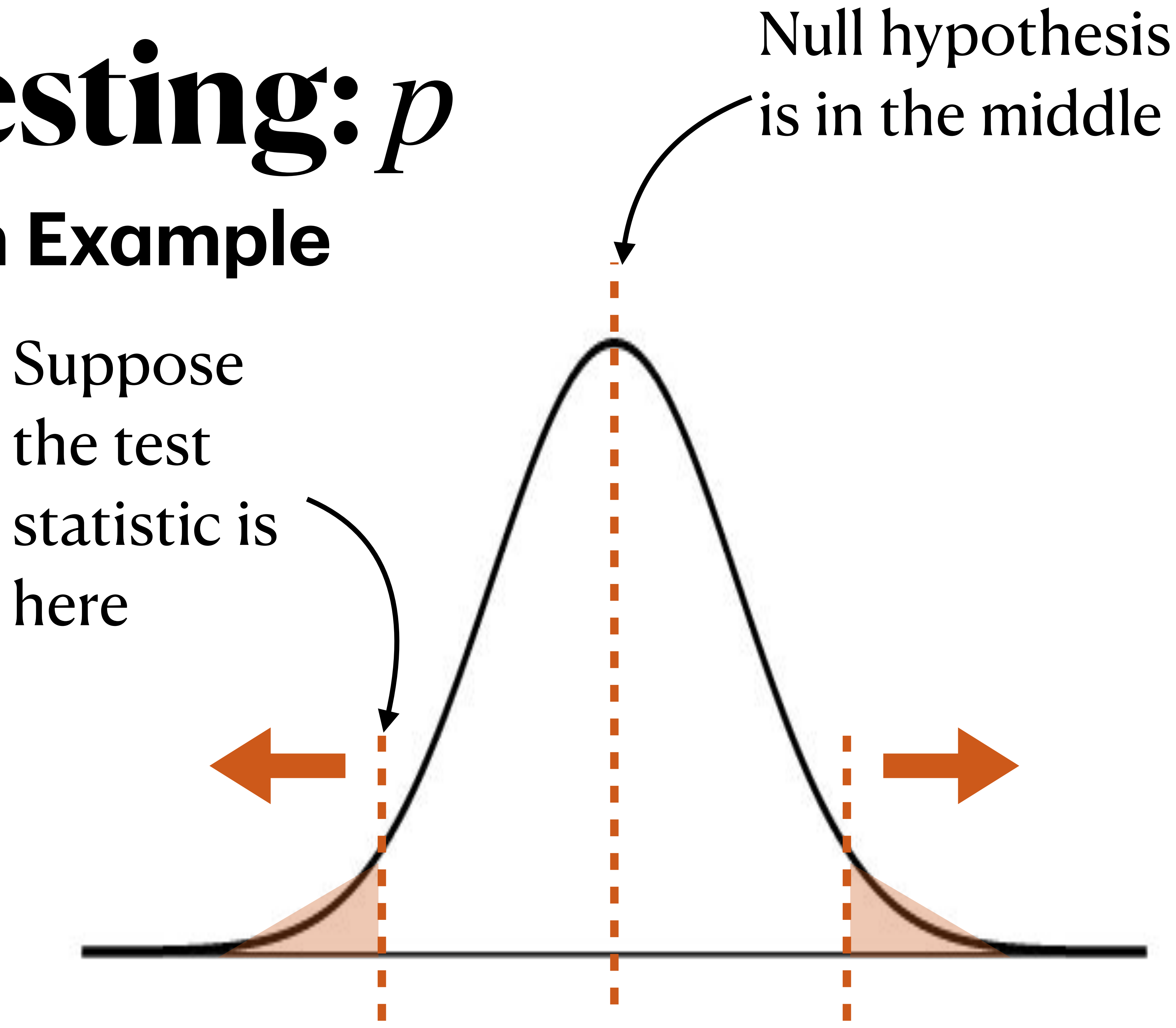

- At a significance level of 5%, do these data provide evidence that the allele is transmitted to affected children at a rate different from 50%?
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$65/150 \approx 0.433$

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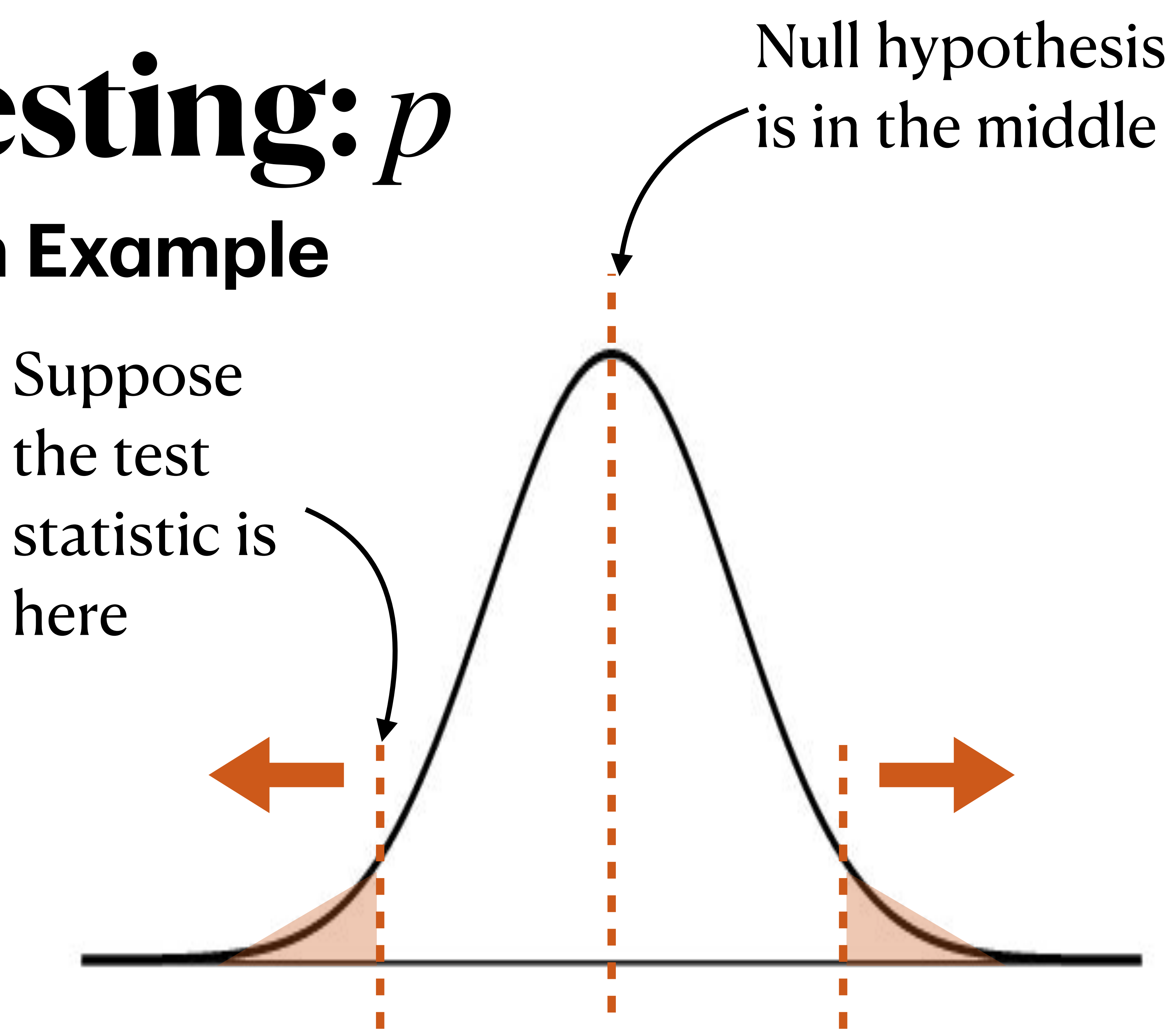

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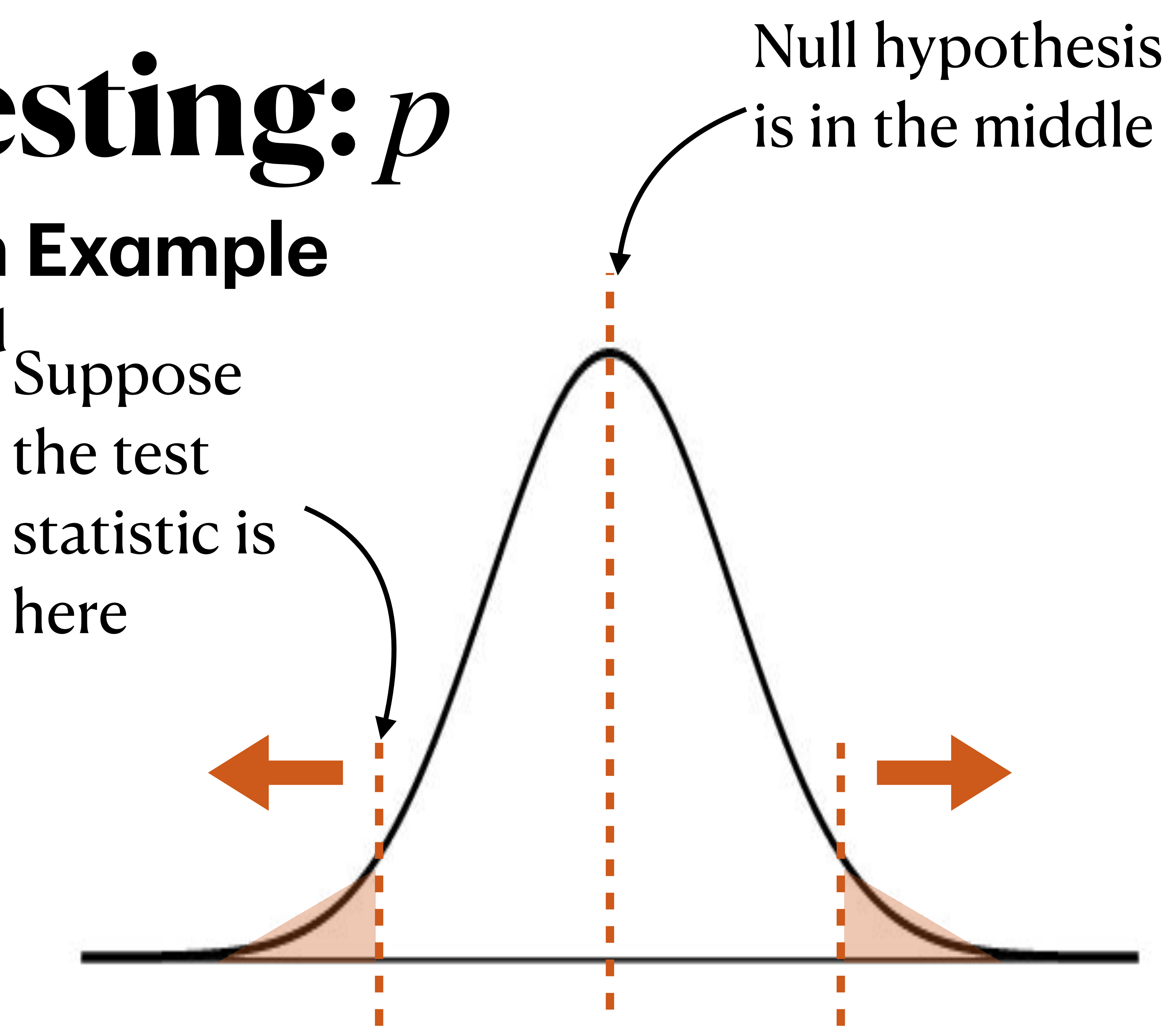

i.i.d.: random sample from large population ✓

Sample size: Under the null hypothesis, $p = 0.5$. **We check the conditions of the CLT under that assumption, too!**

Rule of thumb: $150 \times 0.5 > 10$, $150 \times (1 - 0.5) > 10$ ✓

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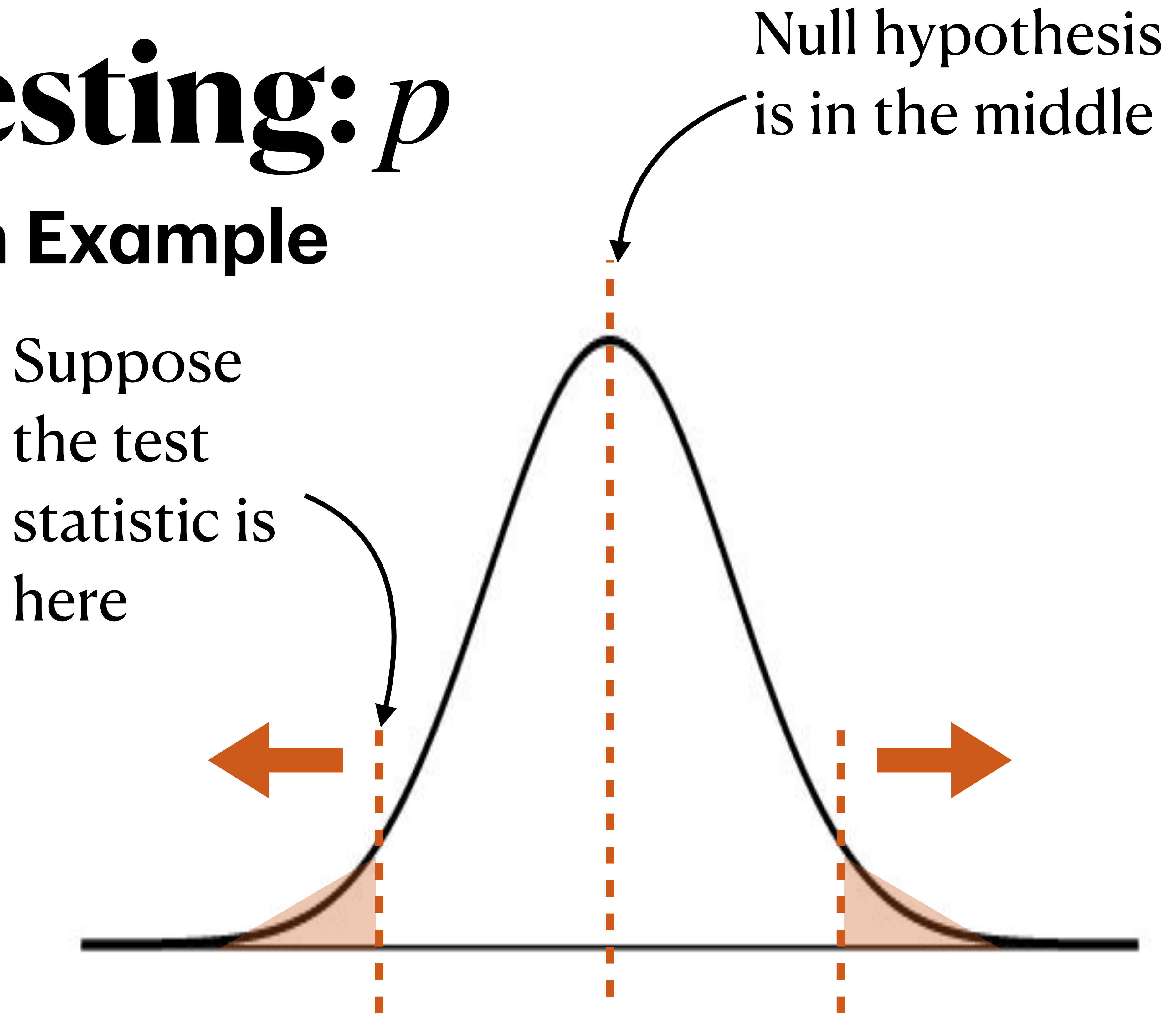

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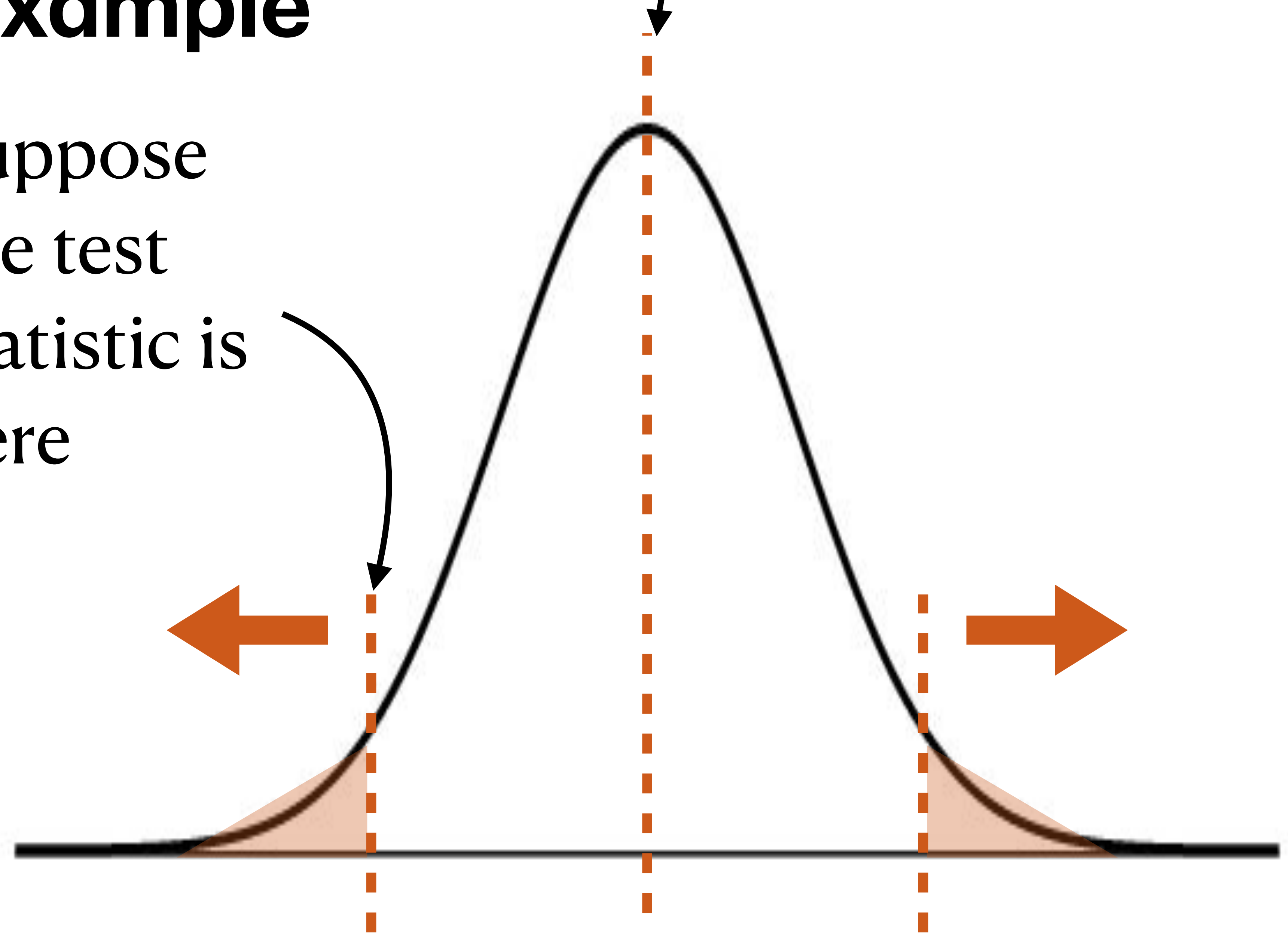
CLT conditions met. Approximately, $\hat{P} \sim N\left(0.5, \sqrt{\frac{0.5(1-0.5)}{150}}\right)$. Note the usage of p from the null hypothesis in the distribution, *not the sample proportion!*

Hypothesis Testing: p

Genetic Transmission Example



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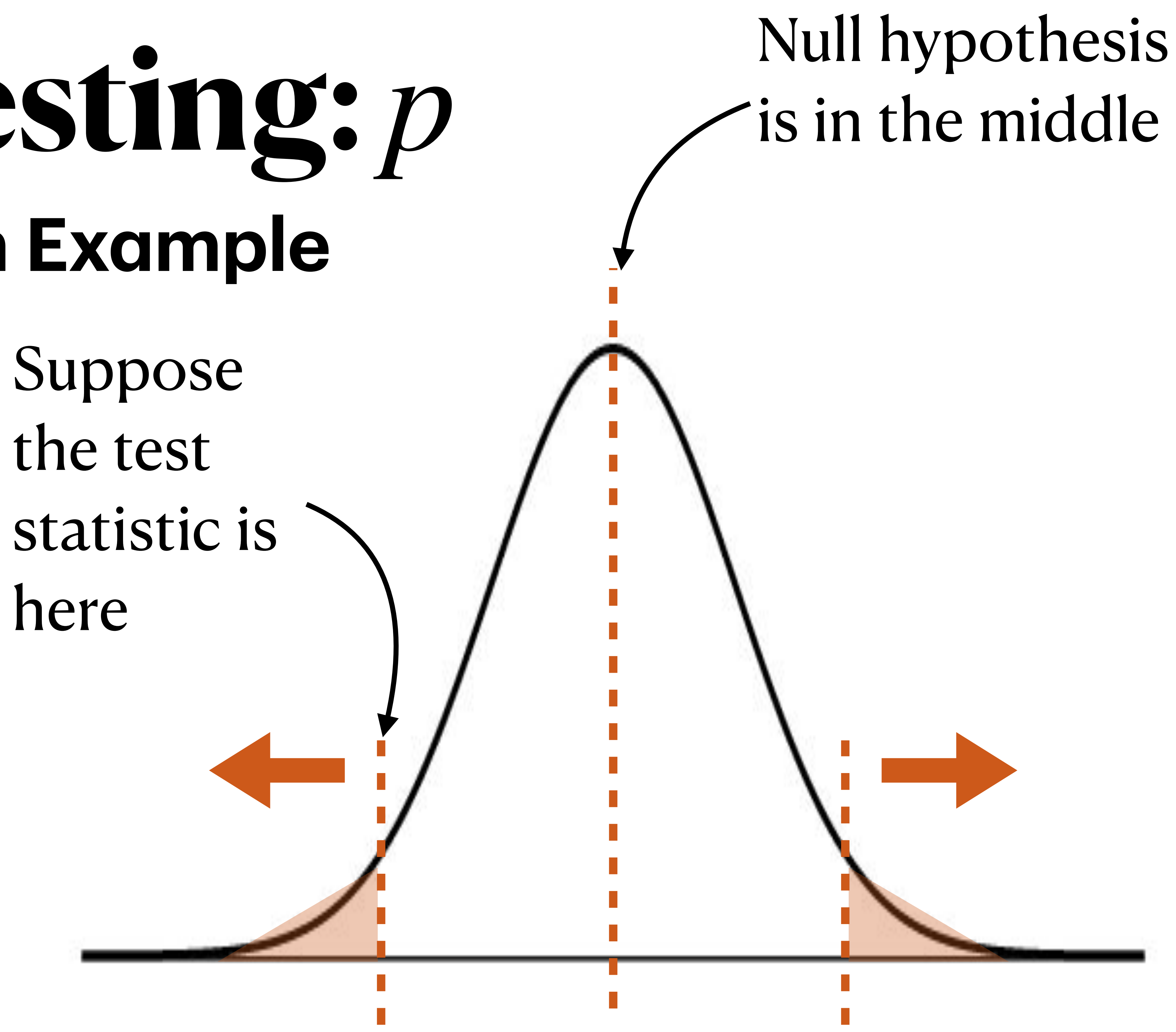

- At a significance level of 5%, do these data provide evidence that the allele is transmitted to affected children at a rate different from 50%?
- What is the test statistic?

We get the test statistic by subtracting the mean and dividing by the standard error, $(0.433 - 0.5)/\sqrt{0.5(1-0.5)/150} \approx -1.64$

Hypothesis Testing: *p*

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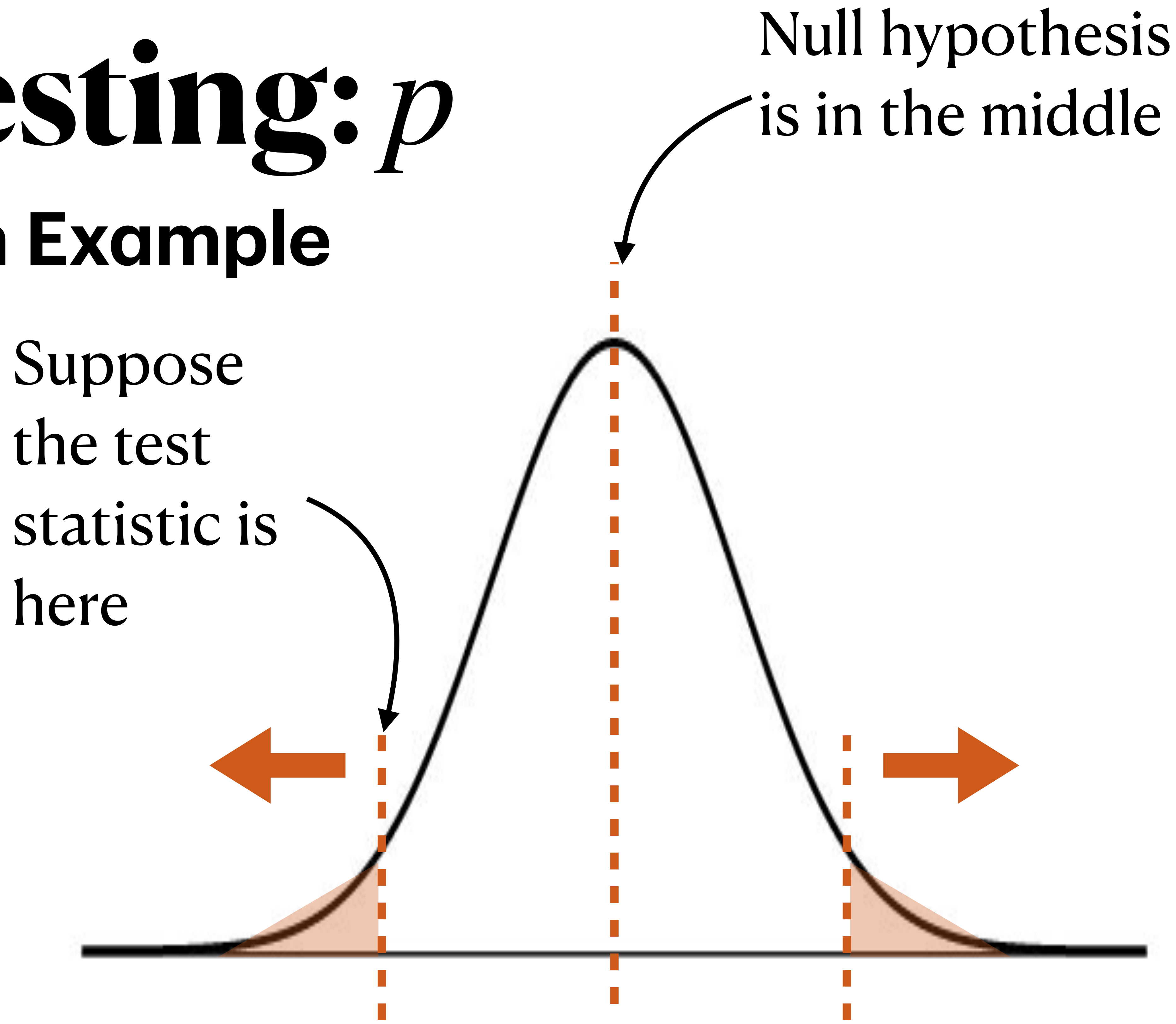

- At a significance level of 5%, do these data provide evidence that the allele is transmitted to affected children at a rate different from 50%?
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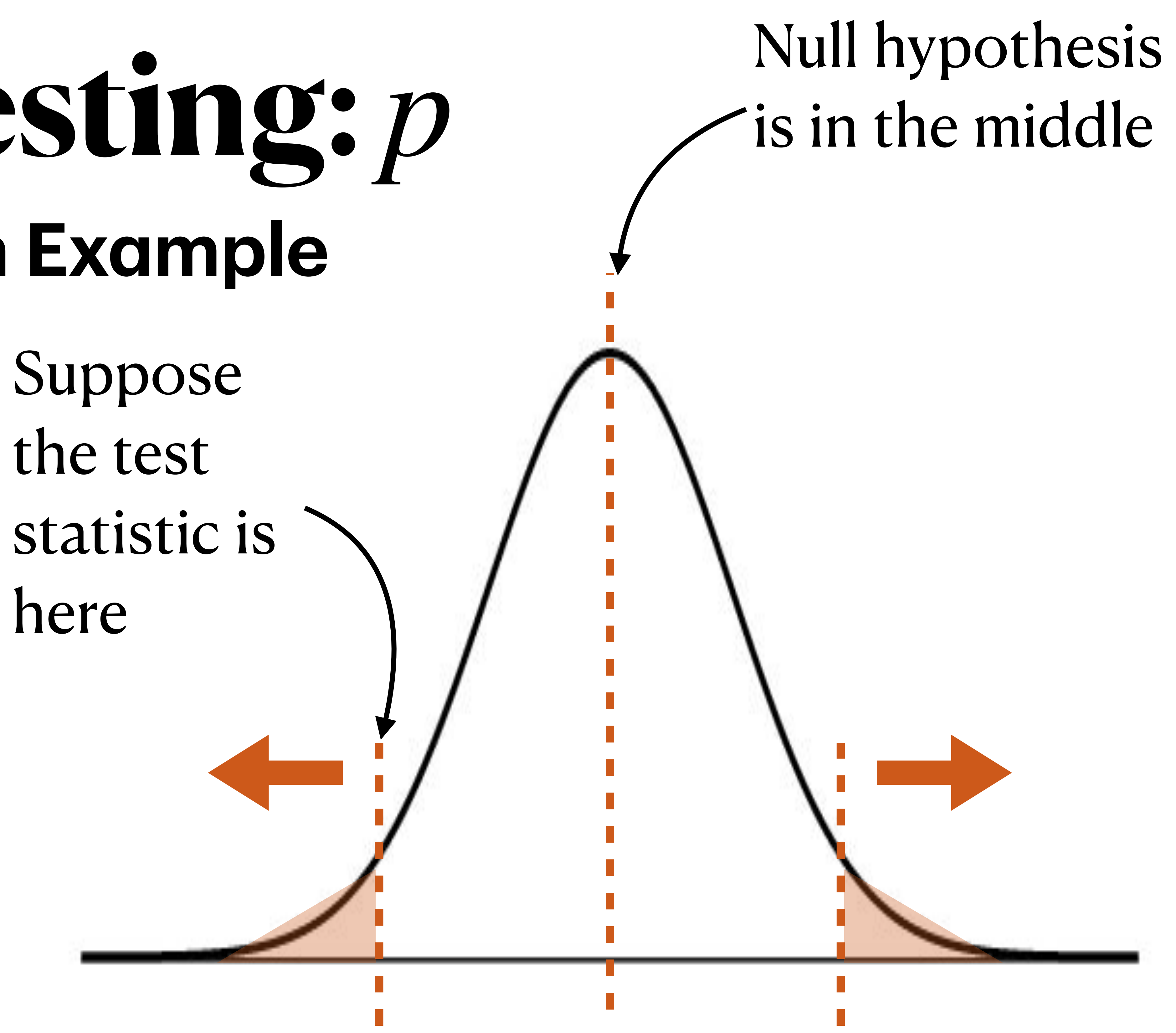

- At a significance level of 5%, do these data provide evidence that the allele is transmitted to affected children at a rate different from 50%?
- What is the p-value?

$2P(Z \leq -1.64) \approx 0.10$ (Using R, `2*pnorm(-1.64)`)

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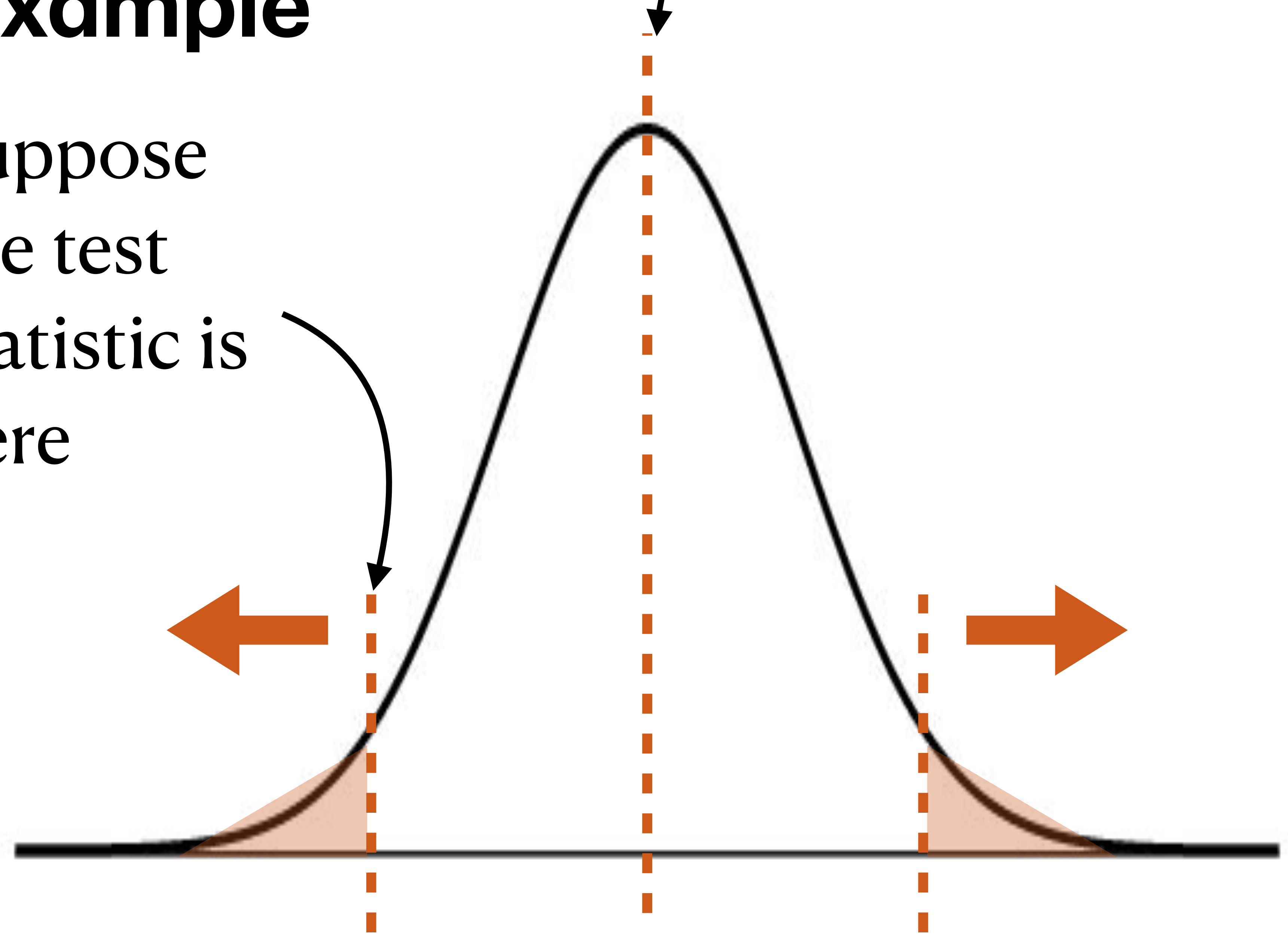

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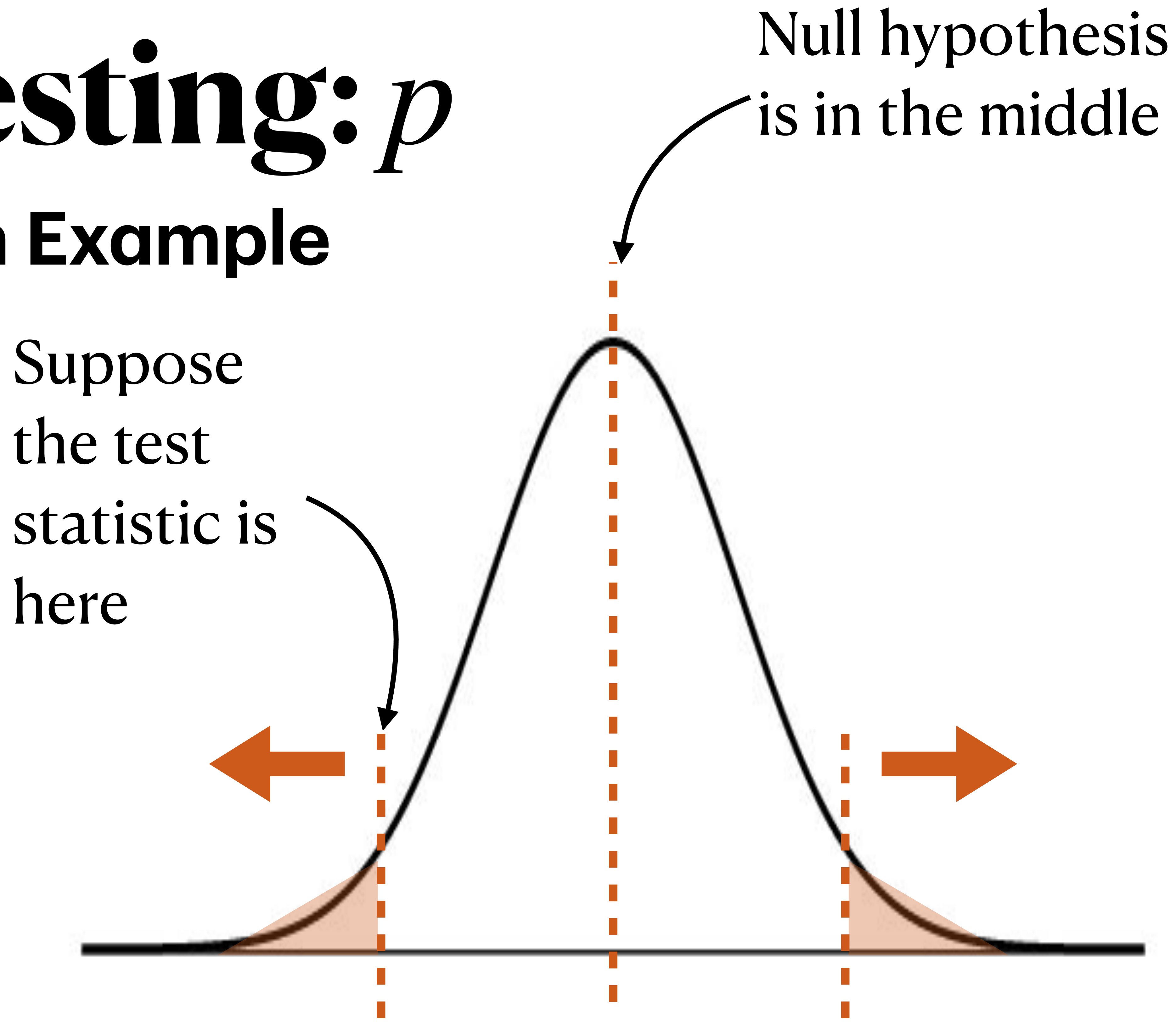


- At a significance level of 5%, do these data provide evidence that the allele is transmitted to affected children at a rate different from 50%?
- What should we conclude?

At a significance level of 5%, we fail to reject the null hypothesis. We do not have evidence to say that the allele is not transmitted to children at a 50% rate.

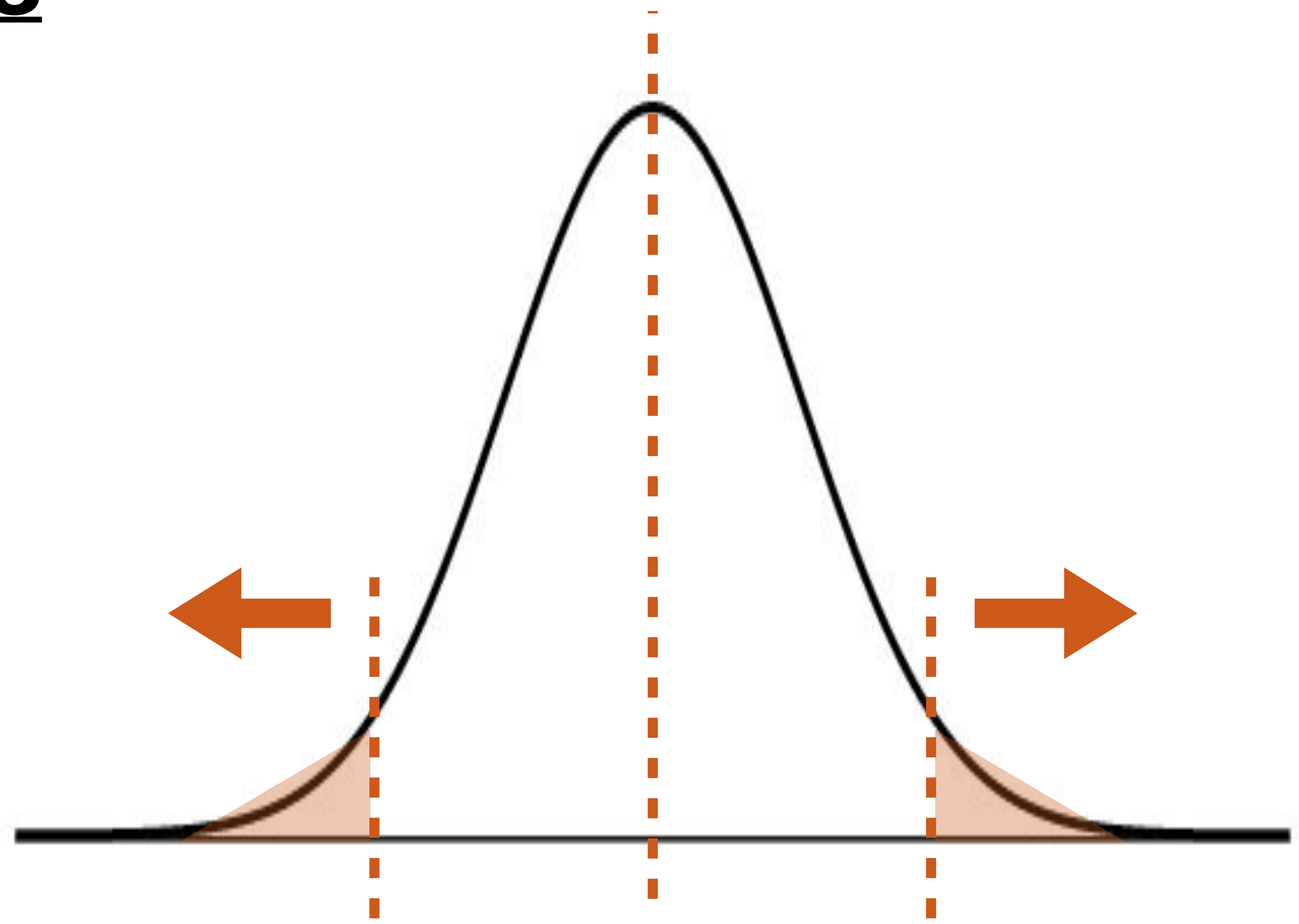
Hypothesis Testing: *p*

PollEv.com/stats8

- In the previous slides, we found that the p-value was 0.10. A student interprets this p-value:

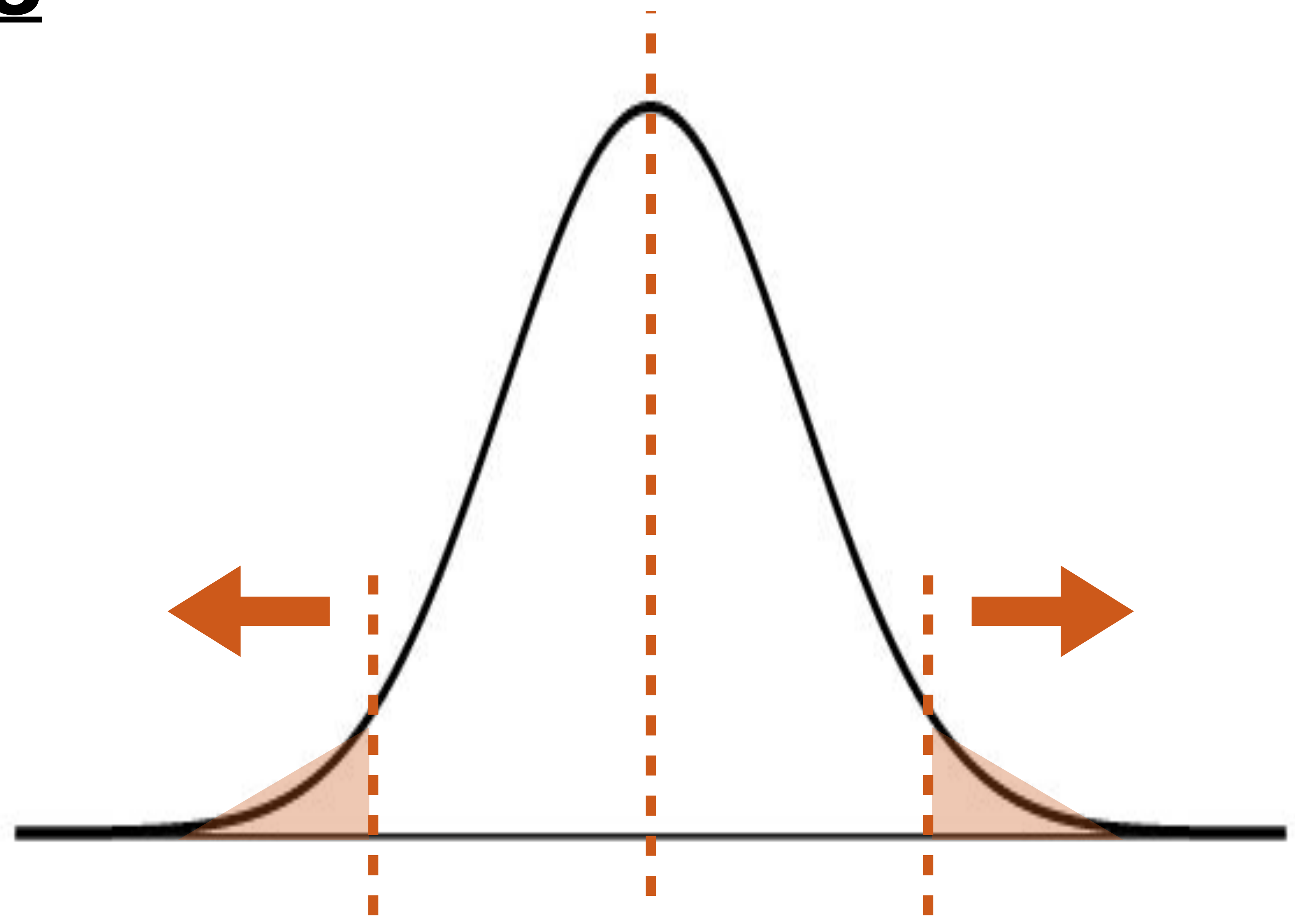
"The probability that the null hypothesis is true is 0.10, or 10%."

Explain why this student is incorrect about this interpretation.



Hypothesis Testing: *p*

PollEv.com/stats8



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"The probability that the null hypothesis is true is 0.10, or 10%."

Explain why this student is incorrect about this interpretation.

The p-value **does not** represent the probability of the null hypothesis — the null hypothesis does not have an associated probability because it is not random. The p-value represents the probability of seeing the data or more extreme data under the assumption that the null hypothesis is true.

Hypothesis Testing for a Proportion

General Procedure (for two-sided tests)

1. Write down the null and alternative hypotheses (statements about the **population parameter**)

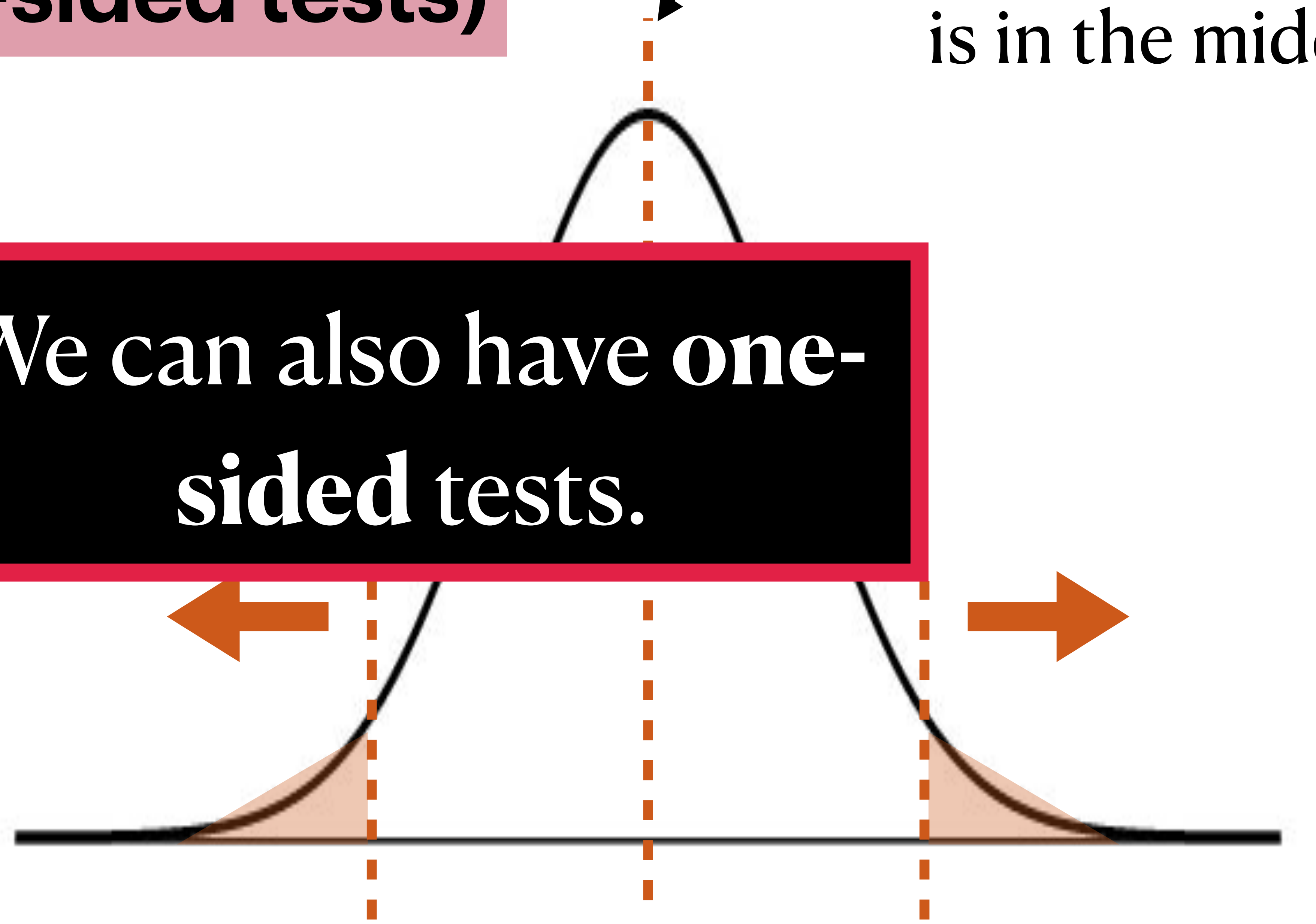
2. Assuming the null hypothesis is true, think about wha the sampling distribution of the statistic is.

   ➡ Are the conditions for CLT met? If so, we can say it is approximately normal.

3. Compute the test statistic. For this class, that means the z-score.

4. Knowing that if the statistic is approximately normal, then the z-score is $N(0,1)$, compute the p-value: $2P(Z \leq -|\text{z-score}|)$.

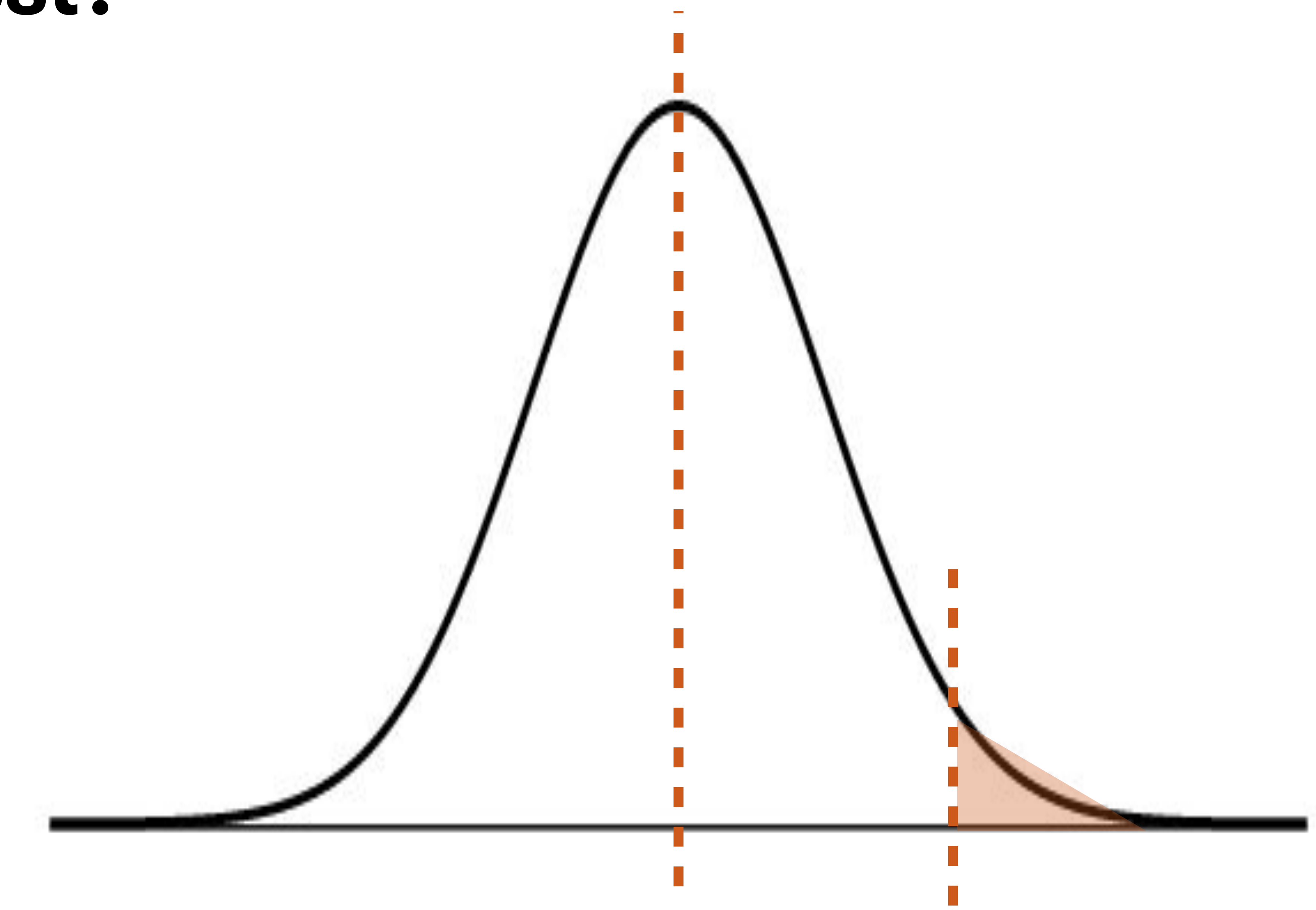
5. Determine whether the p-value is small enough to provide evidence against the null or not.



We can also have **one-sided** tests.

Hypothesis Testing: One-Sided

Why one-sided test?

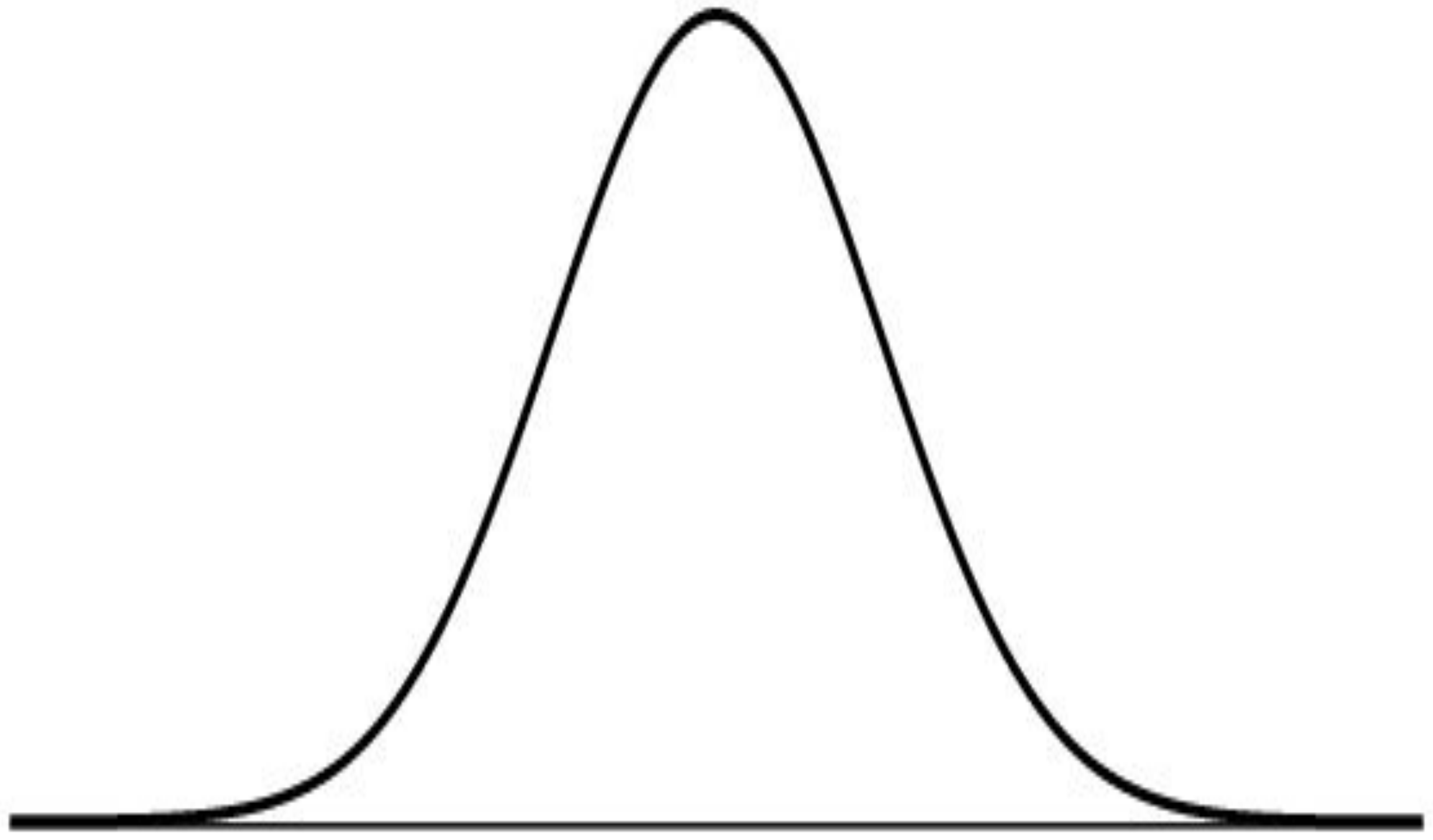


- Sometimes we want to test whether a parameter is larger than some value, or smaller than some value, rather than just "not equal to"
- Ex:
  - Testing for improvement; you want to show something is **better than** a benchmark or not.
  - Testing for harm/safety problems; you want to detect whether something is **worse than** acceptable.
  - Physical or theoretical constraints make one direction meaningless; sometimes only one direction is scientifically plausible or relevant.
- In all of these cases, we only care about one direction

Hypothesis Testing: One-Sided

HPV Vaccine Example

- Healthy People 2030 (source) has a target that at least 80% of adolescents should receive the recommended HPV vaccine doses. In a random sample of 120 adolescents from one clinic, 88 had completed the recommended doses. At a 10% significance level, is there evidence that this clinic is failing to meet the 80% target?



- Notice: we only want to see if there's evidence that the percentage is under 80%. The target isn't *exactly 80%*, it's *at least* 80%.
- With that in mind, what are the null and alternative hypotheses? (Remember, you can only find evidence *against* the null, not *for* the null)

Hypothesis Testing: One-Sided

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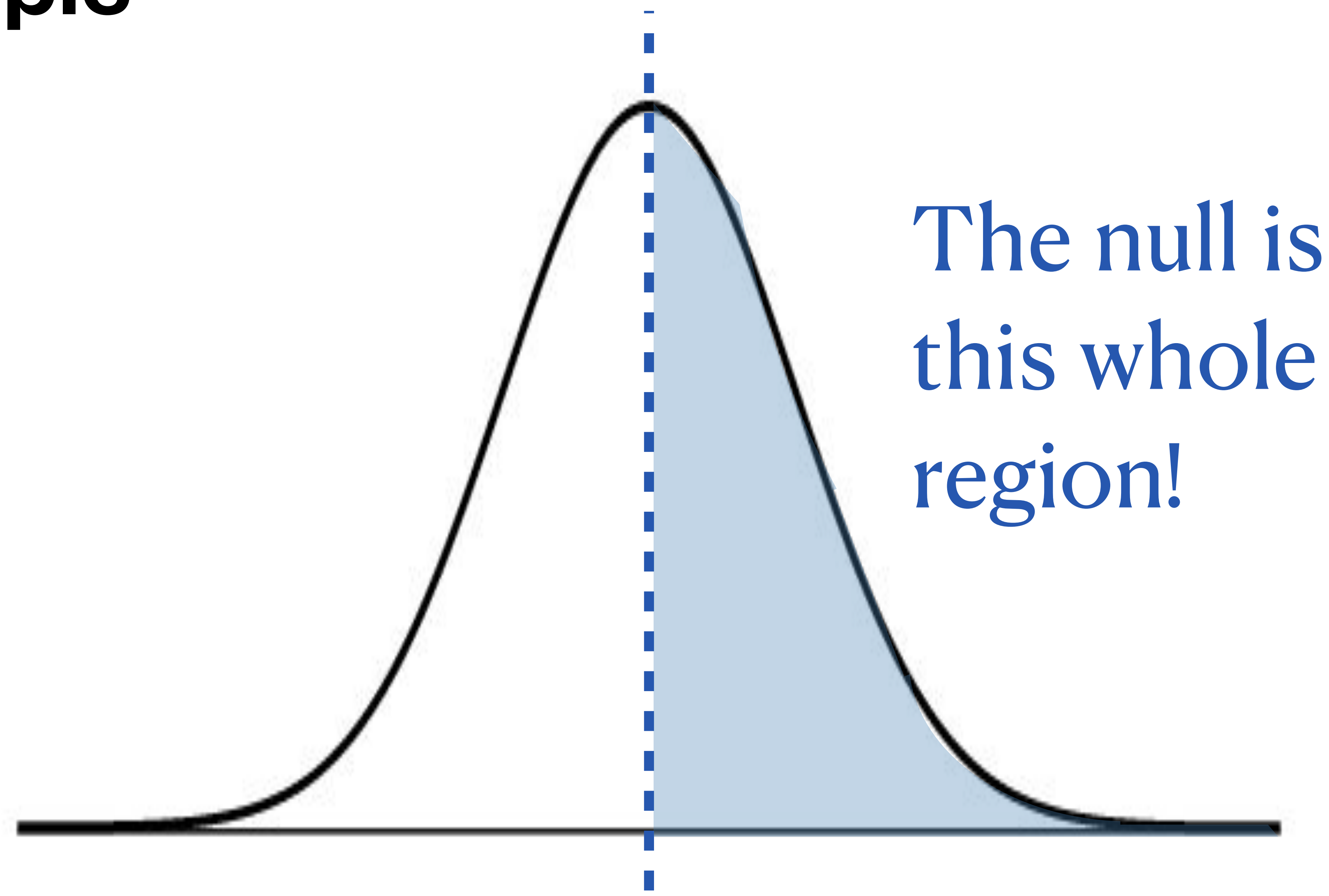
- What are the null and alternative hypotheses?



The null is this whole region!

$$H_0 : p \geq 0.8; \; H_a : p < 0.8$$

Hypothesis Testing: One-Sided

HPV Vaccine Example

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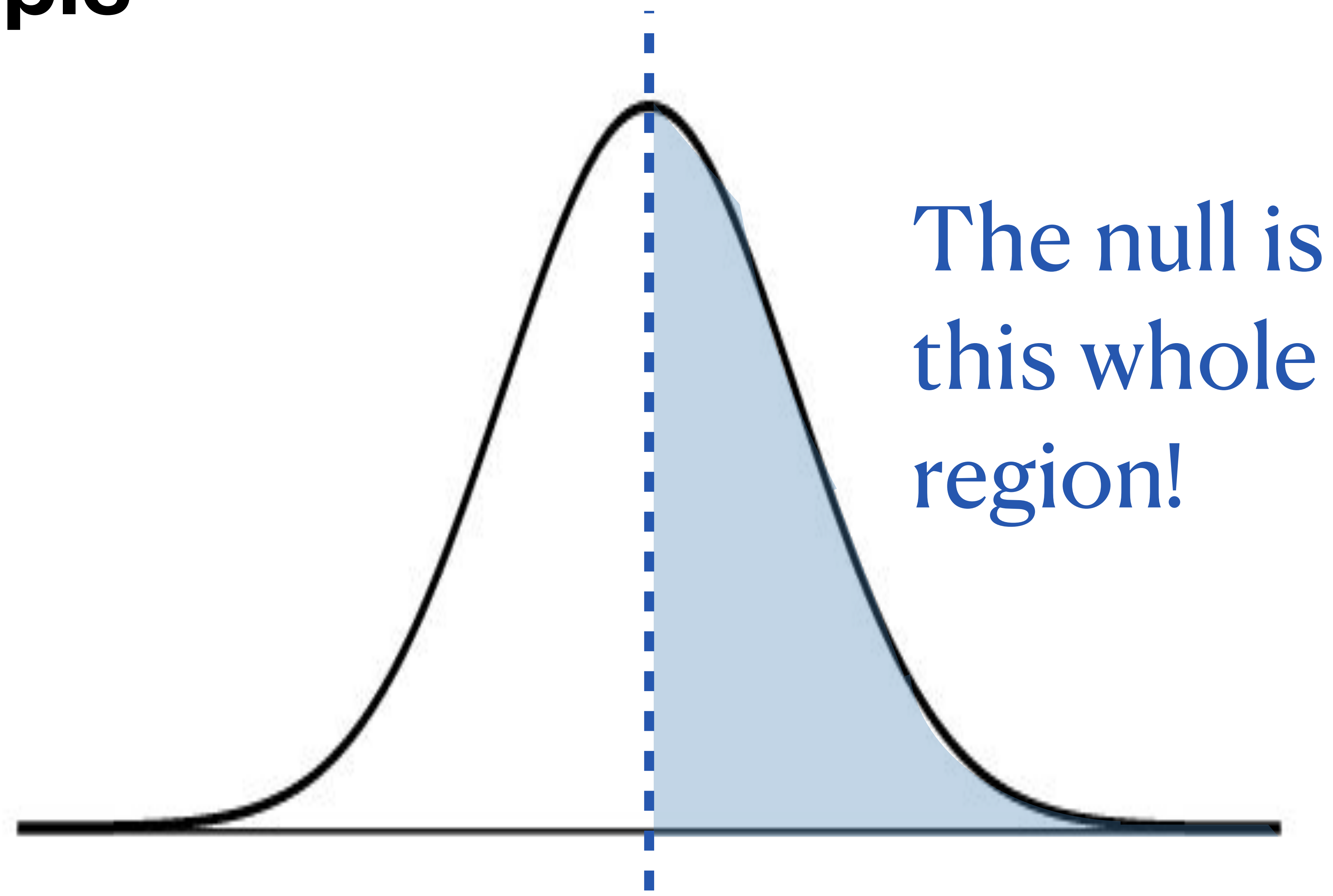

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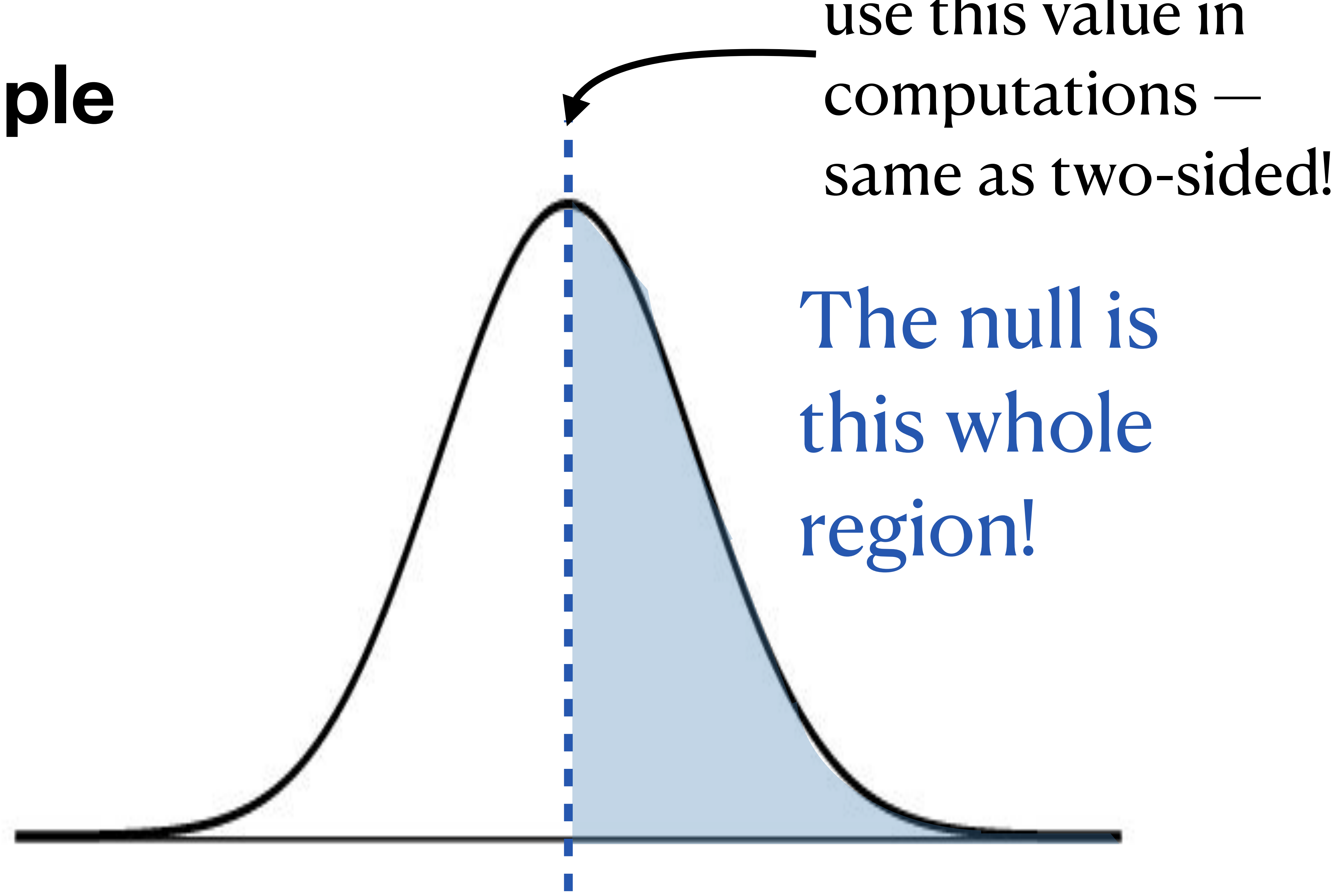

$$H_0 : p \geq 0.8; \; H_a : p < 0.8$$

Now, we proceed under the assumption that H_0 is true. But, under H_0, there are many possible values of p... we compute the test using the boundary value of 0.8, the value of the null that is closest to the alternative. It gives the largest p-value among values in the null — if we can reject at 0.8, we can reject at everything above 0.8, too.

Hypothesis Testing: One-Sided

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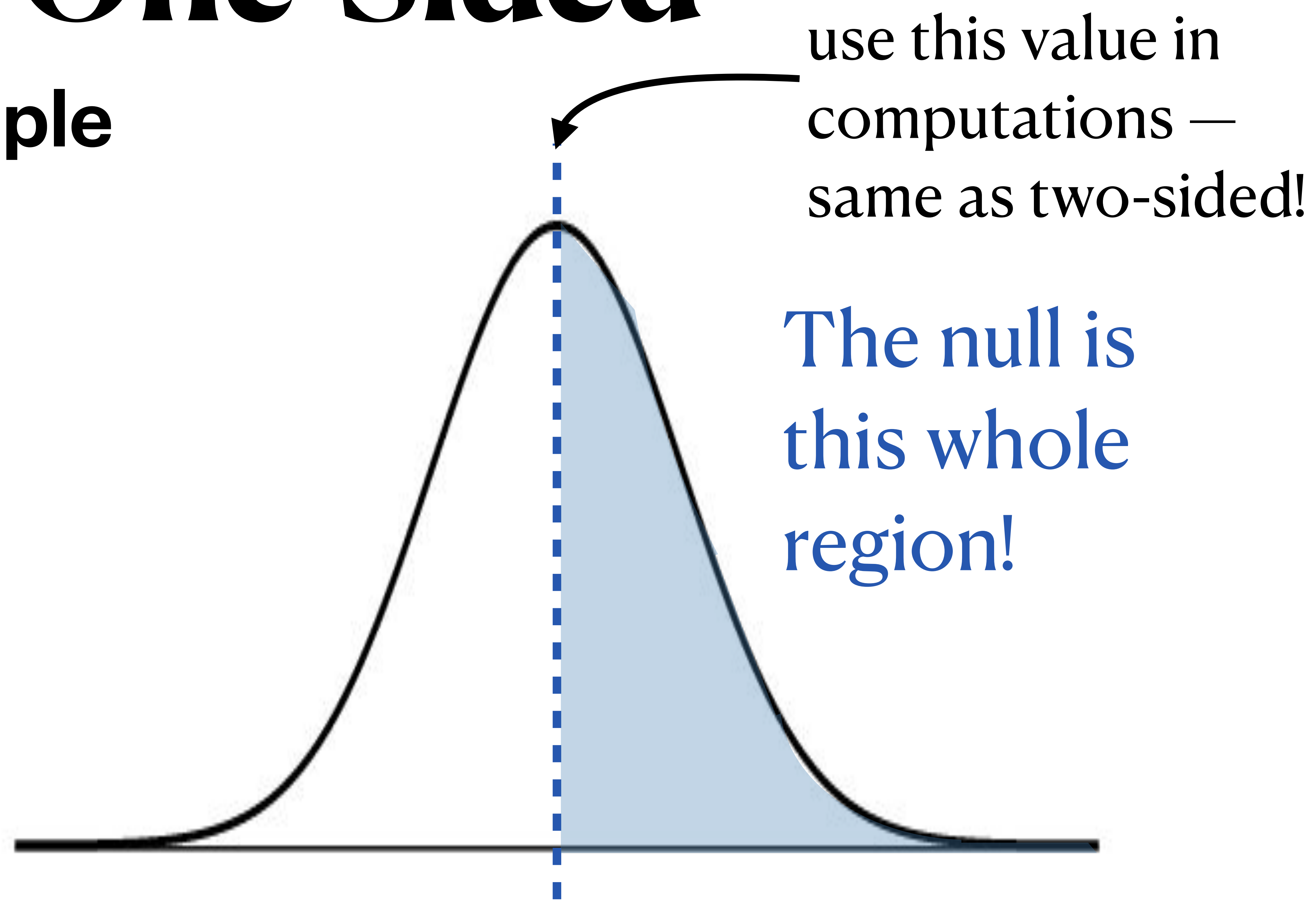
- Assuming $p = 0.8$, what is the sampling distribution of $\hat{P}$?



Hypothesis Testing: One-Sided

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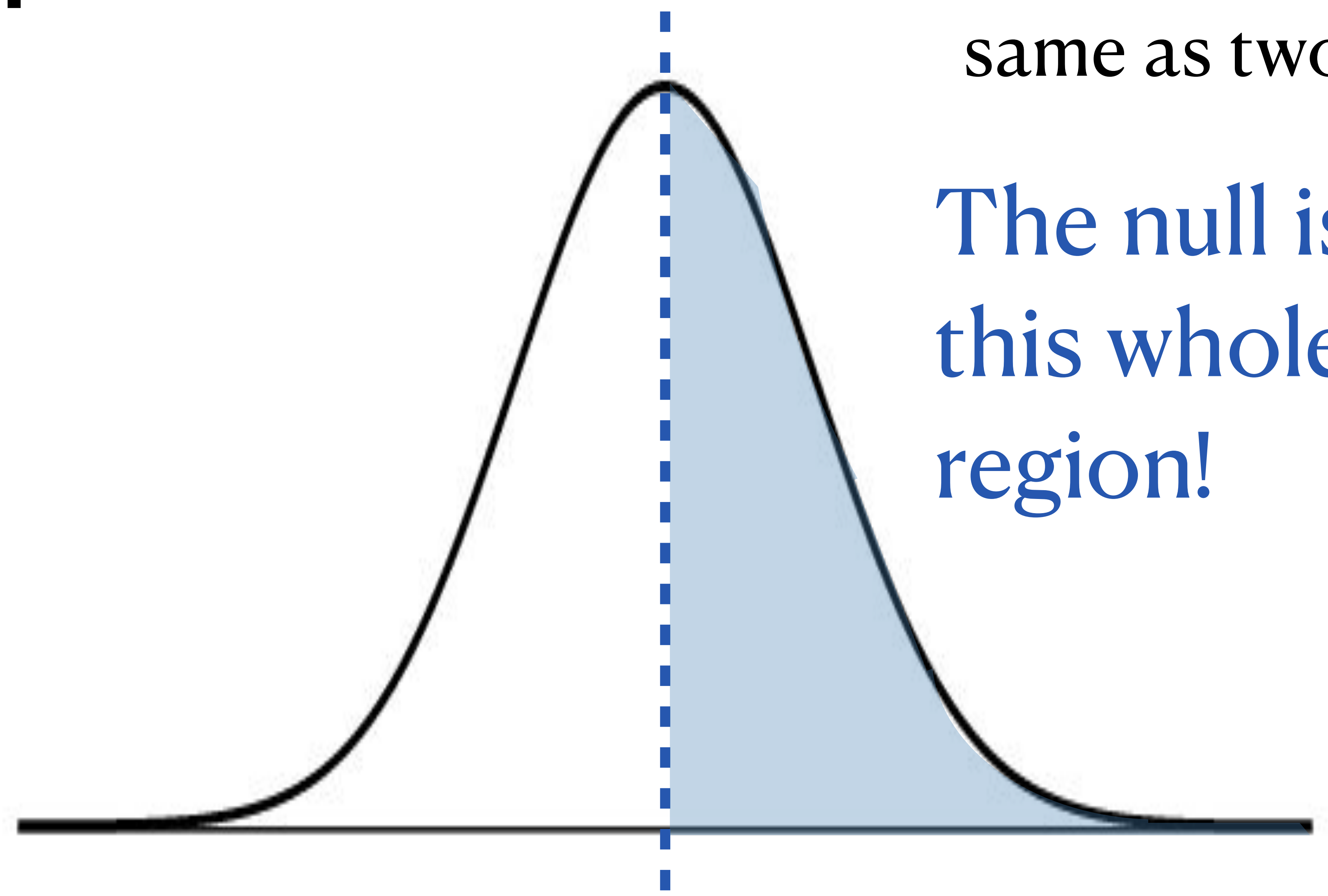
- Assuming $p = 0.8$, what is the sampling distribution of $\hat{P}$?
- i.i.d.: random sample from large population ✓
- Sample size: $120(0.8) > 10$, $120(1 - 0.8) > 10$ ✓
- CLT applies!

use this value in computations — same as two-sided!



The null is this whole region!

Hypothesis Testing: One-Sided

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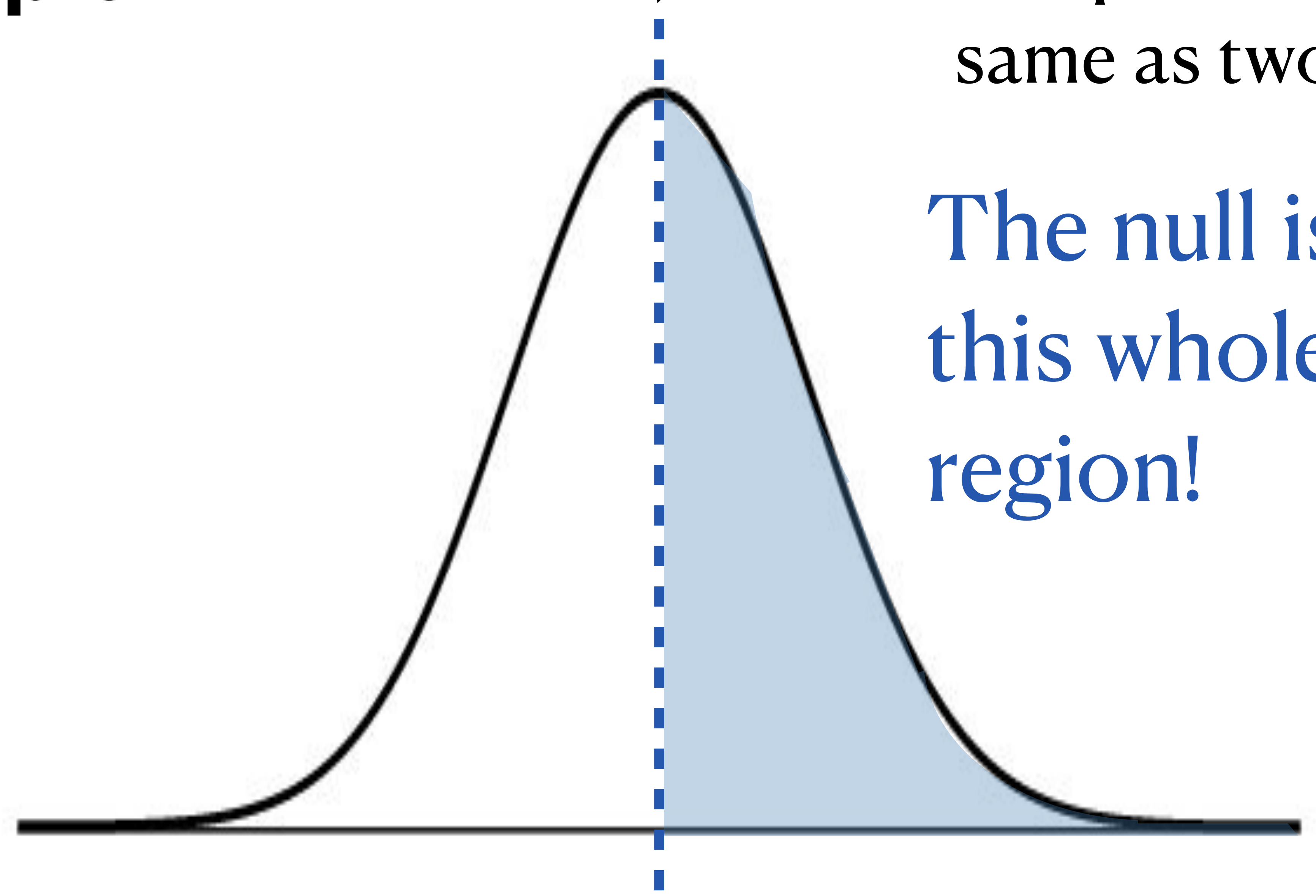
- Assuming $p = 0.8$, what is the sampling distribution of $\hat{P}$?

- Under CLT, we have that approximately:

$$\hat{P} \sim N\left(0.8, \sqrt{\frac{0.8(1-0.8)}{120}}\right)$$

use this value in computations — same as two-sided!



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Hypothesis Testing: One-Sided

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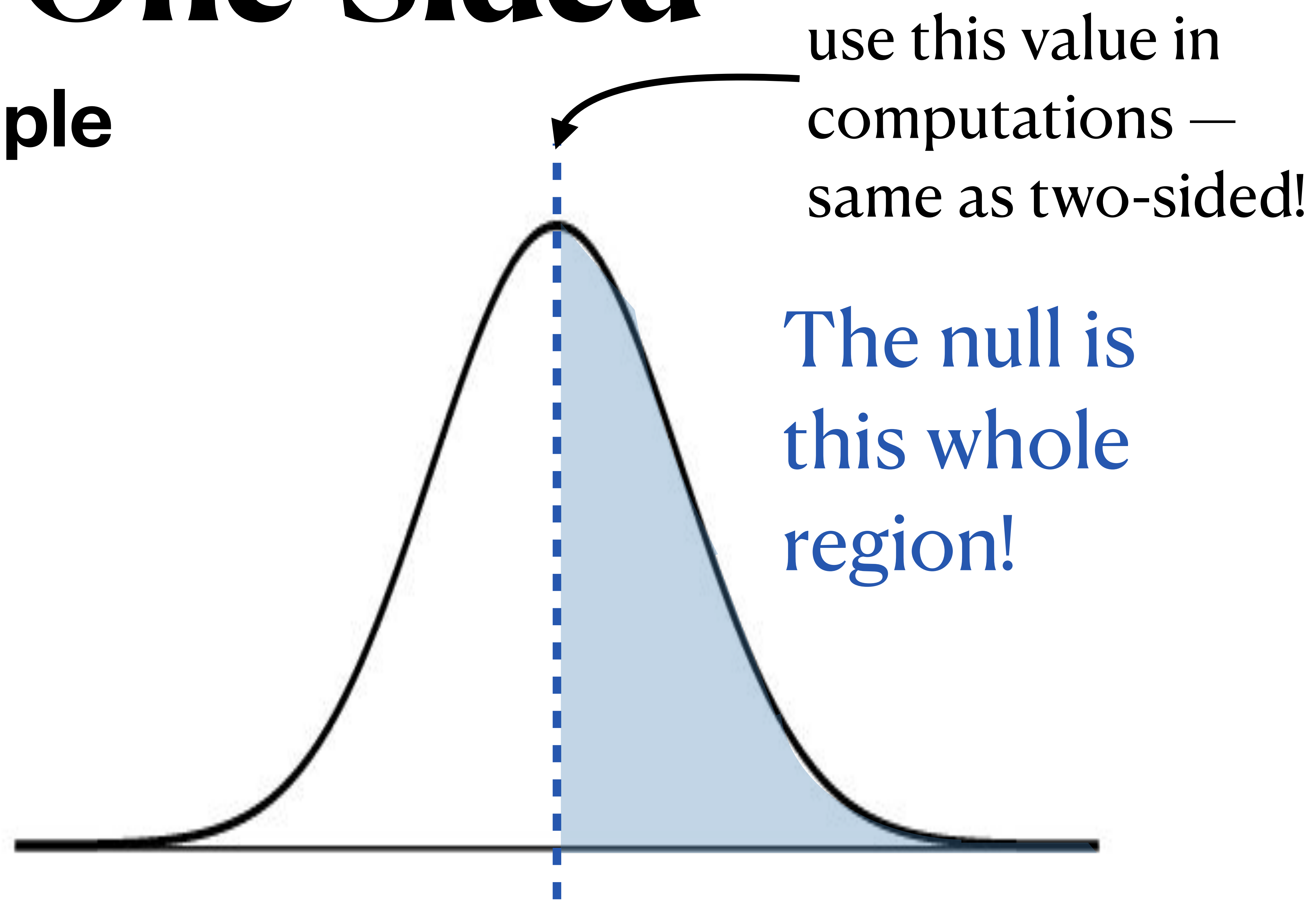
- What is the test statistic?



Hypothesis Testing: One-Sided

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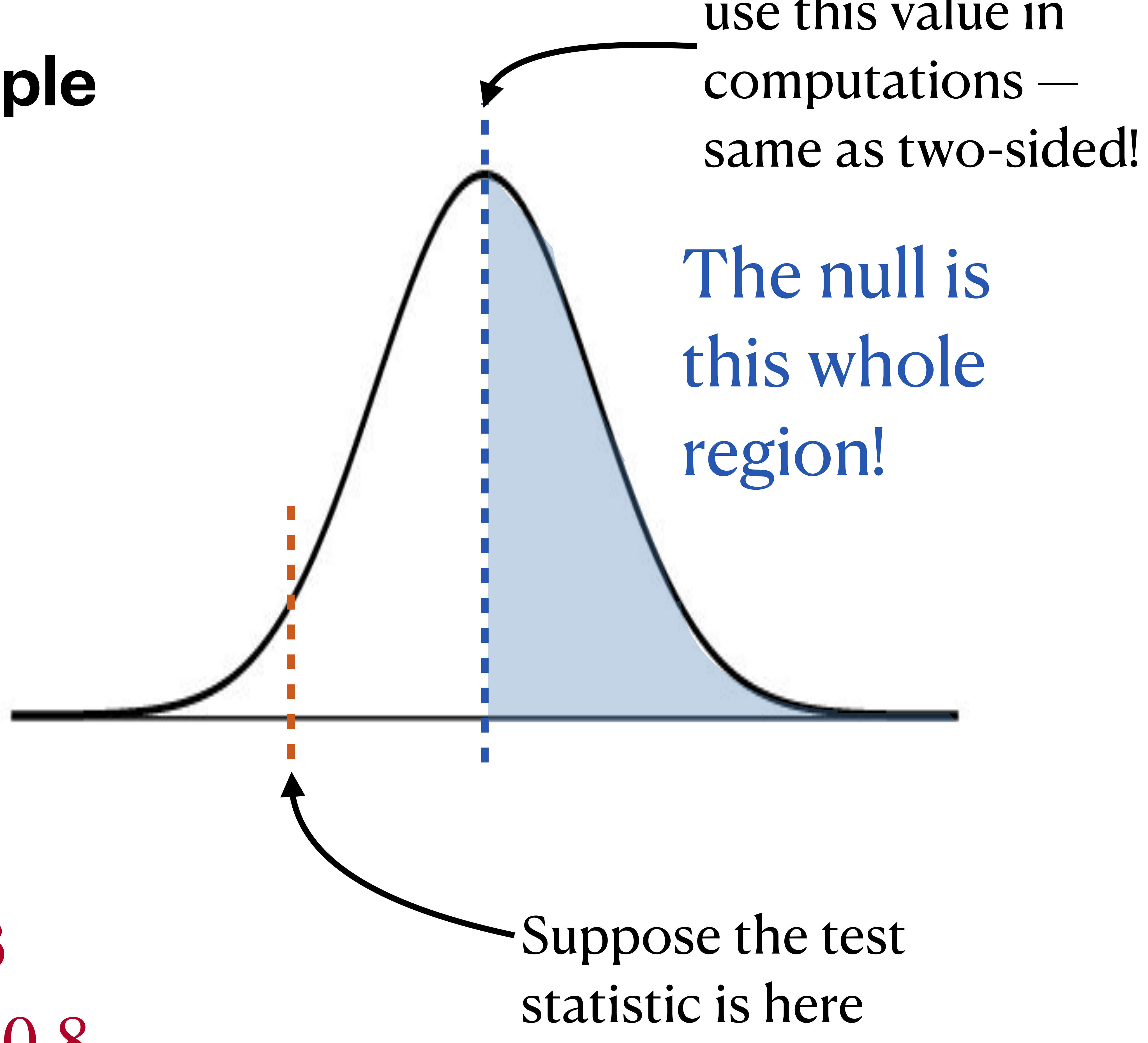

$$\hat{p} = 88/120 \approx 0.733$$

Test statistic (z-score) is: $\dfrac{0.733 - 0.8}{\sqrt{\dfrac{0.8(1-0.8)}{120}}} \approx -1.83$

Hypothesis Testing: One-Sided

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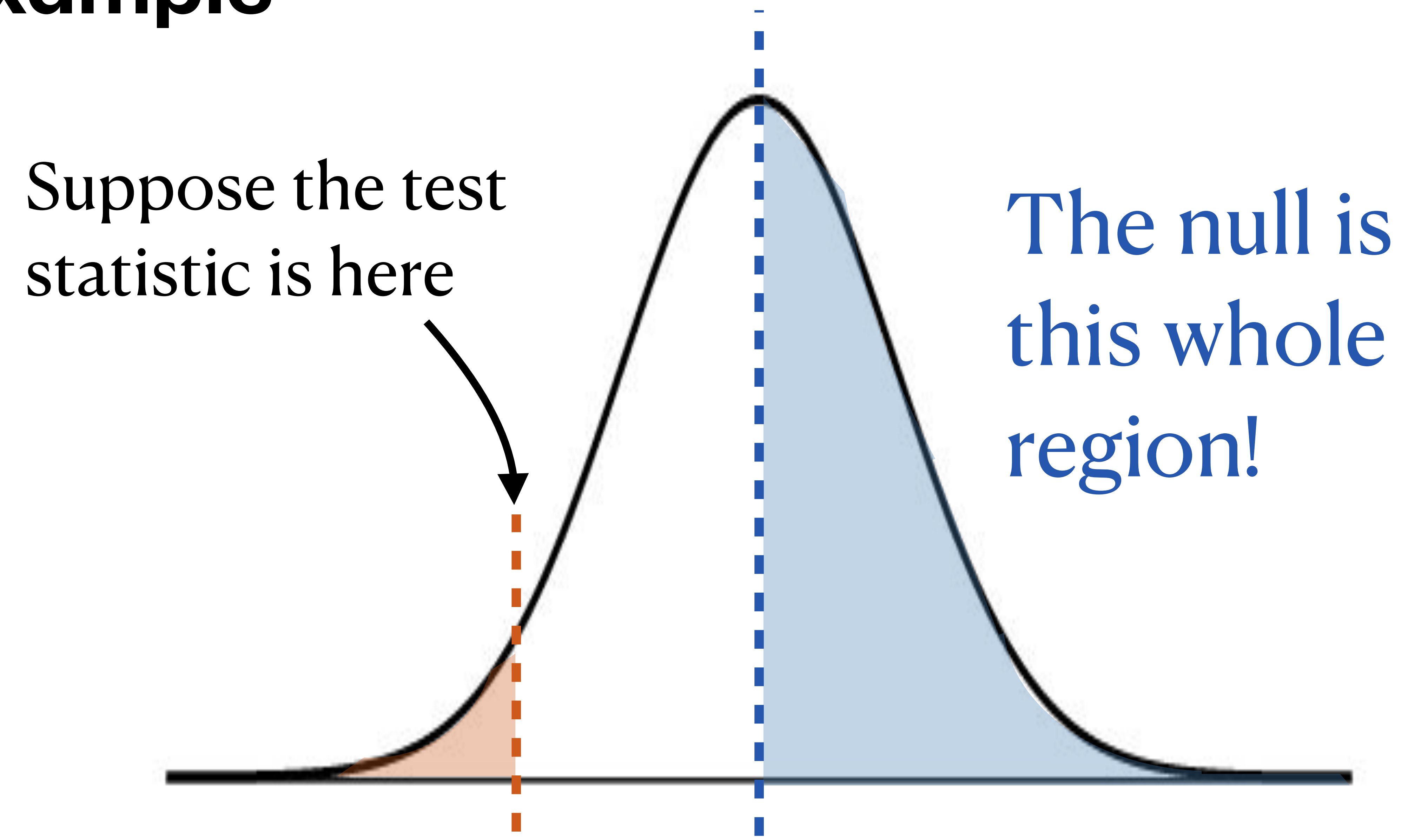

- Using this test statistic, what is the p-value?
- This is different now than it was in the two-sided case. Recall that a p-value is *the probability of the data or more extreme*, where "more extreme" means *further away* from the null hypothesis
- Since the null hypothesis covers the whole upper tail, we only want the probability from the lower tail

Hypothesis Testing: One-Sided

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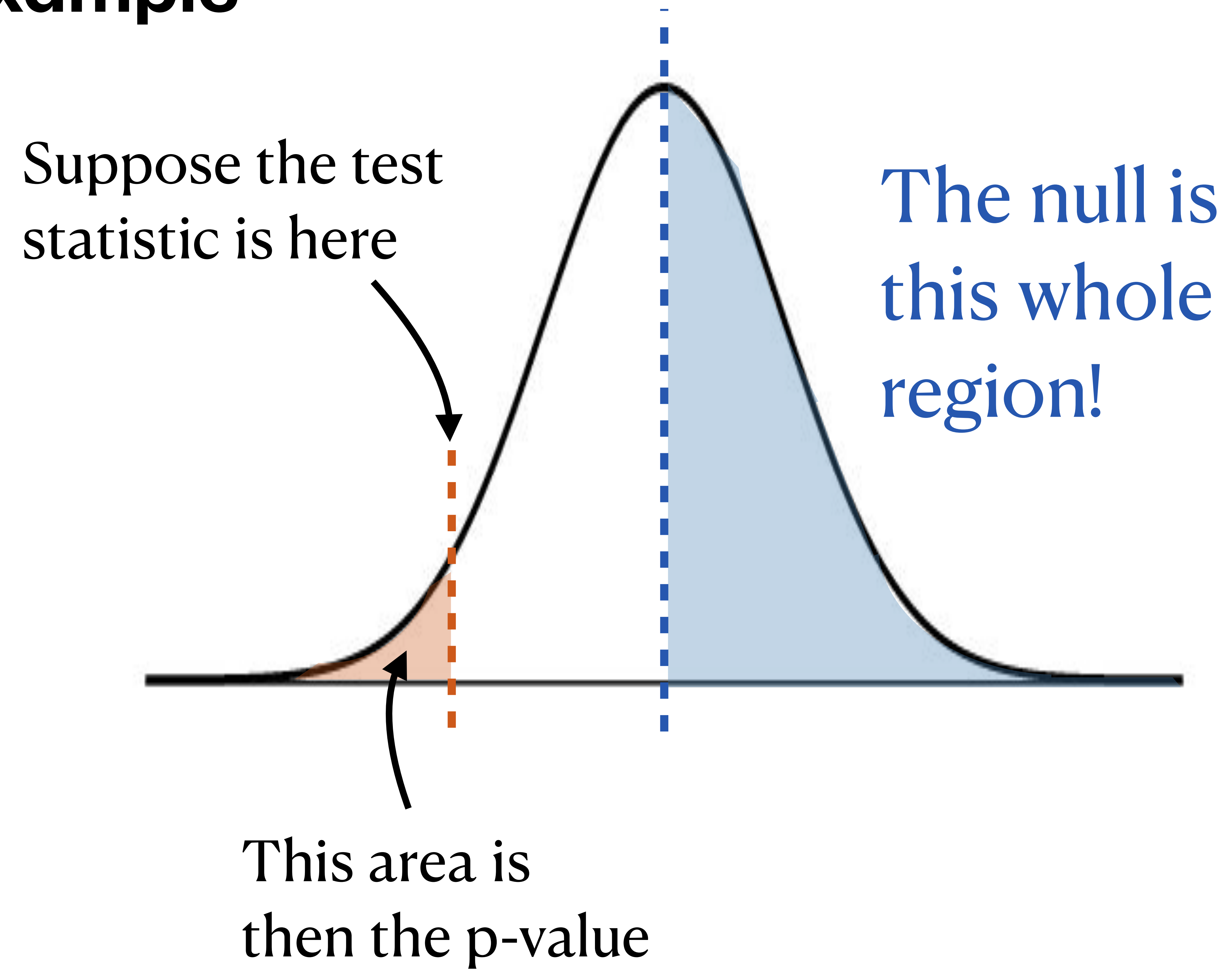
$P(Z < -1.83)$ = pnorm(-1.83) ≈ 0.034



Hypothesis Testing: One-Sided

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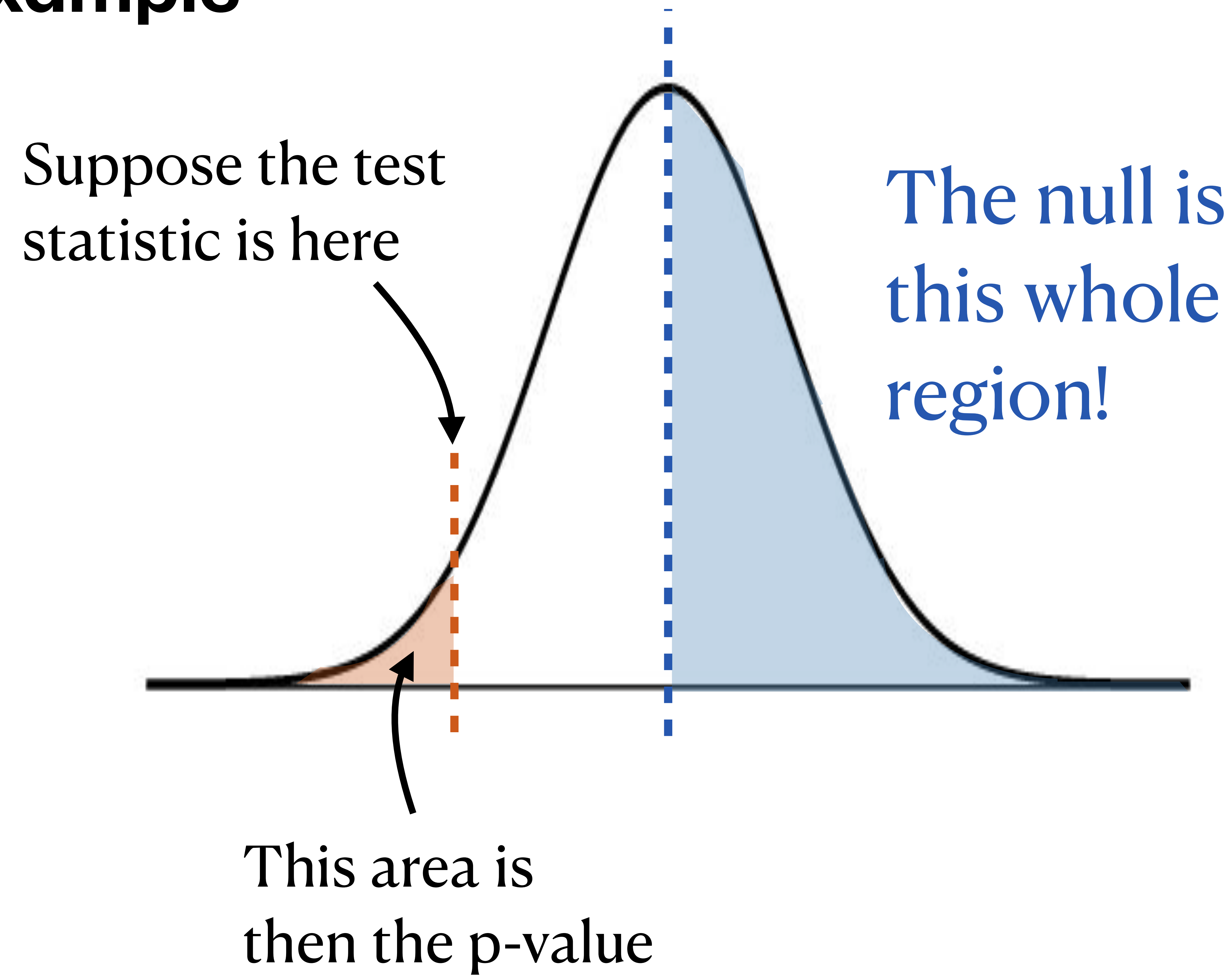
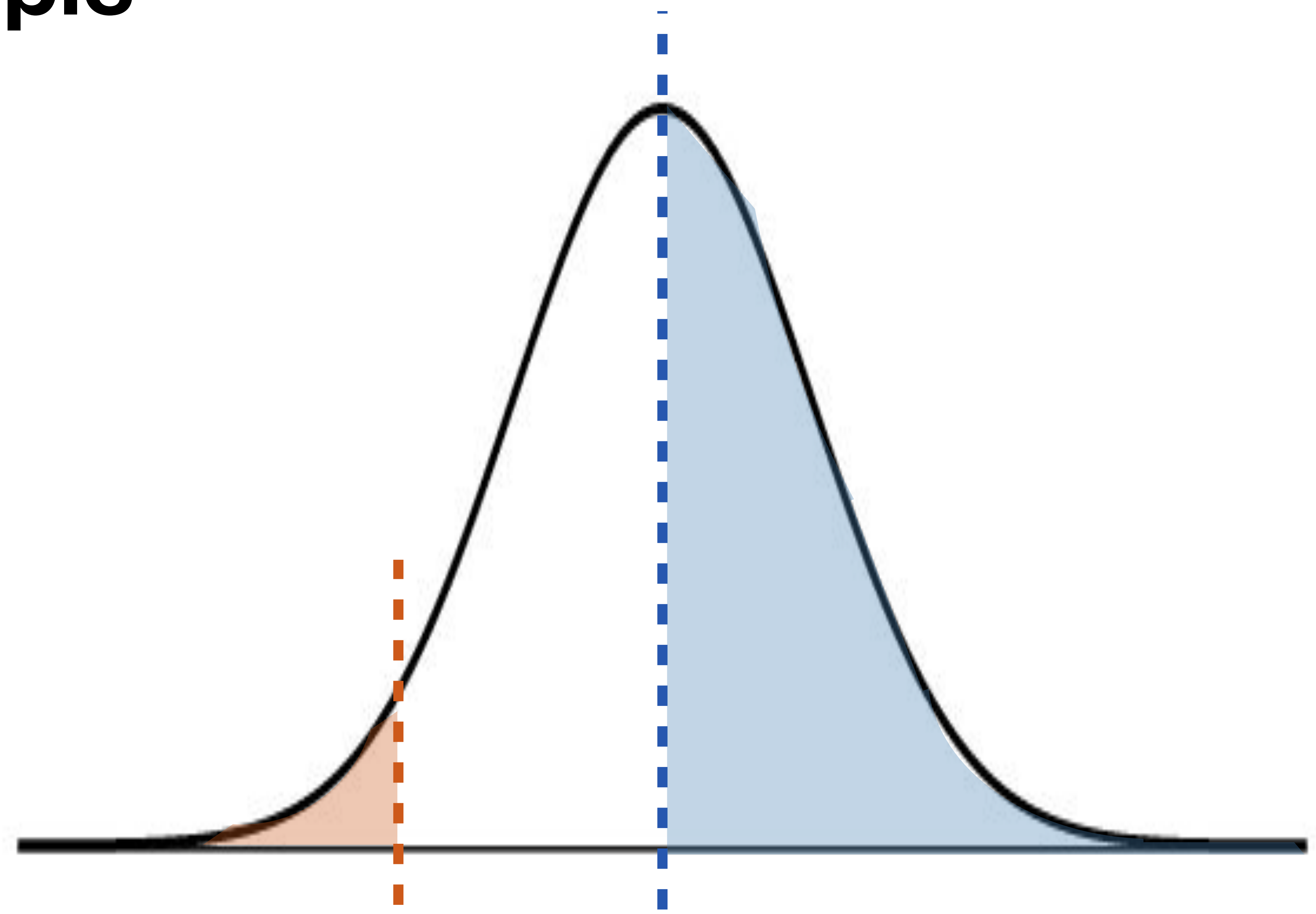
- With a p-value of 0.034, what can we conclude?



Hypothesis Testing: One-Sided

HPV Vaccine Example



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- With a p-value of 0.034, what can we conclude?
- At a significance level of 10%, we reject the null hypothesis (*p-value = 0.034*) that this clinic is meeting the 80% target. We have evidence to say that they are failing to meet the target.

Hypothesis Testing: One-Sided

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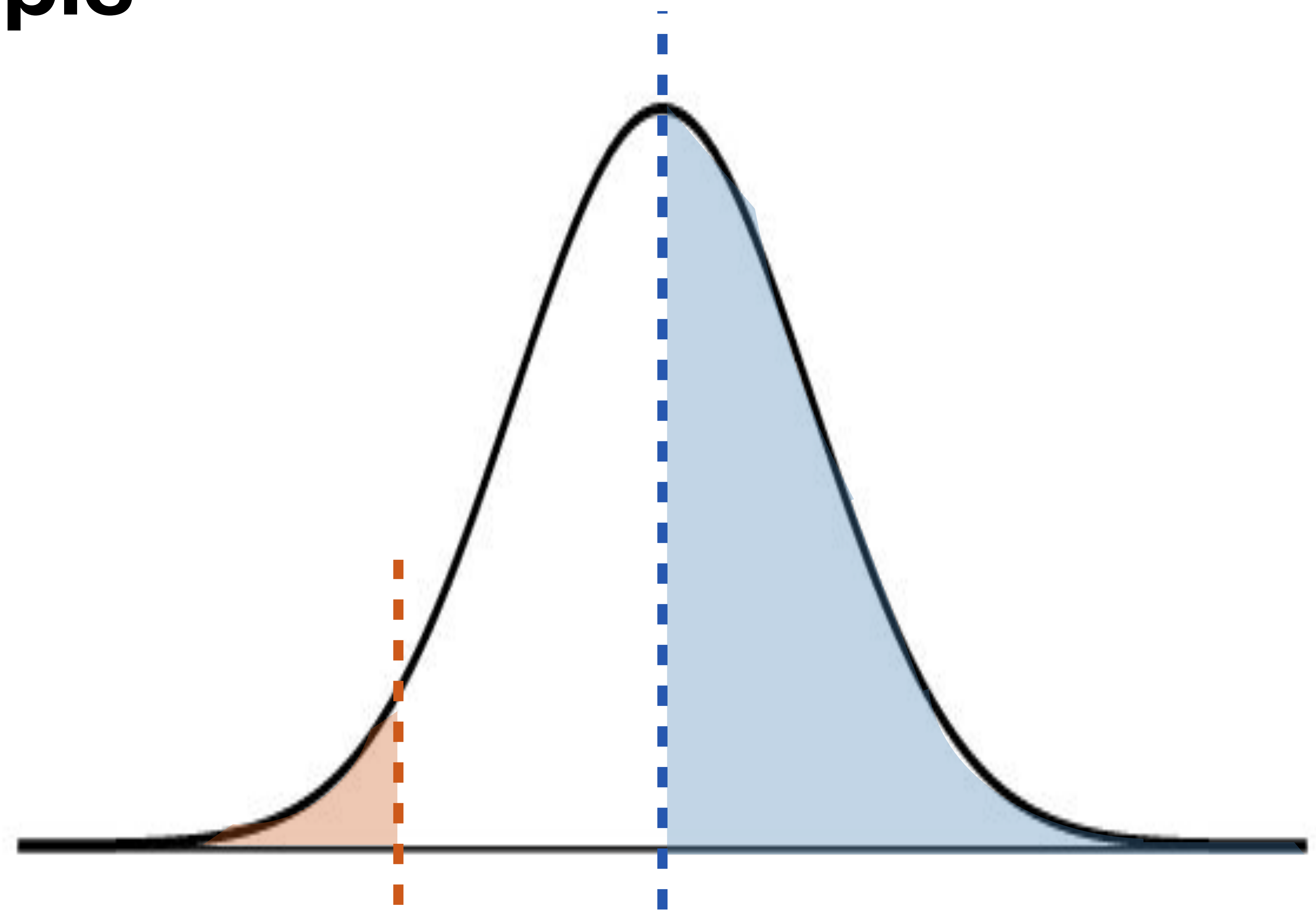

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- At a significance level of 10%, we reject the null hypothesis (*p-value* = *0.034*) that this clinic is meeting the 80% target. We have evidence to say that they are failing to meet the target.
- PollEv.com/stats8: A quick aside: if we had HPV vaccine information on *every* adolescent who goes to that clinic, how would our procedure change?

Type I and Type II Error

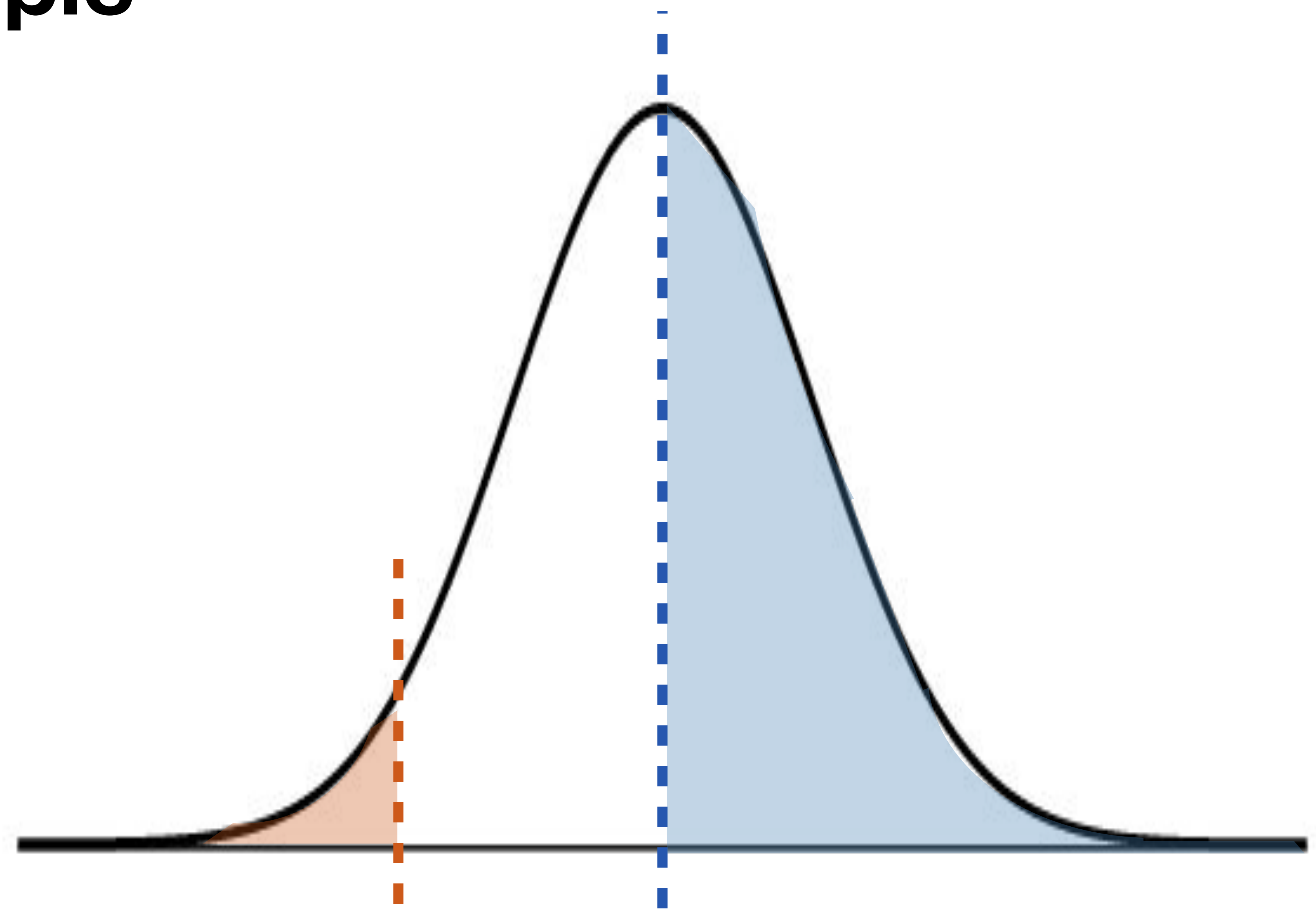
Hypothesis Testing

- The significance level, α, gives the proportion of the time that we can reject the null *even if it is true*
  - Rejecting the null when it is in reality true is called a **Type I error**
  - Failing to reject the null when it is in reality false is called a **Type II error**
  - **We can't know if we are making an error or not!**
- Helpful mnemonic: if we **fail 2 reject**, we could be making a **Type 2 error**

	Null Hypothesis is TRUE	Null Hypothesis is FALSE
Reject null hypothesis	Type I Error (False positive)	Correct Outcome! (True positive)
Fail to reject null hypothesis	Correct Outcome! (True negative)	Type II Error (False negative)

Hypothesis Testing: One-Sided

HPV Vaccine Example



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- At a significance level of 10%, we reject the null hypothesis (*p-value* = *0.034*) that this clinic is meeting the 80% target. We have evidence to say that they are failing to meet the target.
- PollEv.com/stats8: What type of error could we be making, and why?

Practical Significance vs. Statistical "Significance"

- Sometimes, when the sample size is very very large, we get so much precision in our inference that we can reject the null hypothesis when the sample statistic is very close to the null hypothesis
- E.g.: We want to see how effective a new blood pressure drug is, compared to an old one. The old drug reduces blood pressure by 10 mmHg, on average.
- We test the drug on 20,000 people and test whether the new drug is better. The p-value is less than 10^{-16} (tiny!). This must mean that the new drug is *really* good, right?
  - No! A plausible situation:
    - Sample mean reduction was 10.1 mmHg, with a sample standard deviation of 1 mmHg. The standard error of the mean is $1/\sqrt{20000} \approx 0.007$ mmHg.

Hypothesis Testing: How-To

- So, you're given a scenario and you need to perform a hypothesis test. What do you do?
  1. Confirm what **parameter** you want to test (for now, just mean or proportion).
  2. Write down the **null** and **alternative hypotheses**.
  3. Check whether **assumptions/conditions** for the CLT (so, for the interval) are met (i.i.d. + sample size "rules of thumb").
  4. **Identify** the relevant quantities provided; compute the **test statistic** (by taking the sample statistic, subtracting its mean, and dividing by its standard error).
  5. Compute the **p-value** based on the test statistic and the null/alternative hypotheses.
  6. Compare the **p-value** to the pre-specified **significance level**. Make a decision: do you reject or fail to reject the null?
  7. **Interpret** your conclusion in context.
  8. Identify whether you could have made a Type I or Type II error. Consider whether your results are practically significant, or just statistically significant.

The point isn't the procedure…

- I know that because this is a class, it is easy to only care about how to get the question right (memorizing the *procedure*)
- What's more important for later is understanding the logic behind it all
- There are more types of CIs and hypothesis tests (some of which we will learn, most we won't) but the important thing is that once you learn the idea behind each…
    - everything else is the same!
- CIs are always (point estimate $\pm$ margin of error)
- Hypothesis tests are always "assume the null is true, see if the data disagree"

Topic 7: Inference for One Population